Biochemistry

Workbook

For FMG Doctors

Dr. Sayantaann Saha

MD - Gold Medalist

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

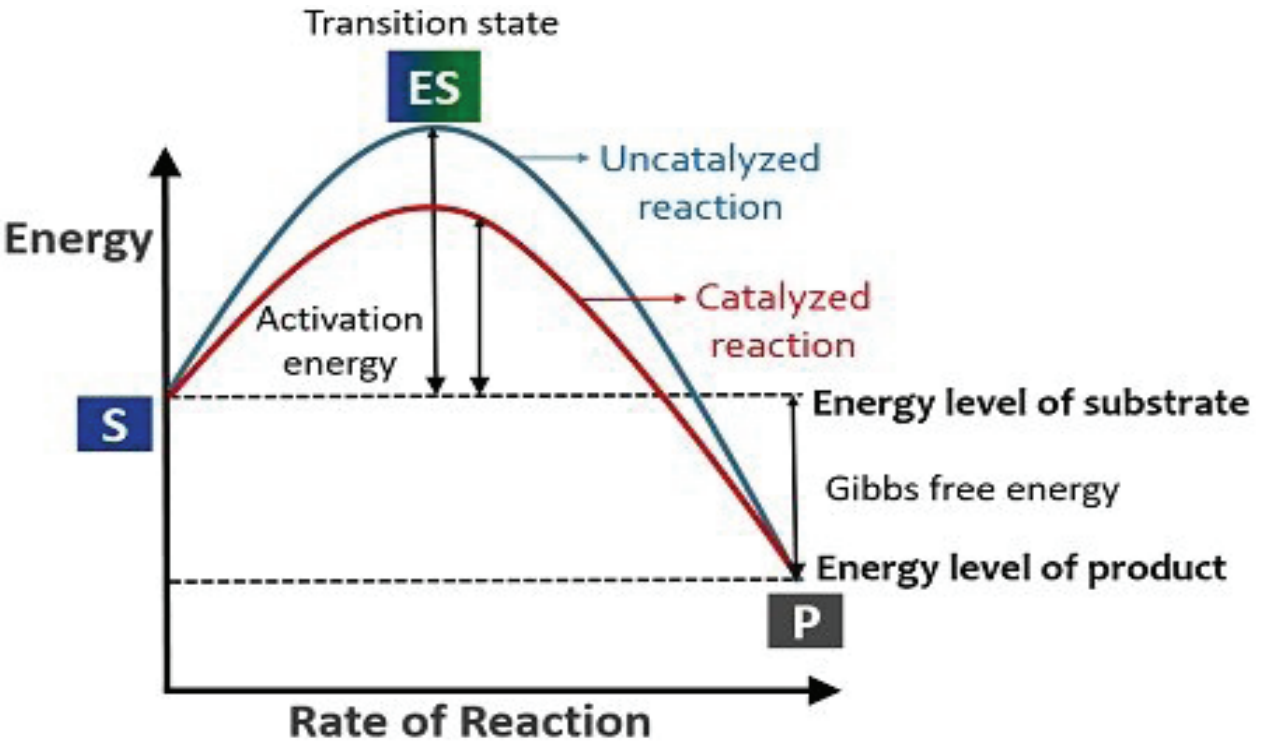
- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

© All Rights Reserved with MIST. No part of this publication shall be reproduced, copied, transmitted, adapted, modified or stored in any form or any means. If found at any point of time, shall be penalized as per the Courts of Law.

Contents

1. Enzymes & its Clinical Importance 4-37
2. Biological Oxidation & ETC coupled to Oxidative Phosphorylation 38-45
3. Carbohydrates Roles & Importance 46-56
4. Carbohydrate Metabolism 57-93
5. Chemistry of Amino Acids & Proteins 94-113
6. General Amino Acid & Protein Metabolism 114-123
7. Individual Amino Acid Metabolism & Associated Disorders 124-144
8. Lipid Chemistry & Metabolism 145-165
9. Lipoprotein Chemistry & Metabolism 166-176
10. Nucleotide Chemistry and Metabolism & Related Disorder 177-180
11. Extended Self Reading on Vitamins 181-194

ENZYMES & ITS CLINICAL IMPORTANCE



Enzymes are **Biocatalyst** that increase the velocity of the reaction by **DECREASING the ACTIVATION ENERGY**.

Enzymes are primarily proteins that are water-soluble+heat labile

$1° \rightarrow 2° \rightarrow 3° \rightarrow 4°$

MEDICAL INSTITUTE FOR SCREENING TEST MIST

REPLICATION --

TRANSCRIPTION --

*Post Transcriptional Modification

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

TRANSLATION --

POST TRANSLATIONAL MODIFICATION--

RIBOZYMES: Ribozymes are special RNAs having Enzymatic activity

E.g. 1. Peptidyl transferase →

2. Ribonuclease-P →

*All Proteins are Enzymes BUT All Enzymes are NOT Proteins

What is ABZYME?

✹ COENZYMES

Protein part of the enzyme is termed as apoenzyme

Coenzyme is non-protein part, called the prosthetic group.

Holoenzyme = apoenzyme +coenzyme

Coenzymes are mostly Water Soluble VITAMIN,

Mode of Action: Accept One of the PRODUCTS of the REACTION.

Bi Bi Reactions:

Vitamin	Coenzyme form aka Active Form	Functions
B1 - Thiamine	Thiamine pyrophosphate (TPP) or Thiamine Diphosphate (TDP)	1. OXIDATIVE DECARBOXYLATION 2. TRANSKETOLATION

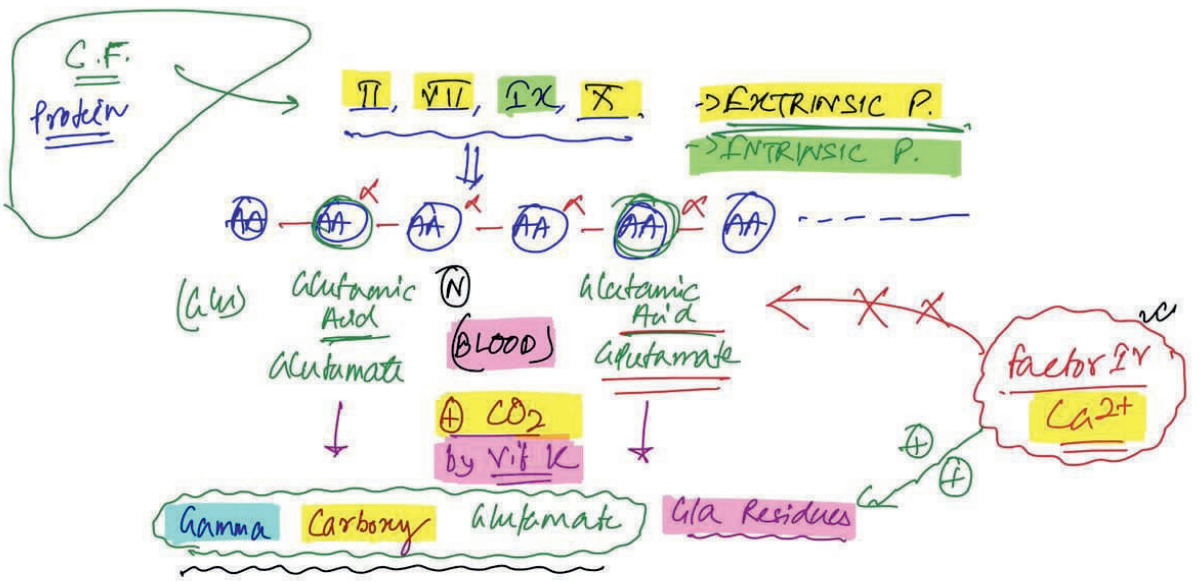
B2 - Riboflavin	Flavin Adenine Dinucleotide	DEHYDROGENATION
B3- Niacin	Nicotinamide Adenine Dinucleotide	

B5-Pantothenic Acid	Coenzyme A (CoA)	ACYL group Carrier

B6- Pyridoxine ONLY W.S.V with SIGNIFICANT TOXICITY called as PERIPHERAL SENSORY NEUROPATHY that presents with	Pyridoxal phosphate (PLP) → 80% of B6 that we eat in our diet is stored in Liver and Muscle as part of Glycogen → Despite of this, in B6 deficiency this STORAGE is USELESS → Its ONLY Available in STARVATION	1. NON-OXIDATIVE DECARBOXYLATION 2. TRANSAMINATION 3. OXIDATIVE DEAMINATION 4. Glycogen Phosphorylase in GLYCOGENOLYSIS

B7- Biotin (.....)	D - Biotin	CARBOXYLATION
B9- Folic Acid	Tetrahydrofolate (THF)	ONE CARBON POOL or One carbon group transfer viz. CH3, CH2, CH, HCHO but Except →
B12- Cobalamin	Methyl cobalamin	TRANSMETHYLATION Reactions
Vitamin C	L- Ascorbate	HYDROXYLATION Reactions viz. Mostly in COLLAGEN Synthesis

Vitamin K ONLY Fat-Soluble vitamin with a precise COENZYME function	Mainly 3 Forms ◊ Phylloquinone in plants. ◊ Menaquinone produced by the intestinal bacteria and also found in animals. ◊ Menadione → ONLY water soluble form. Importance:	GAMMA CARBOXYLATION or POST TRANSLATIONAL MODIFICATION of GLUTAMATE (Glutamic Acid) of Clotting Factors II, VII, IX, X, Protein C & Protein S to form GAMMA CARBOXY GLUTAMATE (Gla Residues) that act as Chelation Site for Binding of Calcium (Factor IV)



EXTRA EDGE:

NON VITAMIN Coenzyme:

SAM LIPOIC ACID, BIOPTERIN

✹ COFACTORS

METAL COUNTERPART OF COENZYME → METALLOENZYMES

*What is Metal activated enzyme?

1. Zinc Containing Enzymes:

Super Carbon P A G A L

ADA : ADENOSINE DEAMINASE

• -ADA deficiency leads to severe combined immunodeficiency (SCID) for which 1st successful gene therapy was done. • -In SCID there is combined deficiency of T & B lymphocytes and NK cells • MCC of death	• A value of CSF ADA > 8-10 IU/L denotes TB meningitis. • Whereas A value of Ascitic Fluid ADA ≥ 36 IU/ L denotes TB Peritonitis • However, in Both cases Dx must be confirmed by presence of at least 1 Acid fast bacilli in ZN stain

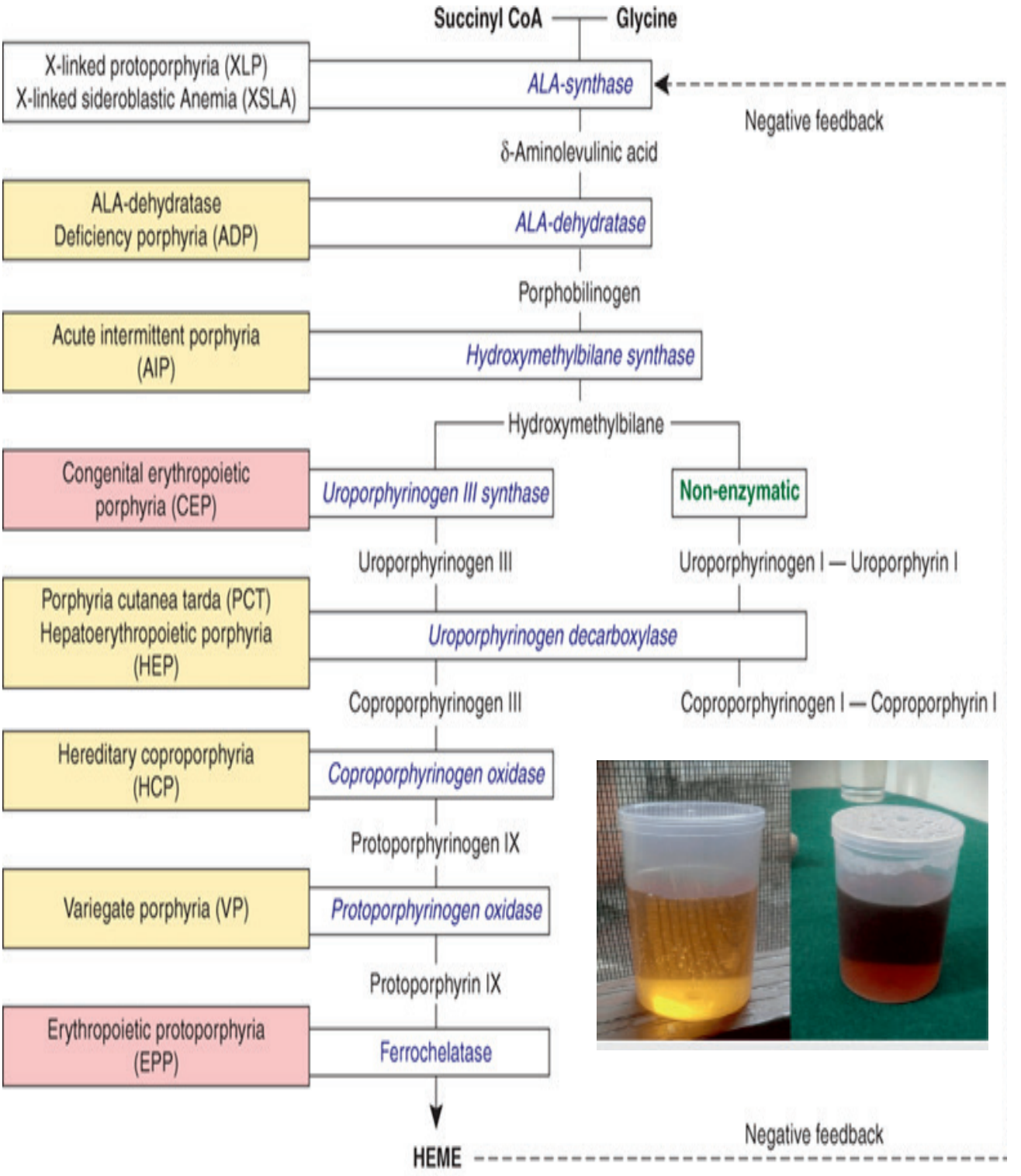
2. Copper Containing Enzymes

P C M S A L T D

❑ ALA SYNTHASE:

SUCCINYL CoA + GLYCINE .....→ δ Amino Levulenic Acid → → → → → HEME

HEME SYNTHESIS AND PORPHYRIAS



GOLDEN MCQ POINTS

◊ Heme synthesis is in Liver and Erythroid Cells viz. Bone marrow

◊ Occurs partly in Cytosol & partly in mitochondria.

◊ 1st step is in Mitochondria → Then till Coproporphyrinogen III is in Cytosol → Then remaining again in Mitochondria

◊ ALA synthase reaction is PLP / B6 dependent and is the RLE & Committed step.

→This enzyme deficiency DOESN'T lead to porphyria usually but causes Pyridoxine responsive X-linked sideroblastic anemia (XLSA)

◊ ALA dehydratase in Zn dependent & inhibited by Lead.

◊ Porphobilinogen Deaminase is AKA Hydroxymethylbilane Synthase → 4 Molecules of PBG produce 1 HMB

◊ Ferrochelatase incorporates IRON (Fe2+) in Heme and inhibited by Lead.

◊ VAMPIRE's disease: Congenital Erythropoietic Porphyria, a RARE disorder that presents with Red-brown teeth discoloration aka ERYTHRODONTIA, hypertrichosis and PHOTOSENSITIVITY (Occassional BLISTERS)

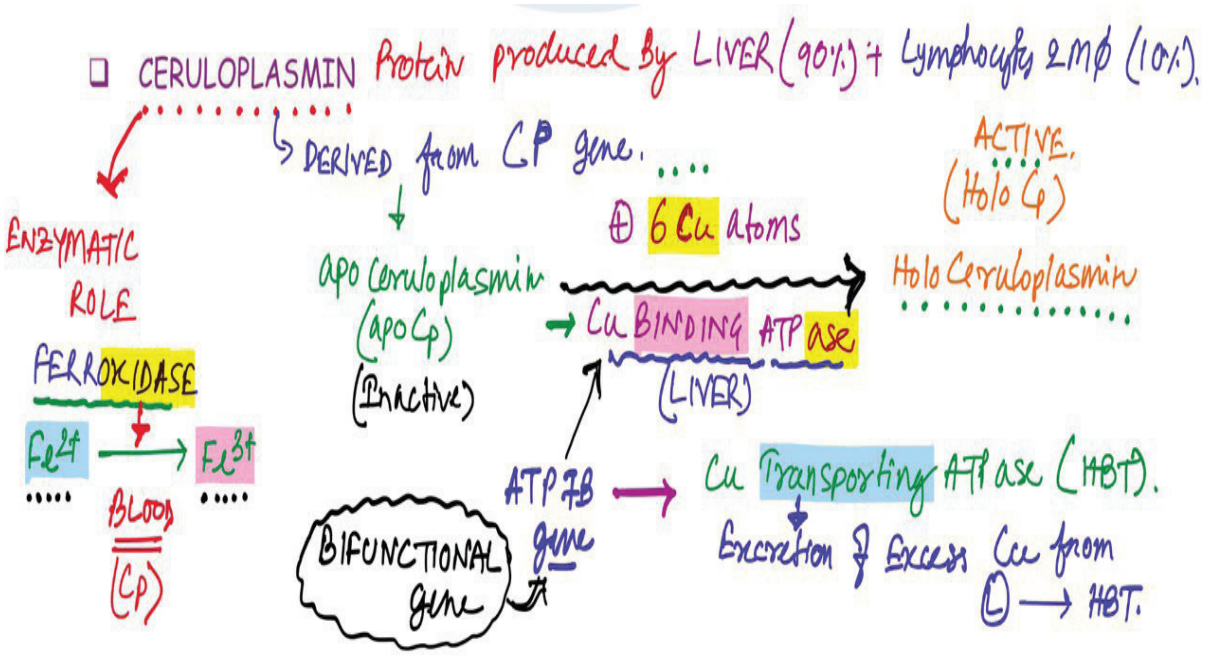
◊ MC Porphyria in INDIA is PORPHYRIA CUTANEA TARDA that presents with PHOTOSENSITIVITY

◊ MC Acute Porphyria in India is ACUTE INTERMITTENT PORPHYRIA that presents with Acute abdominal pain and NEUROPSYCHIATRIC Symptoms

◊ Dx of Acute porphyrias is usually done by estimation of δ ALA & PBG in urine

◊ Glucose administration relieves acute attack of porphyria.

◊ PORPHYRINS ARE COLORED & FLUORESCENCE (measured at SORET band near 400 nm) d/t conjugated double bonds & have a deep red or purple colour



EXTRA EDGE: IRON ABSORPTION & Its HOMEOSTASIS

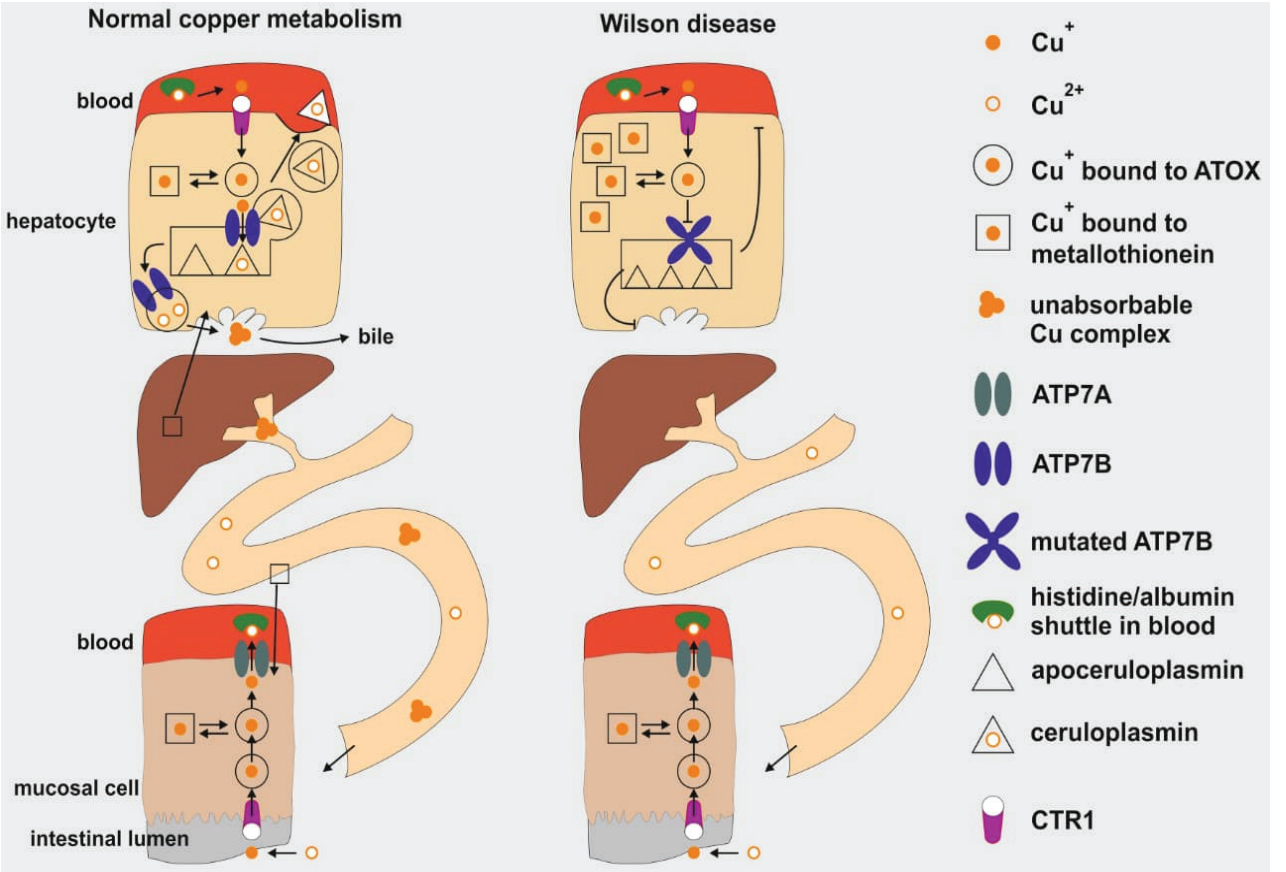
Iron is absorbed by upper part of duodenum in Fe++ (ferrous) form (reduced form) ONLY [Calcium, copper, lead and phosphates will inhibit iron absorption]. Homeostasis is maintained by regulation at the level of absorption and not by excretion – ONE WAY ELEMENT

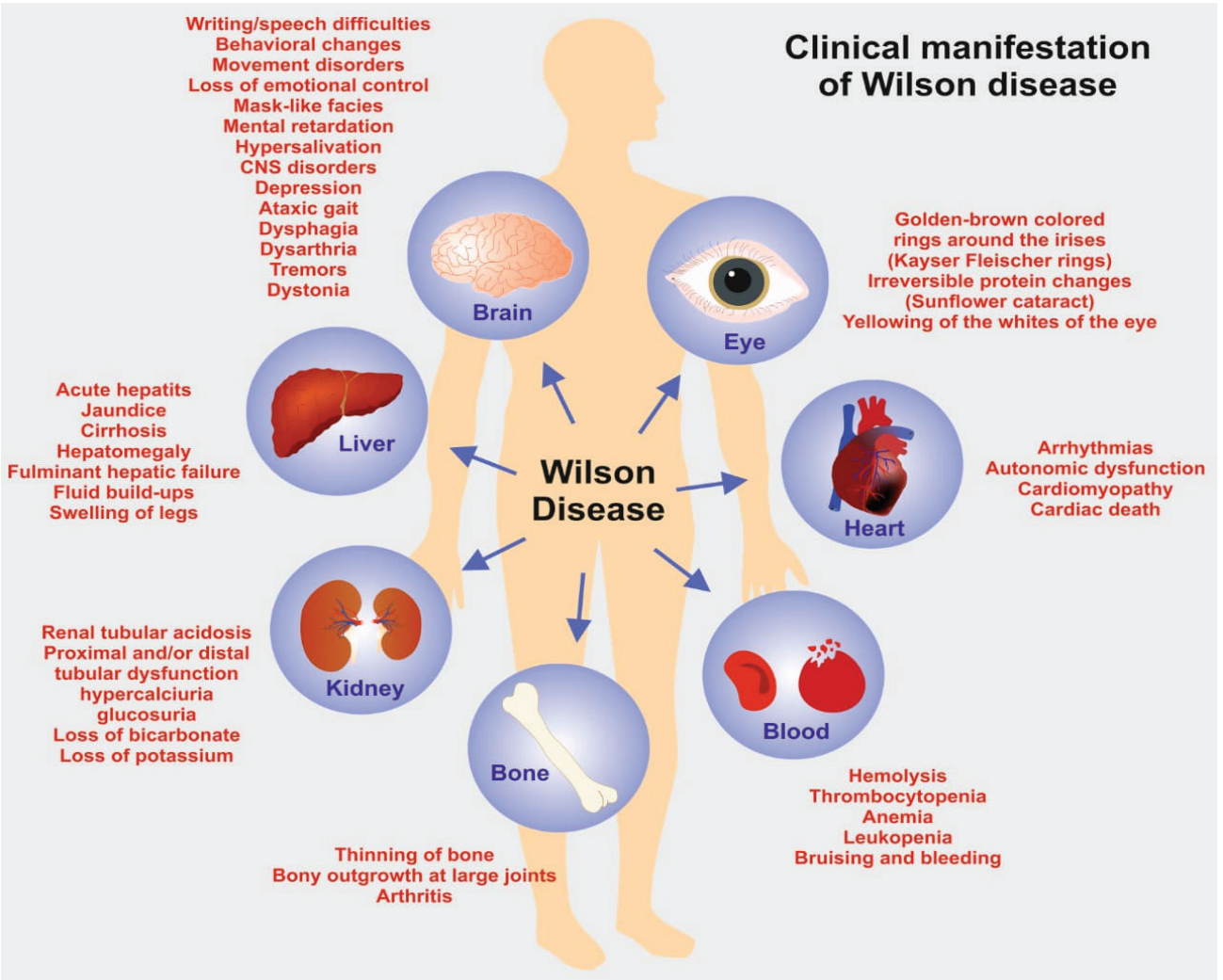
- Food in Iron is in Fe3+ form. Changed to Fe^{2+} by Vitamin C & ferric reductase (aka Duodenal Cytochrome b - DcytB (present in enterocytes).
- Ferrous iron in the intestinal lumen binds to mucosal cell protein divalent metal transporter-1 (DMT-1)
- Inside the mucosal cell, iron is oxidized to ferric state by intrinsic catalytic activity of apoferritin and is complexed with apoferritin to form ferritin →The STORAGE form | Iron in ferritin when released crosses the mucosal cell in ferrous form with the help of a transport protein called ferroportin → In blood, ceruloplasmin is the ferroxidase, which oxidizes ferrous to ferric state and iron is further transported by transferrin.

Intestinal uptake of heme iron occurs with the help of heme carrier protein (HCP1).

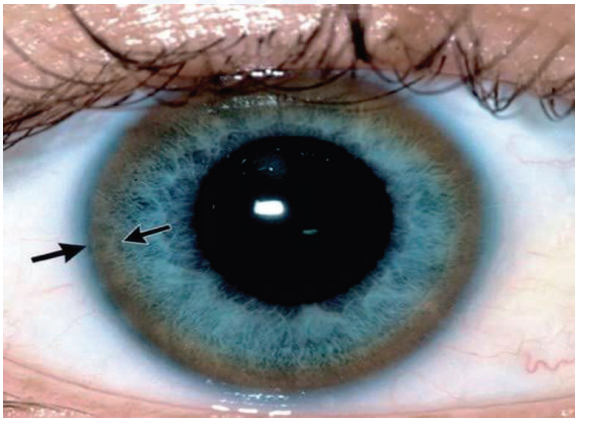
- Chromium →potentiates the action of insulin, hence impaired glucose tolerance is seen in Chromium deficiency

- Zinc →Synthesis, storage, stabilization & Secretion of insulin.


Normal copper metabolism
Wilson disease
blood
hepatocyte
bile
blood
mucosal cell
intestinal lumen
Cu+
Cu2+
Cu+ bound to ATOX
Cu+ bound to metallothionein
unabsorbable Cu complex
ATP7A
ATP7B
mutated ATP7B
histidine/albumin shuttle in blood
apoceruloplasmin
ceruloplasmin
CTR1


Clinical manifestation of Wilson disease
Writing/speech difficulties
Behavioral changes
Movement disorders
Loss of emotional control
Mask-like facies
Mental retardation
Hypersalivation
CNS disorders
Depression
Ataxic gait
Dysphagia
Dysarthria
Tremors
Dystonia
Brain
Golden-brown colored rings around the irises (Kayser Fleischer rings)
Irreversible protein changes (Sunflower cataract)
Yellowing of the whites of the eye
Eye
Acute hepatits
Jaundice
Cirrhosis
Hepatomegaly
Fulminant hepatic failure
Fluid build-ups
Swelling of legs
Liver
Wilson Disease
Heart
Arrhythmias
Autonomic dysfunction
Cardiomyopathy
Cardiac death
Renal tubular acidosis
Proximal and/or distal tubular dysfunction
hypercalciuria
glucosuria
Loss of bicarbonate
Loss of potassium
Kidney
Blood
Hemolysis
Thrombocytopenia
Anemia
Leukopenia
Bruising and bleeding
Bone
Thinning of bone
Bony outgrowth at large joints
Arthritis

Sunflower or Sunburst Cataract in Wilson's disease due to Copper deposition in lens



EXTRA EDGE:

Superoxide Dismutase: A free radical Scavenger

CYTOSOLIC MITOCHONDRIAL

SOD1 Gene mutation → Amyotrophic Lateral Sclerosis aka LOU GEHRIG's disease

3. Selenium Containing Enzymes aka SELENOCYSTEINE containing Enzymes

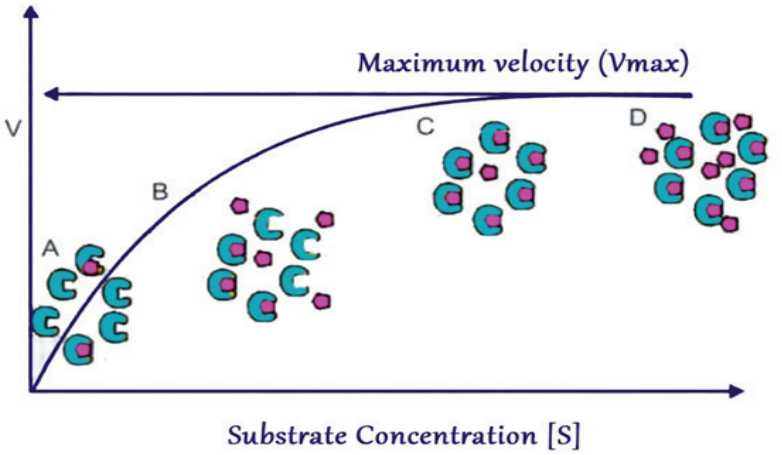
- GLUTATHIONE PEROXIDASE
- THIOREDOXIN REDUCTASE
- DE-IODINASE (Conversion of T4 → T3)

D1 vs D2 vs D3

4. Molybdenum Containing Enzymes

✹ Important FACTORS affecting Enzyme Activity

- Substrate concentration

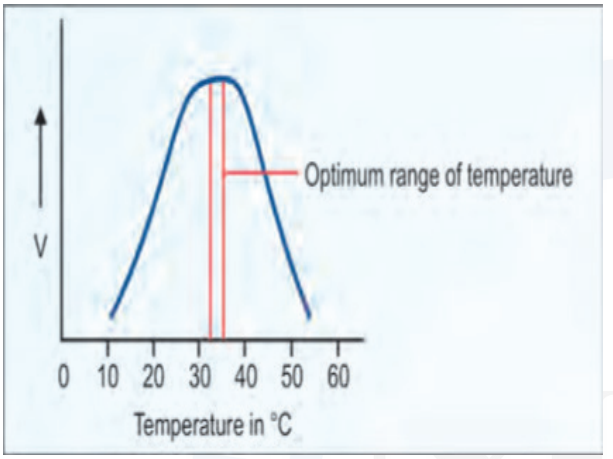


* Parabolic graph

Km – Substrate concentration at half maximal velocity

- ONLY 50% of enzymes are saturated with substrate at that.....
- Km is independent of enzyme concentration,
- Km denotes signature of enzyme → 2 different enzymes can't have same Km value
- Km is inversely proportional to affinity

- Temperature PCR



- TAQ POLYMERASE (TAQ – Thermus Aquaticus)

EXTRA EDGE:

Horse radish peroxidase: Used as a Tool in Molecular Biology techniques

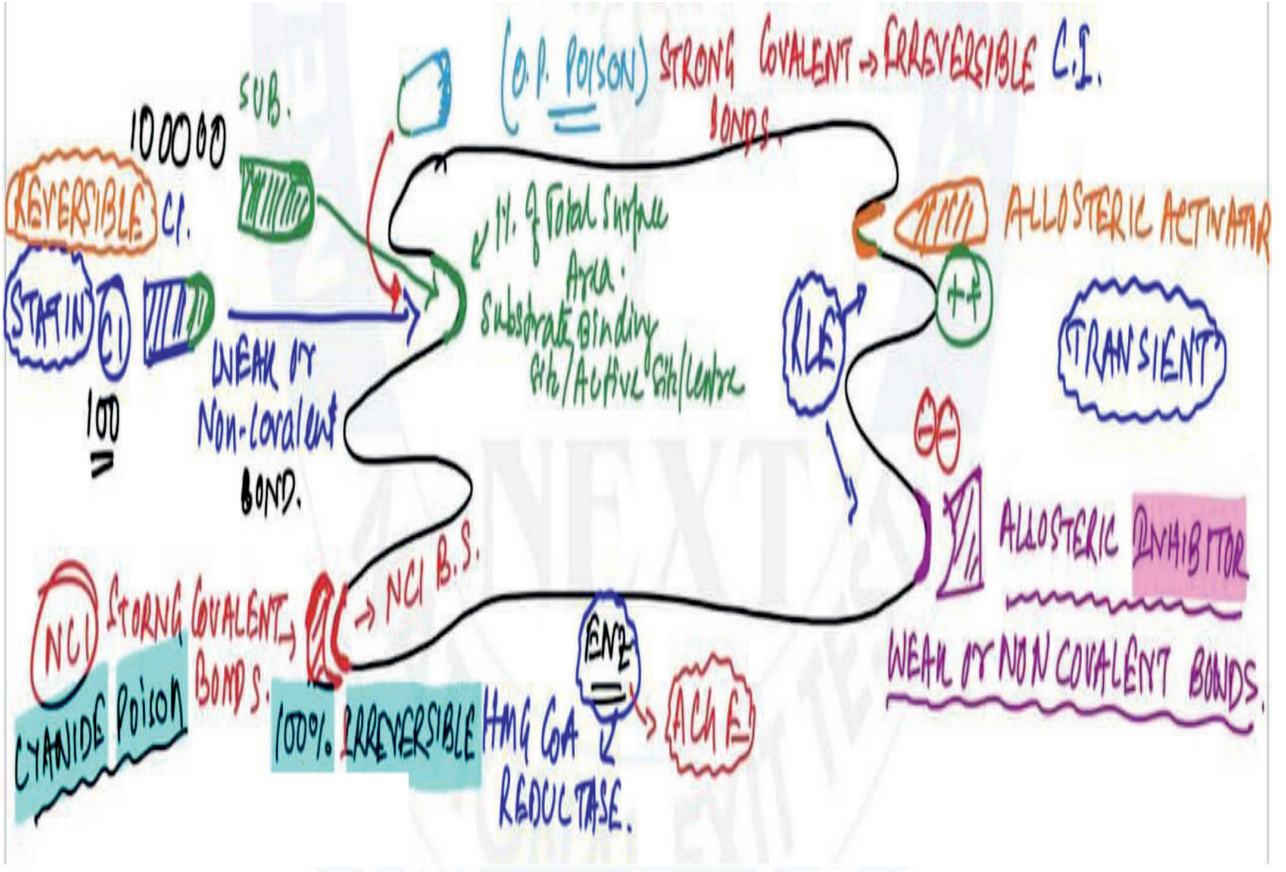
1. BLOTTING TECHNIQUES

S N O W D R O P

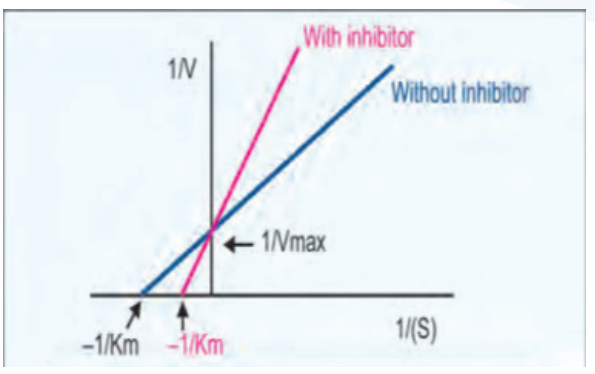
EASTERN BLOT -

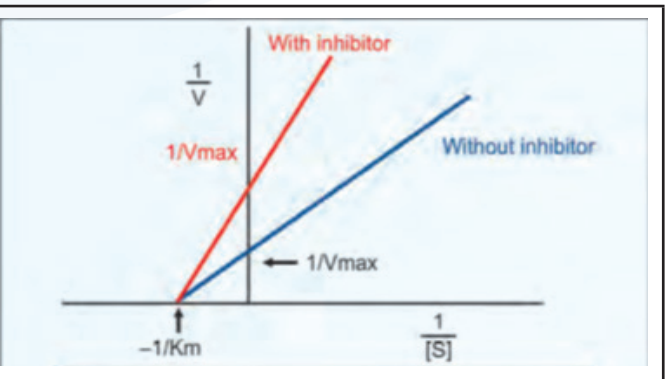
- Presence of Inhibitors

✹ Enzyme inhibition



	Competitive inhibition	Non- competitive inhibition
Acting on	Substrate binding site or Active site	Non-competitive inhibitor binding site
Structure of inhibitor	Substrate analog	Unrelated molecule
On adding EXCESS substrate Inhibition is	Reversible	Irreversible
Excess substrate	Inhibition relieved	No effect
Km	Increased	No change
Vmax	No change	Decreased
Significance	Drug action	Toxicological or POISON





EXTRA EDGE:

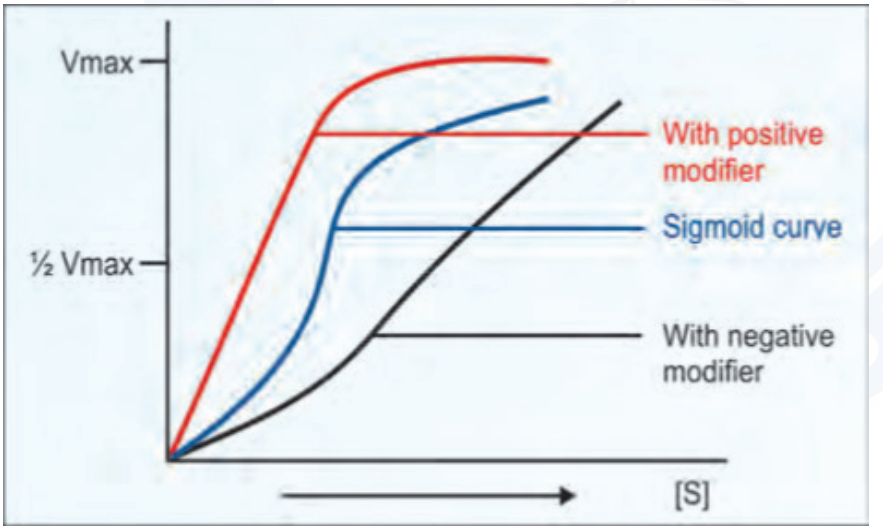
Allosteric Regulation:

-The inhibitor is not a substrate analog.

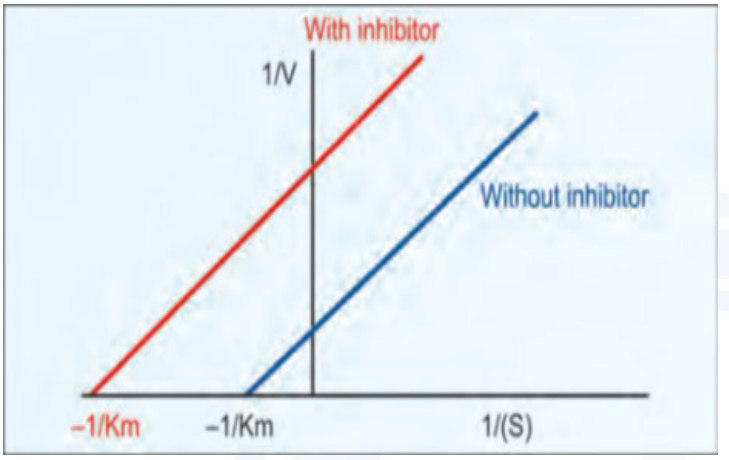
-It is partially reversible, when excess substrate is added

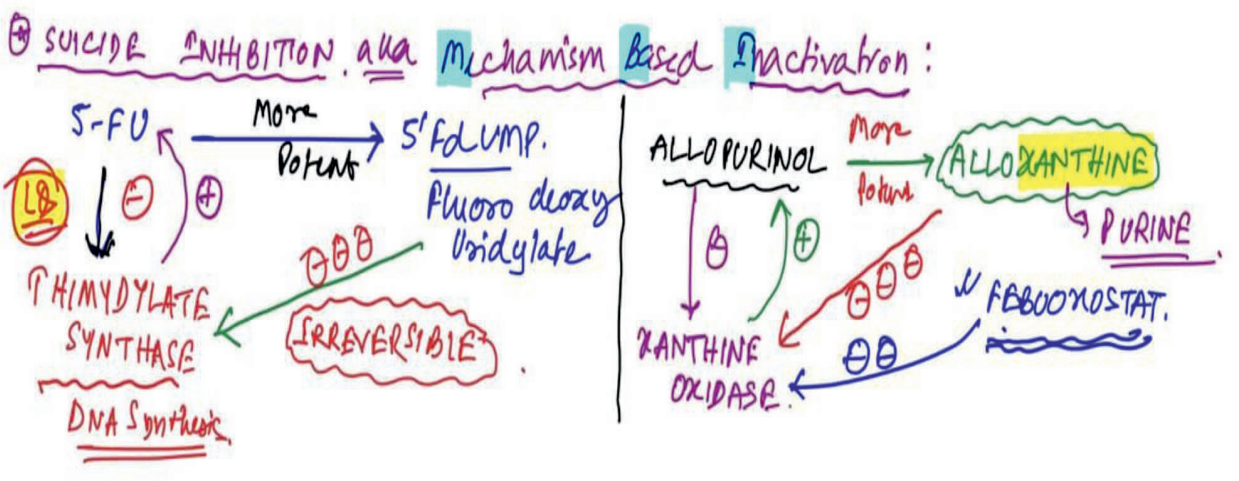
-Km is →

-Vmax →



Uncompetitive inhibition:





- Presence of repressor or de repressor

→ Repressor acts at the gene level. Slow process E.g. ALA synthase is autoregulated by heme by means of repression

Covalent modification

PHOSPHO-DEPHOSPHORYLATION

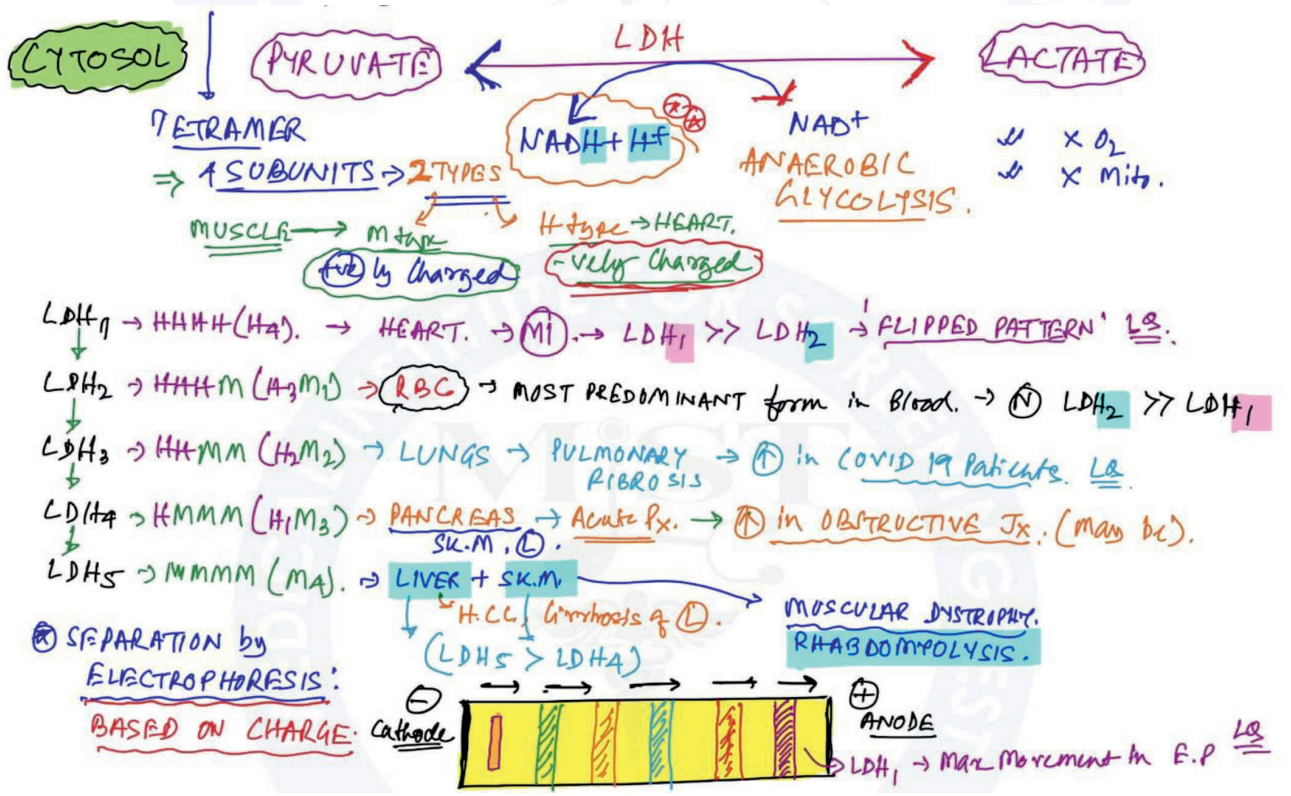
ACTIVE in PHOSPHORYLATED STATE	ACTIVE in DEPHOSPHORYLATED STATE
a. Glycogen phosphorylase	Glycogen synthase
b. ATP Citrate Lyase	Acetyl-CoA carboxylase
c. Hormone Sensitive Lipase	Pyruvate dehydrogenase
d. DNA Dependent RNA Polymerase II (RNAP-II or RNAP-B)	PFK2
	Pyruvate Kinase
	HMG CoA reductase

✹ CLINICALLY IMPORTANT ISOENZYMES

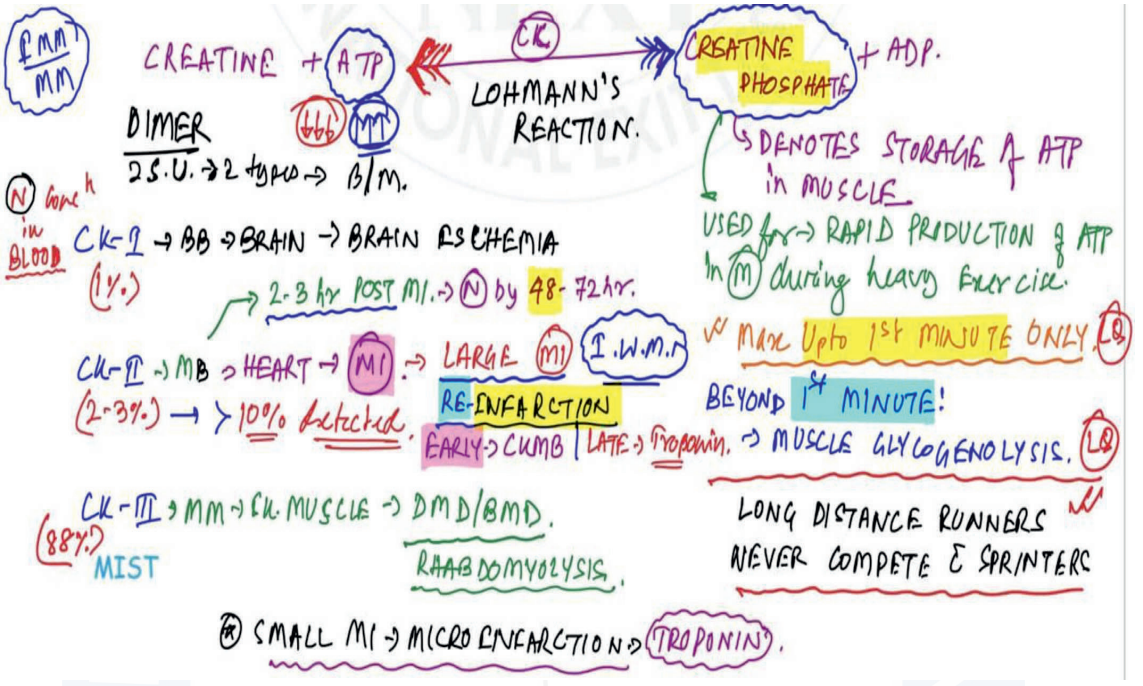
- Multiple (Physically distinct) forms of an enzyme catalyzing the same reaction are isoenzymes or isozymes BUT at .....
- TISSUE SPECIFIC in Origin

- Differ in their physical and chemical properties which include the structure, electrophoretic and immunological properties, Km and Vmax values, pH optimum, relative susceptibility to inhibitors and degree of denaturation.

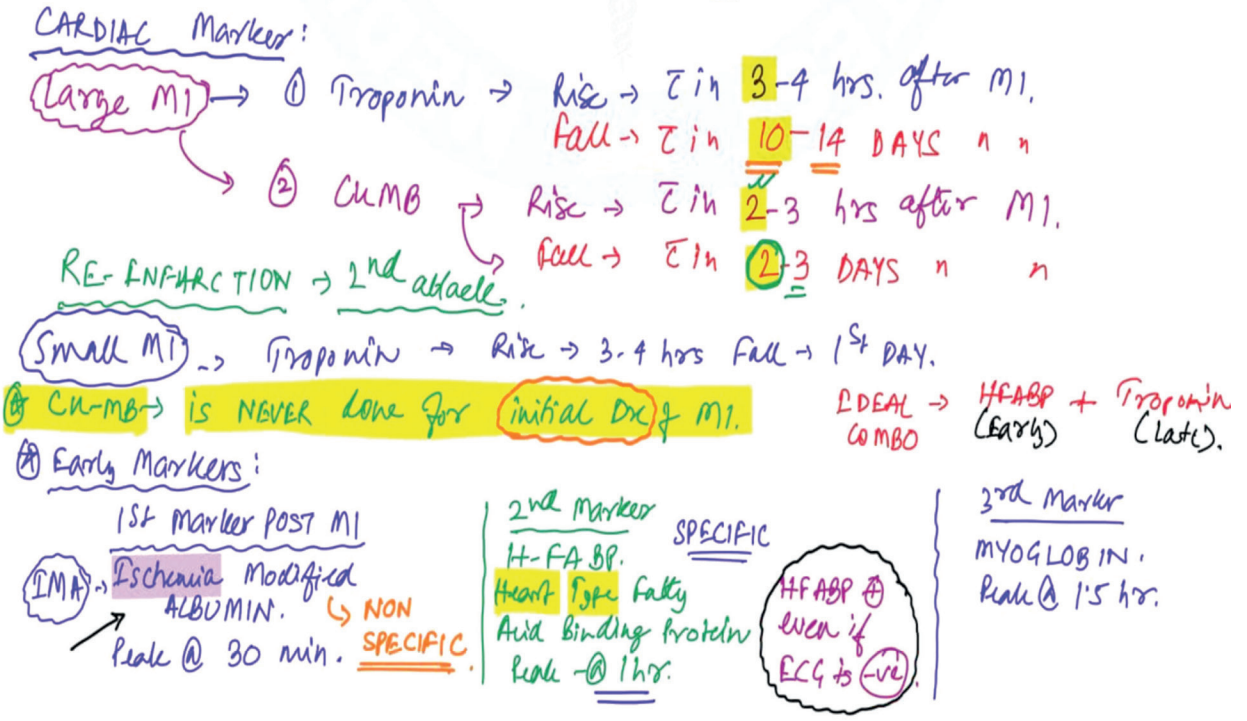
Lactate dehydrogenase-LDH isoforms



Creatine kinase -CK isoforms



Cardiac markers in a nutshell



Marker Enzymes:

Subcellular organelle	*Pellet formed at the centrifugal force of*	*Marker enzyme*
Nucleus	600–750 x g, 10 min	
Mitochondria	10,000–15,000 x g, 10 min	Inner membrane: ATP Synthase
Lysosome	18,000–25,000 x g, 10 min	Cathepsin
Golgi complex	35,000–40,000 x g, 30 min	Galactosyl transferase
Microsomes	75,000–100,000 x g, 100 min	Glucose-6-phosphatase
Cytoplasm	Supernatant	Lactate dehydrogenase

PRACTICE MCQS:

1. In skeletal muscles, isoform of LDH is

   a. LDH-5

   b. LDH-4

   c. LDH-3

   d. LDH-2

2. All are true about isoenzymes EXCEPT

   a. Different Km value

   b. Act on different substrate

   c. Have different electrophoretic mobility

   d. Have different physical properties

3. The following are salient features of allosteric enzymes, EXCEPT

   a. They are usually multiple subunit enzymes

   b. They are involved in metabolic regulation

   c. They do not show any change in Km and Vmax in the presence of modifiers

   d. On binding the inhibitor, usually Km is increased and Vmax is decreased

4. All of the following metalloenzymes contain copper EXCEPT

a. Superoxide dismutase

b. Tyrosinase

c. Glutathione peroxidase

d. Cytochrome oxidase

5. Zinc is not an essential component of the following enzyme

a. Carbonic anhydrase

b. Alkaline phosphatase

c. Xanthine oxidase

d. Ethanol Dehydrogenase

6. Km of an enzyme is always

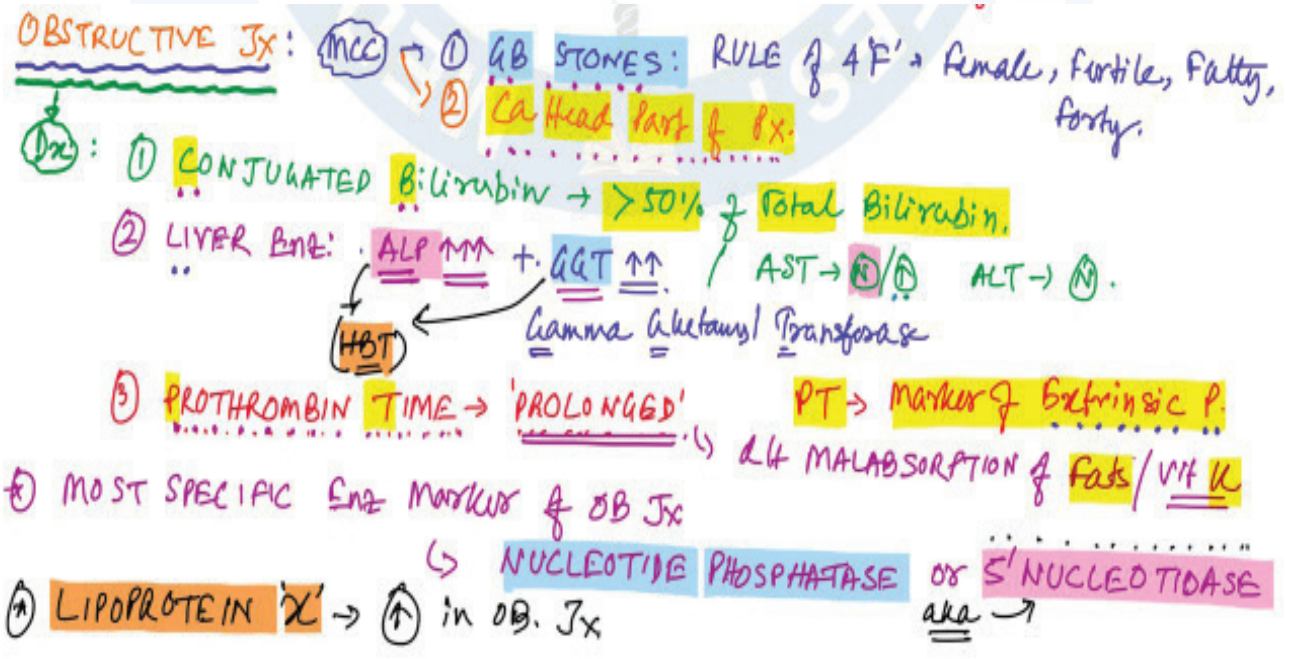
a. One half of the VMAX

b. Dissociation constant

c. Normal substrate concentration

d. Substrate concentration that gives half maximal velocity

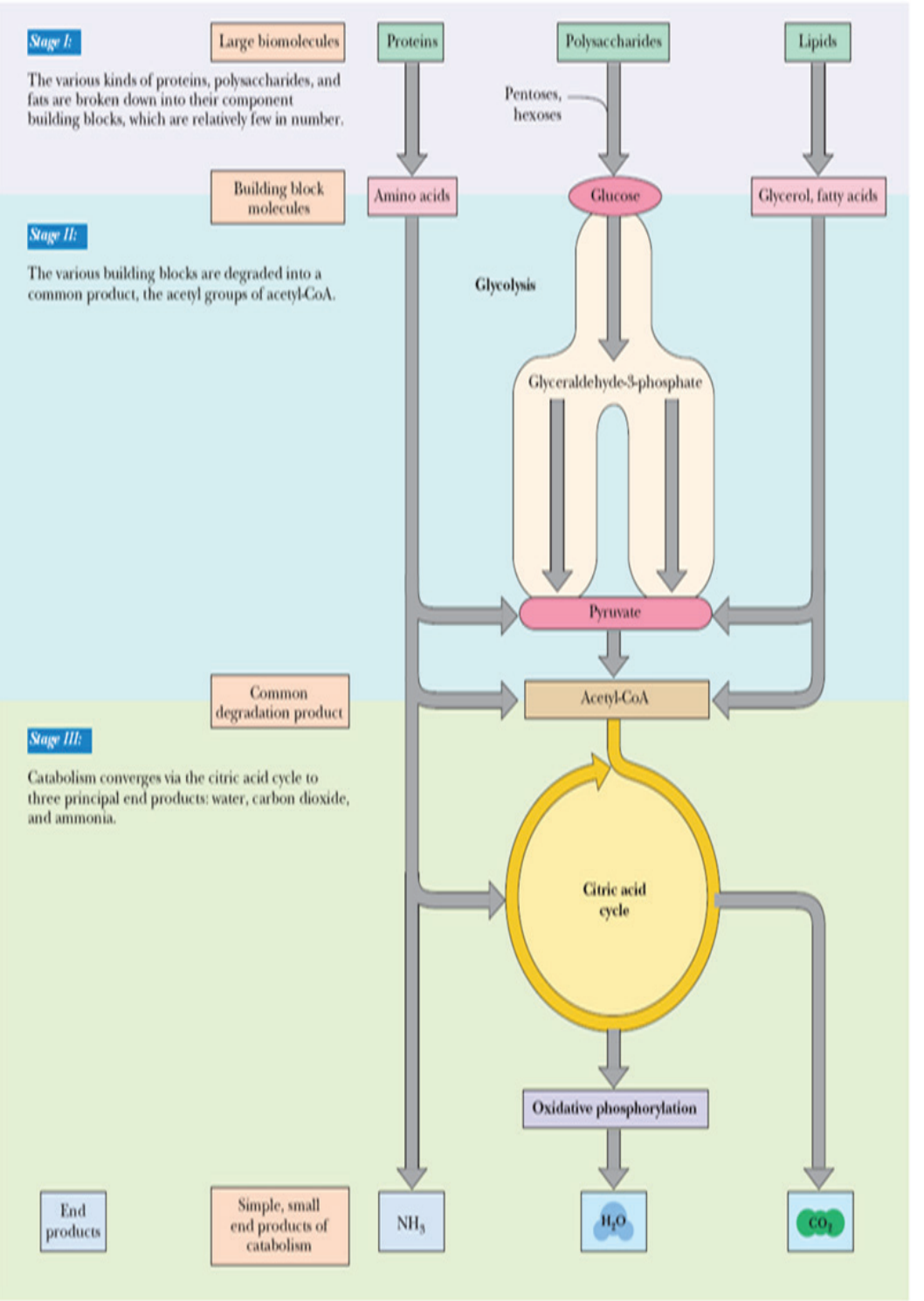
Workbook Sheet



Workbook Sheet

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

BIOLOGICAL OXIDATION & ETC COUPLED TO OXIDATIVE PHOSPHORYLATION



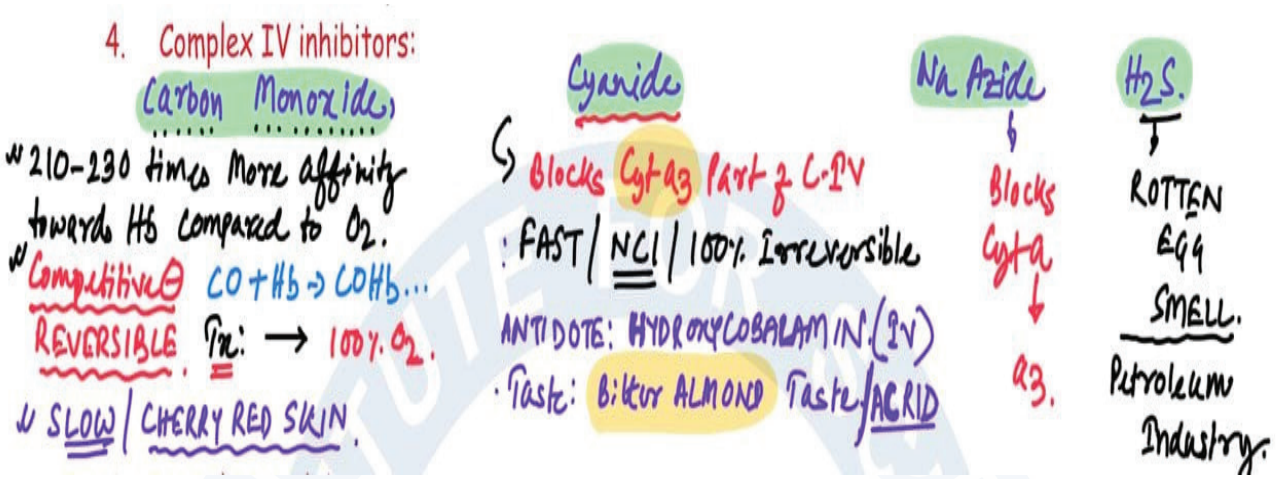
ELECTRON TRANSPORT CHAIN (ETC) & OXIDATIVE PHOSPHORYLATION

INHIBITORS and UNCOUPLERS OF ETC & OXIDATIVE PHOSPHORYLATION

1. Complex I inhibitors:

2. Complex II inhibitors:

3. Complex III inhibitors



5. Complex V inhibitors:

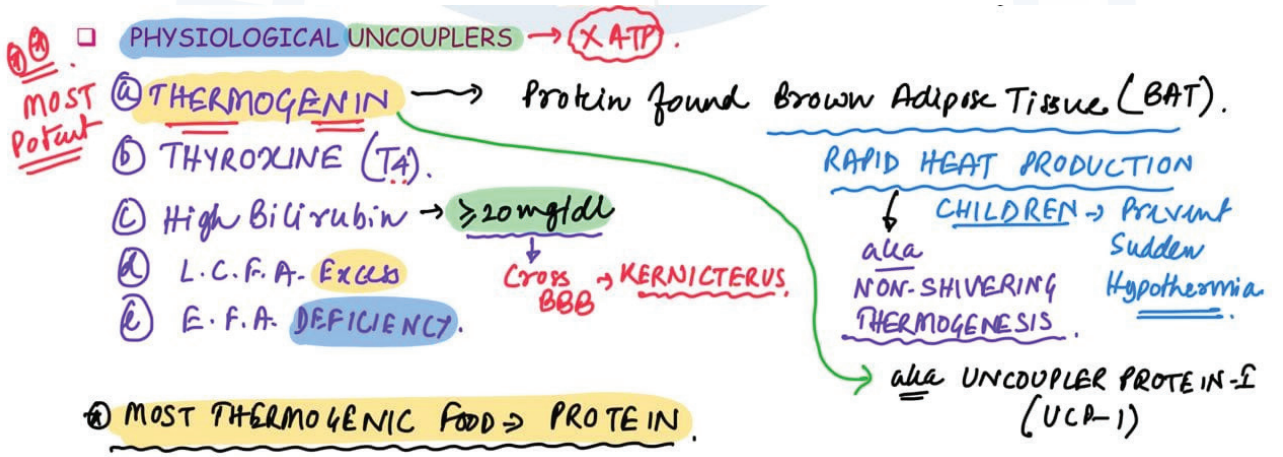
Extra Edge:

- Transport inhibitors:

❑ Uncouplers:

❑ Ionophores:

❑ PHYSIOLOGICAL UNCOUPLERS



- Mitochondrial DNA is inherited cytoplasmically and is therefore transmitted maternally

- DISEASES ASSOCIATED WITH MITOCHONDRIAL OXIDATION AKA OXPHOS DISEASES

  - Fatal infantile mitochondrial myopathy and renal dysfunction

  - Mitochondrial encephalomyopathy lactic acidosis stroke like episodes (MELAS):

  - Lethal infantile mitochondrial ophthalmoplegia

  - Leber's Hereditary Optic Neuropathy (LHON)

  - Myoclonic epilepsy associated with ragged red fibers (MERRF)

PRACTICE MCQS:

Which of the following are reduced coenzymes

a. NADH+H+ and FADH2

b. NAD+ and FAD

c. ATP and GTP

d. Coenzyme A and ubiquinone

Rapid source of energy for exercising muscles after 1st minute is

a. Blood glucose

b. Glycogenolysis

c. Stored ATP

d. Stored Phosphocreatine

Which is not a high energy compound

a. ATP

b. Carbamoyl phosphate

c. Glucose 6 phosphate

d. Arginine phosphate

Inhibition of the synthesis of ATP during oxidative phosphorylation by oligomycin is thought to be due to which of the following events?

a. Blocking of the proton gradient between NADH-Q reductase and QH2

b. Blocking of the proton gradient between cytochrome c1 and cytochrome c

c. Dissociating of cytochrome c from mitochondrial membranes

d. Inhibition of mitochondrial ATP Synthase

About oxidative phosphorylation, true is

a. Generation of ATP

b. Generation of ADP

c. Utilization of ATP

d. Utilization of NADH

MEDICAL INSTITUTE FOR SCREENING TEST MIST

Workbook Sheet

High Anion Gap Metabolic Acidosis (HAGMA)

M: methanol
U: uremia
D: diabetic ketoacidosis (DKA)
P: phenformin, paraldehyde
I : INH, iron
L: lactic acid
E: ethanol, ethylene glycol
S: salicylates

G: glycols (propylene glycol and ethylene glycol)
O: 5-oxoproline (associated with acetaminophen use)
L : L-lactate
D: D-lactate (short bowel syndrome)
M: methanol
A: aspirin
R: renal failure
K: ketoacidosis (diabetic/alcohol/starvation)

MEDICAL MISTAKES TESTING

CARBOHYDRATES – ROLES & IMPORTANCE

Extra Edge:

HbA_1C →

Normal Levels: Prediabetic Range: Diabetic:

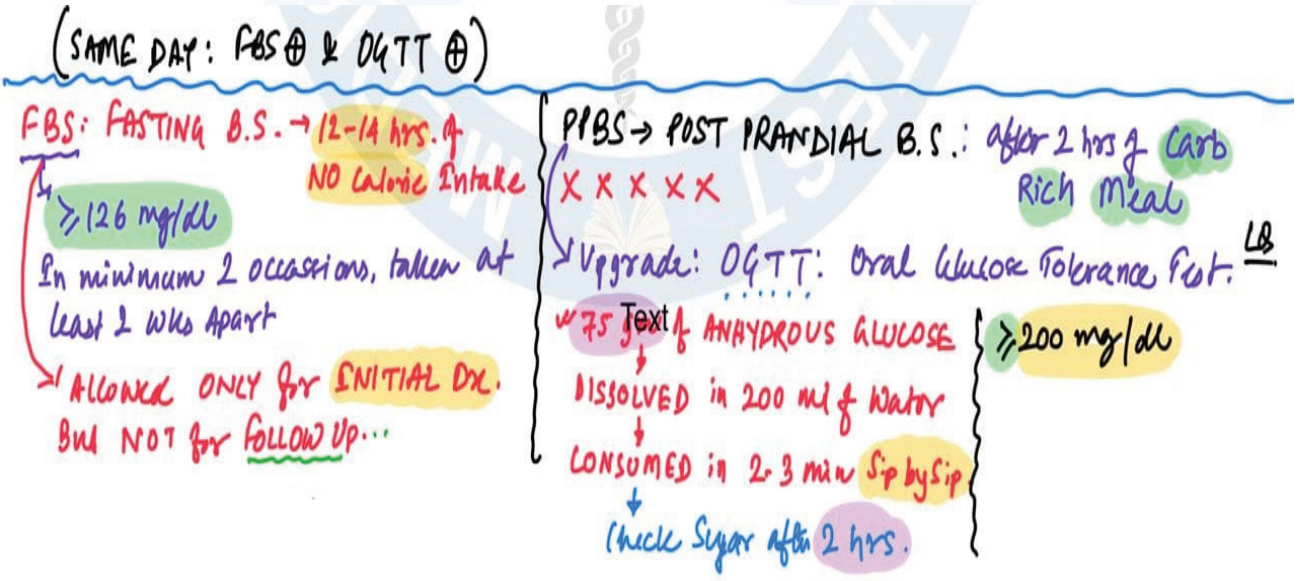
DIABETIC STATUS DIABETIC CONTROL

LATEST ADA GUIDELINES:

eAG (estimated Average Glucose in mg/dL) =

Best Technique to measure HbA1c →

What is AGE?



Continous Glucose Monitoring

- Specific Glycosaminoglycans of Physiological Significance

❑ Hyaluronic acid-(D-glucuronate + GlcNAc)n

Occurrence: synovial fluid, ECM of loose connective tissue. Also found in bacteria.

→ .....

→ .....

❑ Dermatan sulfate - (L-Iduronate + GalNAc sulfate)n

Occurrence: skin, blood vessels, ...

❑ Keratan sulfate - (Gal + GlcNAc sulfate)n

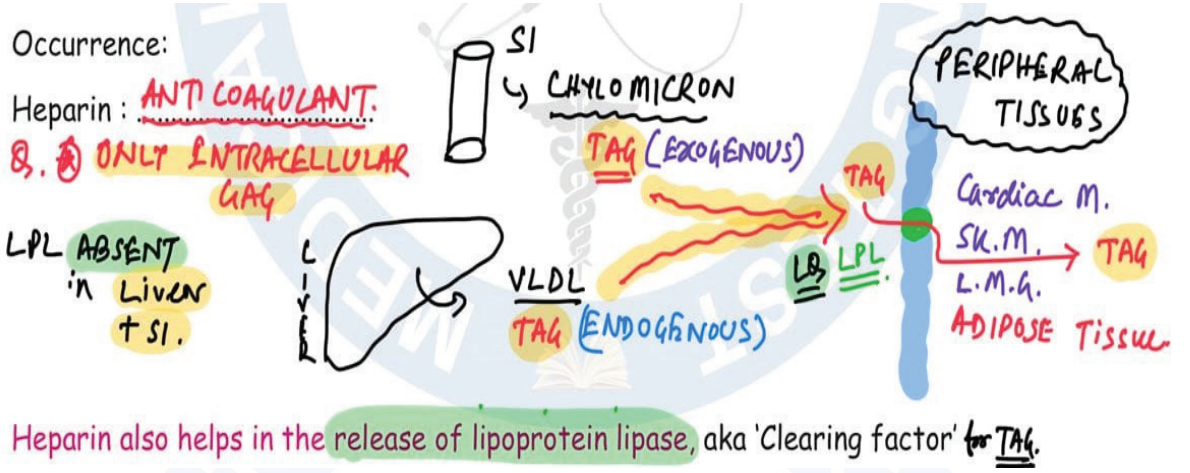
Occurrence: cornea, sclera, bone, cartilage, tendons

→ .....

→ .....

❑ Heparin - (D-glucuronate sulfate + N-sulfo-D-glucosamine)n

❑ Heparans have less sulfate groups than heparins



- Heparan sulfate: ..... component of cell surfaces.

- Chondroitin sulfate - (D-glucuronate + GalNAc sulfate)n

Occurrence: cartilage, tendons, ligaments, heart valves and aorta.

.....

✹ MUCOPOLYSACCHARIDOSIS – (GAG STORAGE DISORDERS) → Total 7 types

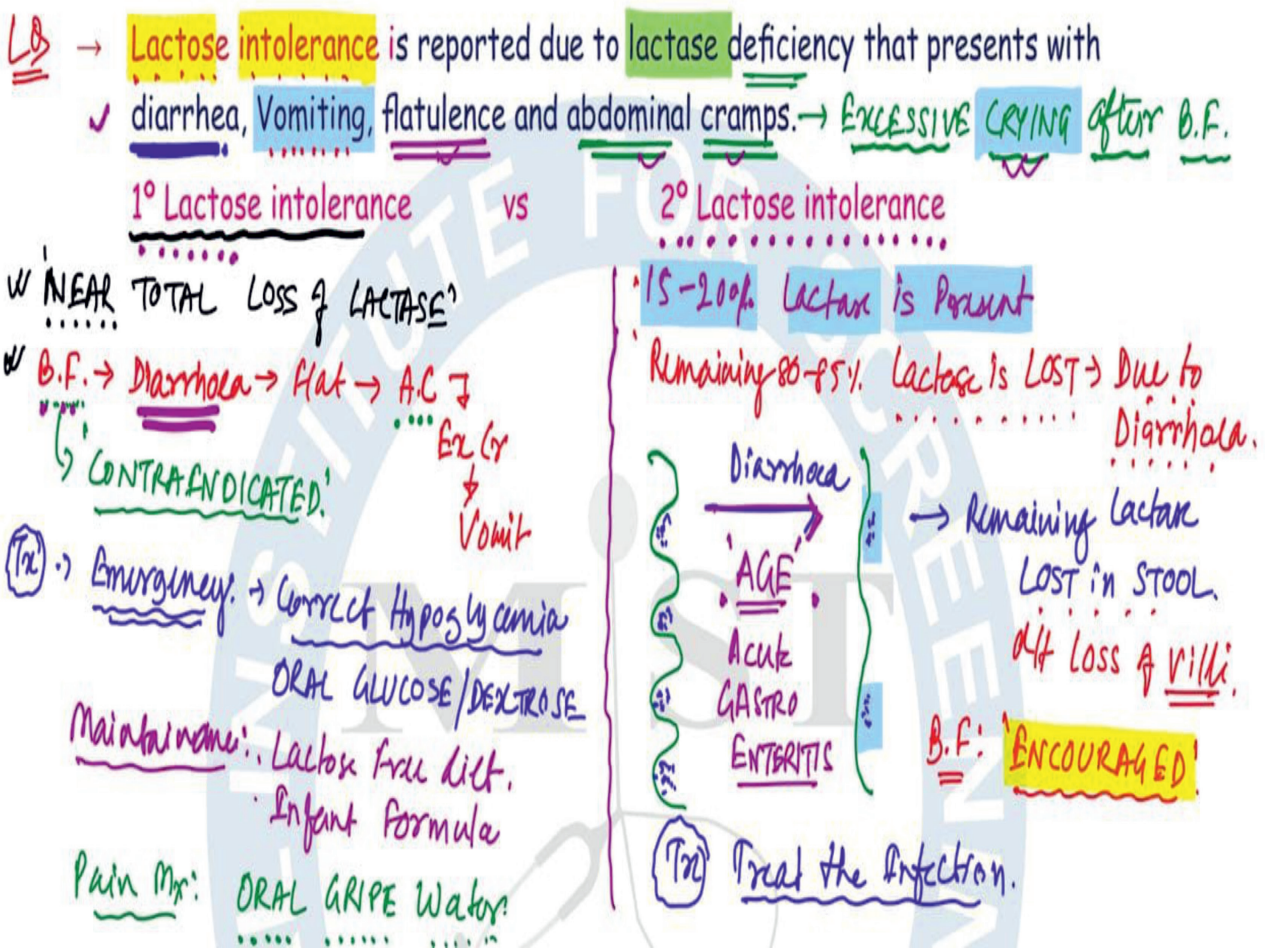
Type	Name	Enzyme Deficiency	Clinical Feature
I	Hurler Syndrome	L-Iduronidase	DS /HS in urine, Coarse facial hair, CORNEAL CLOUDING, AR
II	Hunter Syndrome	Iduronate Sulfatase	DS /HS in urine, Coarse facial hair, Sensorineural Hearing Loss, X- LINKED RECESSIVE

MPS	Name	Increased GAGs	Inheritance	Enzyme deficiency	Gene location
I	Hurler (IH), Hurler-Scheie (IHS) or Scheie (IS)	HS + DS	Autosomal Recessive	α-Iduronidase	4p16.3
II	Hunter	HS + DS	X-linked recessive	Iduronate sulfatase	Xq28
III A	Sanfilippo A	HS	Autosomal recessive	Heparan-N-sulfatase	17q25.3
III B	Sanfilippo B	HS	Autosomal recessive	α-N-acetylglucosaminidase	17q21.2
III C	Sanfilippo C	HS	Autosomal recessive	AcetylCoA α- glucosamine acetyltransferase	8p11.21
III D	Sanfilippo D	HS	Autosomal recessive	N-acetylglucosamine 6-sulfatase	12q14.3
IV A	Morquio A	KS	Autosomal recessive	Galactosamine-6-sulfate sulfatase	16q24.3
IV B	Morquio B	KS	Autosomal recessive	β-Galactosidase	3p22.3
VI	Maroteaux-Lamy	DS	Autosomal recessive	N-acetylgalactosamine 4-sulfatase	5q14.1
VII	Sly	HS + DS	Autosomal recessive	β-Glucuronidase	7q11.21
IX	Natowicz	Hyaluronan	Autosomal recessive	Hyaluronidase 1	3p21.31

Notes:
(1) Scheie syndrome was initially classified as MPS V, but when the enzyme defect was identified it was recognized as the attenuated end of the MPS I spectrum; to avoid confusion, MPS V was no loger used.
(2) An enzyme defect was found and proposed as MPS VIII, but shortly thereafter this was recognized as a laboratory mistake, and the new type of MPS was withdrawn; also, to avoid confusion, MPS VIII was no longer used.

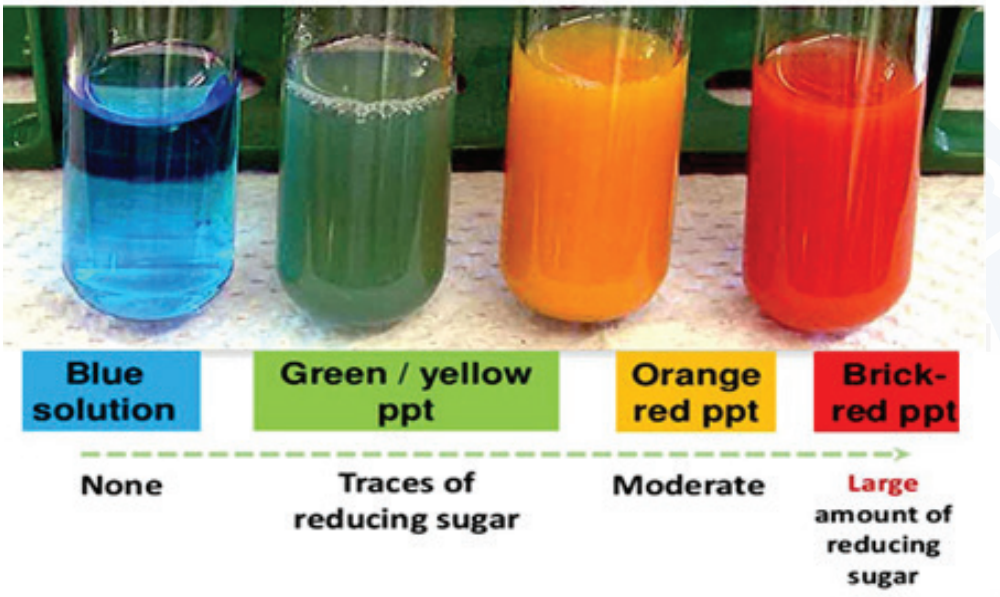
- What is LACTULOSE?
- Lactose is made up of Galactose + Glucose in ß 1 → 4 glycosidic linkage.

Extra Edge:



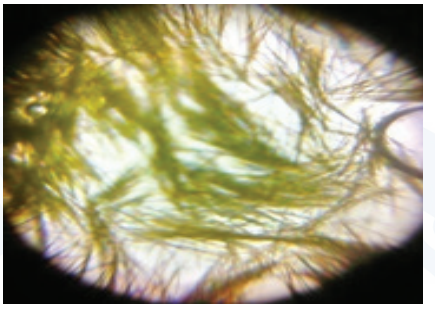
Extra Edge:

→ Benedict's test .....

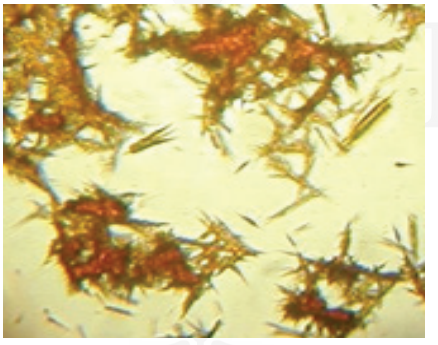


→ Osazone tests (Positive for Monosaccharides BUT used mainly for Confirmation for disaccharides) * However, Sucrose DOES NOT form any crystals

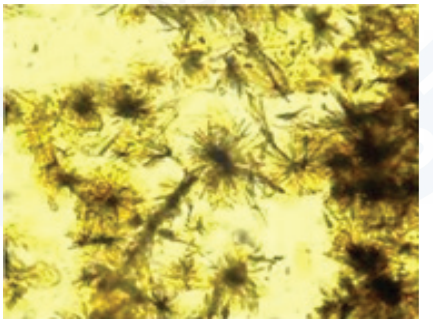
Green or Yellow coloured NEEDLE or BROOM STICK SHAPED CRYSTALS



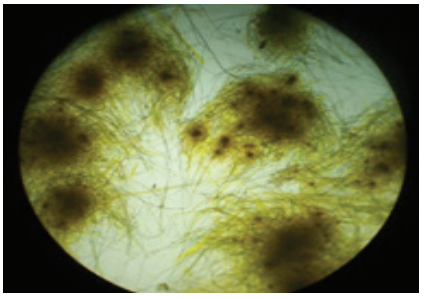
GOLDEN BROWN coloured RHOMBIC PLATES SHAPED CRYSTALS



SUNFLOWER or PETAL SHAPED CRYSTALS



POWDER PUFF or HEDGE HOG SHAPED CRYSTALS



PRACTICE MCQS:

Formation of L & D sugar is based on

a. Glyceraldehyde

b. Fructose

c. Glucose

d. Glycogen

C4 Epimer combination is

a. D-glucose & D-fructose

b. D-glucose & D-talose

c. D-glucose & D-gulose

d. D-galactose & D-glucose

Glucose as a structural unit is contained in which of the following

a. Lactose

b. Sucrose

c. Maltose

d. All of the above

Negative benedict test given by

a. Sucrose

b. Fructose

c. Mannose

d. Galactose

GAG present in cornea

a. Keratan Sulfate

b. Hyaluronic acid

c. Chondroitin Sulfate

d. Dermatan Sulfate

Composition of hyaluronic acid

a. N- acetyl glucosamine + d-glucosamine acid

b. N- acetyl glucosamine + d- glucuronic acid

c. N- acetyl glucosamine + sulfated glucosamine acid

d. N- acetyl glucosamine + iduronic acid

CARBOHYDRATE METABOLISM

✹ OVERVIEW OF ALL CARBOHYDRATE METABOLIC PATHWAYS

GLYCOLYSIS - Glucose oxidation; Glucose → Pyruvate (Aerobic), Lactate (Anaerobic)

GLUCONEOGENESIS – New Glucose synthesis from non-carbohydrate sources

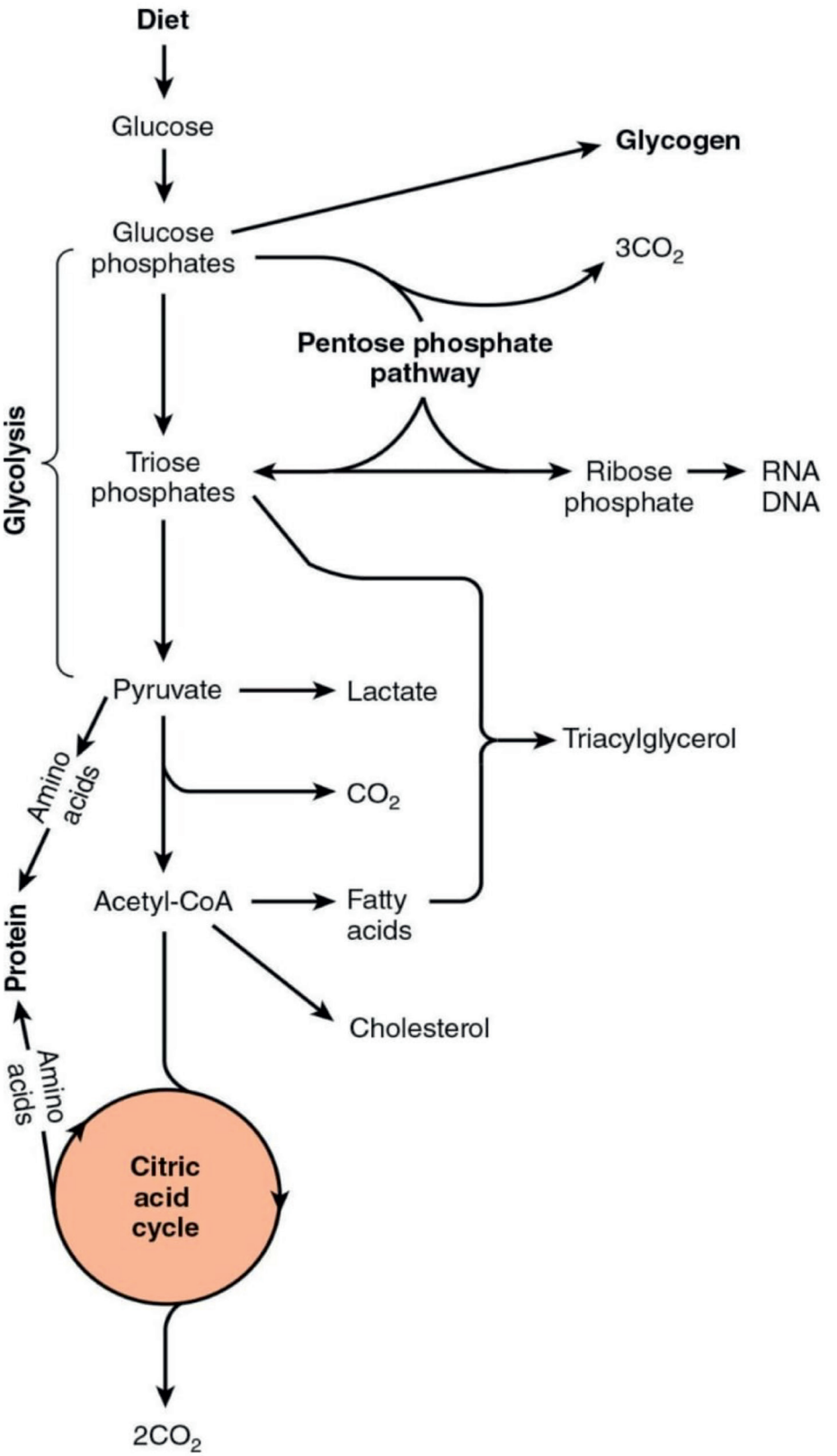
LINK REACTION - Pyruvate → Acetyl CoA

TCA / KREB's CYCLE - ATP, FADH2, [NADH+H+], CO_2

GLYCOGENESIS – Glycogen synthesis from Glucose in well fed state

GLYCOGENOLYSIS – Glycogen break down to Glucose during fasting / starvation

HMP-SHUNT – Alternative pathway of glucose oxidation


Diet
Glucose
Glycogen
Glucose phosphates
3CO2
Pentose phosphate pathway
Glycolysis
Triose phosphates
Ribose phosphate
RNA
DNA
Pyruvate
Lactate
Triacylglycerol
Amino acids
CO2
Acetyl-CoA
Fatty acids
Protein
Cholesterol
Amino acids
Citric acid cycle
2CO2

✹ GLYCOLYSIS (EMP PATHWAY) EMBDEN MEYERHOF PARNAS PATHWAY

SITE: ONLY in CYTOSOL ORGANS: ALMOST In ALL TISSUES (MAX →

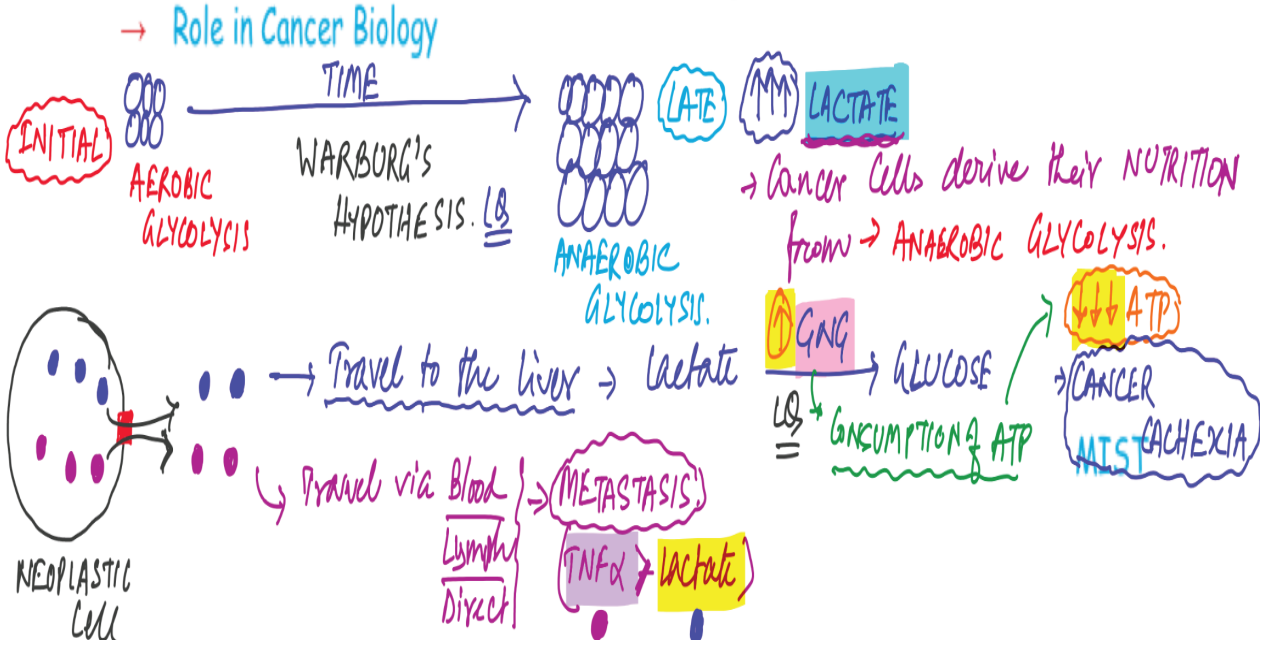
BRAIN: Prefers glucose as fuel than fat and protein specially in a WELL FED STATE and consumes about 100 gm of glucose per day. However, IN STARVATION- Brain is primarily dependent on →

RBC: ONLY FUEL is GLUCOSE.

So, END Product of AEROBIC GLYCOLYSIS in RBC →

→ CORI CYLE for RBC

Extra Edge:



→ GOLDEN MCQ POINTS

a) Rate Limiting Enzyme →

b) 1st Committed Step →

c) Junction of several carbohydrate metabolic pathways →

d) SLP of glycolysis →

e) Incorporation of Inorganic Phosphate & Production of NADH+H+ →

f) Irreversible reactions (3 out of 10) of glycolysis are →

g) Rate Controlling Enzymes →

h) Only reversible kinase →

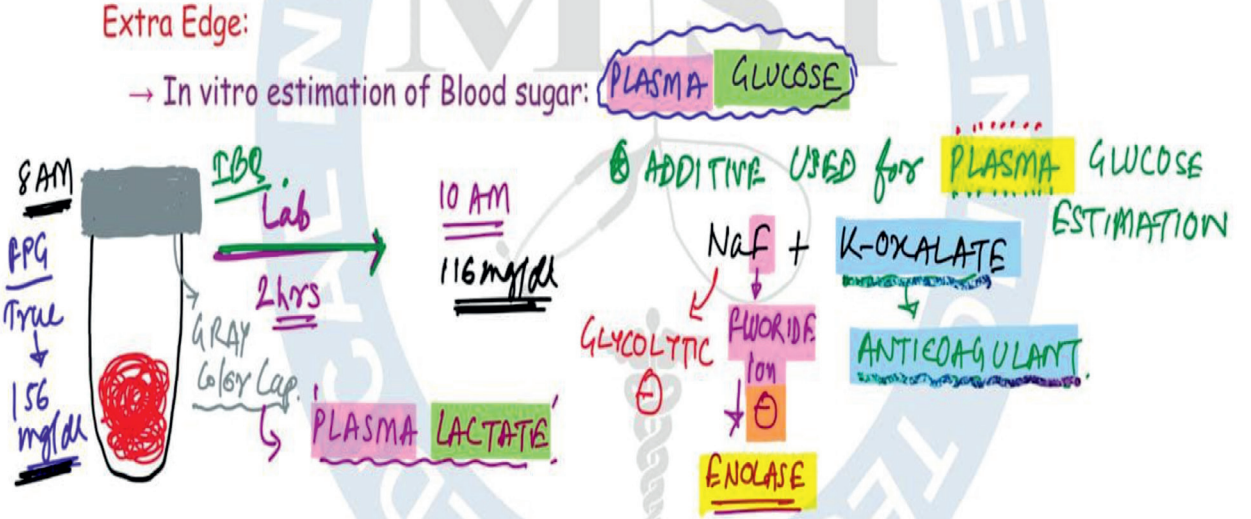
i) High energy compounds generating ATP

→

→

Extra Edge:

Extra Edge:



Tube	Color	Name	Additive	Test used for
	Blood Culture Bottle	Culture Bottle	Sodium Polyanethol sulfonate (anticoagulant) and Growth media for microorganisms	Two bottles are typically collected, in one blood draw; one for aerobic organisms and one for anaerobic organisms.
	Light Blue	Sodium Citrate	3.2% Sodium citrate (anticoagulant)	Coagulation tests
	Red	Red or plain	No additive, No Anticoagulant	Immunology, Serological examination
	Gold	Serum Separating Tube	Serum Separating Gel and clot activator	All Biochemistry test
	Light Green	Heparin Tube	Sodium heparin or Lithium heparin (anticoagulant)	prevent clotting, Chromosome testing, HLA typing, ammonia, lactate
	Purple/Lavender	EDTA	Ethylene diamine tetra acetic acid (EDTA) (Anticoagulant)	Hematological examination like complete Hemogram
	Pink		Ethylene diamine tetra acetic acid (EDTA) (Anticoagulant) Used only for Whole Blood sample being send to transfusion lab	Blood typing and cross-matching, direct Coombs test for autoimmune haemolytic anemia, HIV viral load, Group and save (G & S) These tubes are preferred for blood bank tests.
	Grey	Sodium fluoride	Sodium fluoride (glycolysis inhibitor) Potassium oxalate (anticoagulant)	Glucose, lactate testing
	Yellow	Acid-citrate-dextrose (anticoagulant)		Tissue typing, DNA studies, HIV cultures

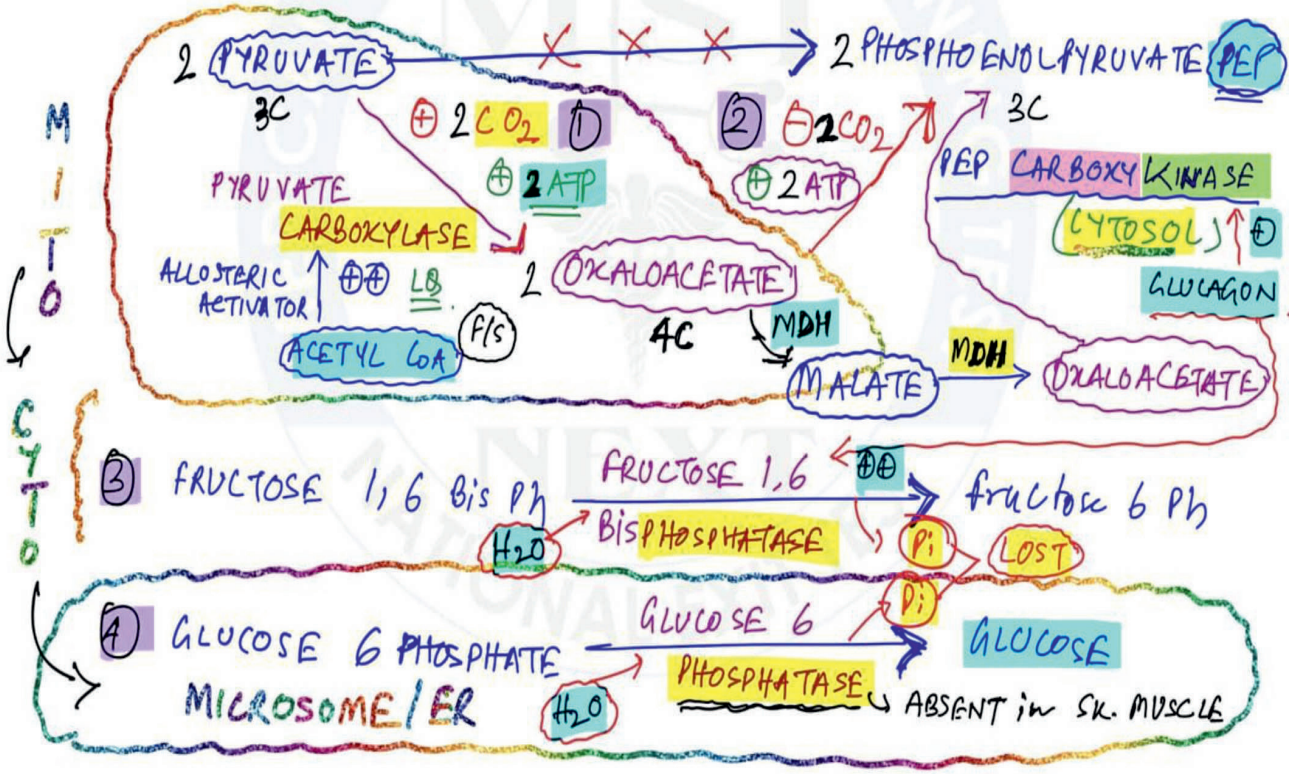
Comparison between Hexokinase & Glucokinase

	Hexokinase (II, I, III)	Glucokinase (HK IV)
Specificity		
Distribution	All tissues	Liver, pancreatic ß-cells, **Hypothalamus**, Intestine & Neuroendocrine cells
Product Inhibition	inhibited by Glucose-6-P	Not inhibited by Glucose-6-P
Km for glucose	Low Km i.e. (High af finity)	High Km i.e. (Low affinity)
Effect of Insulin	Not affected	Activated
Activity	Active at low glucose to meet needs of brain cells and RBCs when the BSL falls below 50 mg/dl	In liver- must handle increase in glucose after a meal. Activated once the BSL is >100 mg/dl (Diabetes mellitus)
Overall Importance		

- CLINICAL CORRELATION of Pyruvate kinase deficiency:

  In RBC – Deficiency can cause HEMOLYTIC ANEMIA

- GLUCONEOGENESIS:



→ RLE of GNG:

→ Principal organs responsible for gluconeogenesis -

→ Why No gluconeogenesis in Muscle?

→ Compartmentalization of GNG

→ Substrates for GNG:

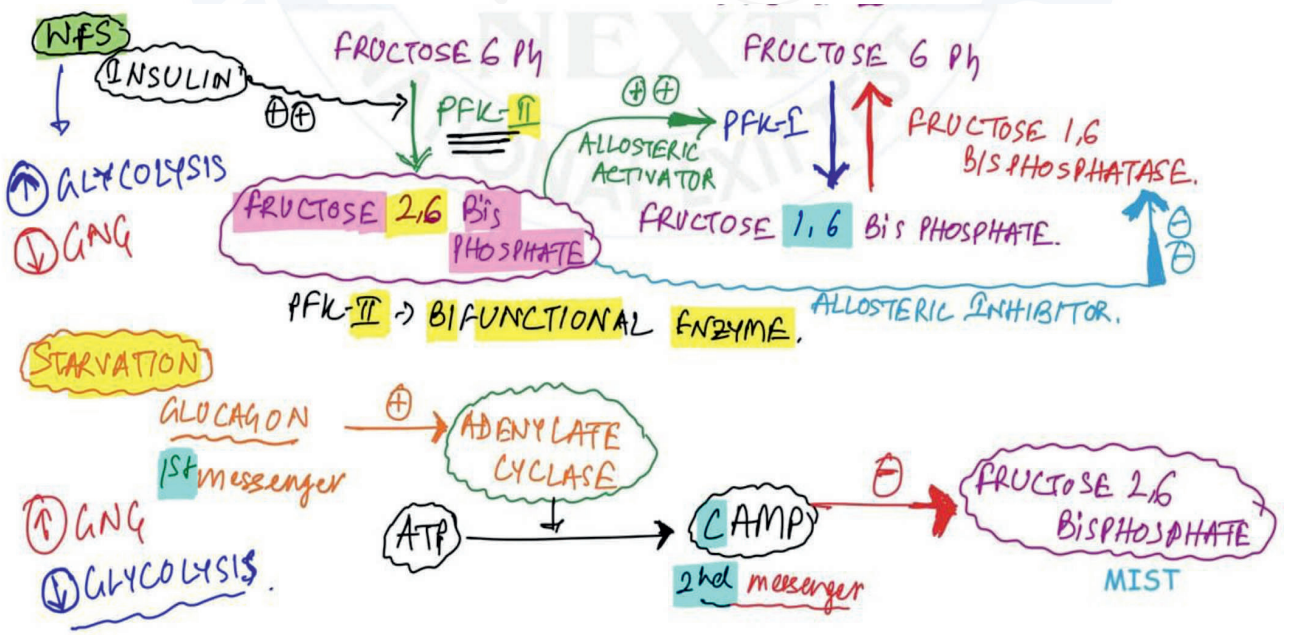
→ ENERGETICS of GNG:

2 molecules of Pyruvate or 2 molecules of lactate can produce 1 molecule of glucose by utilizing → ATP

→ Hormonal Control of GNG:

Corticosteroids and glucagon secreted during starvation stimulate glucose – 6-Phosphatase and PEP carboxy kinase and increases gluconeogenesis. Insulin decreases gluconeogenesis in a well-fed state

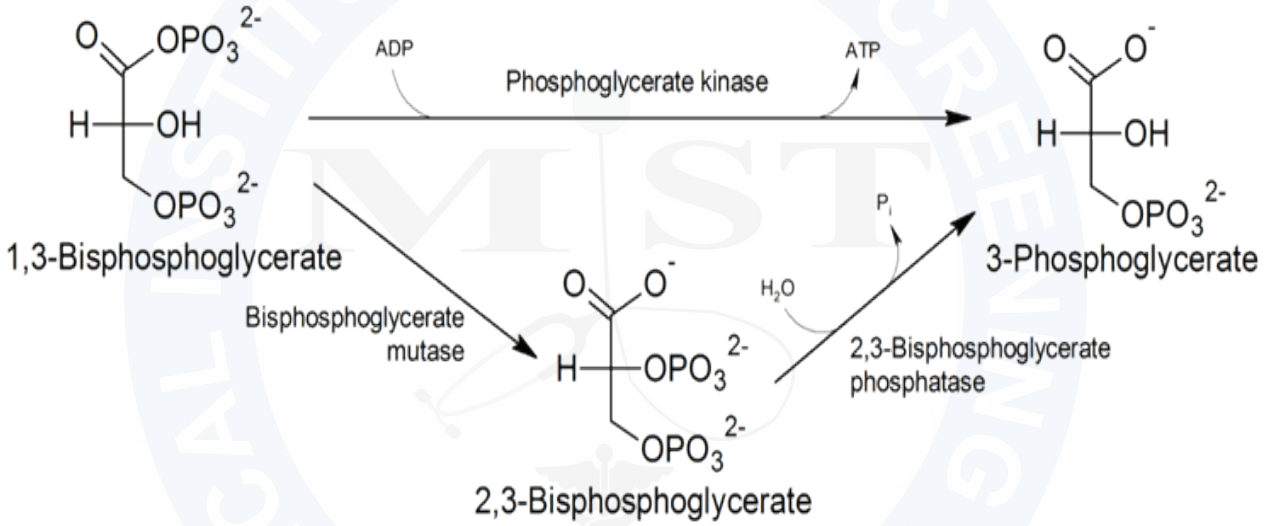
CONTROL MECHANISM between Glycolysis & GNG:



✹ MALATE ASPARTATE SHUTTLE & GLYCEROPHOSPHATE SHUTTLE:

✹ SPECIAL PATHWAY OF GLYCOLYSIS IN RBC:

◊ Rapoport Luebering Cycle (BPG Shunt)



He Is

C: Increase in CO2

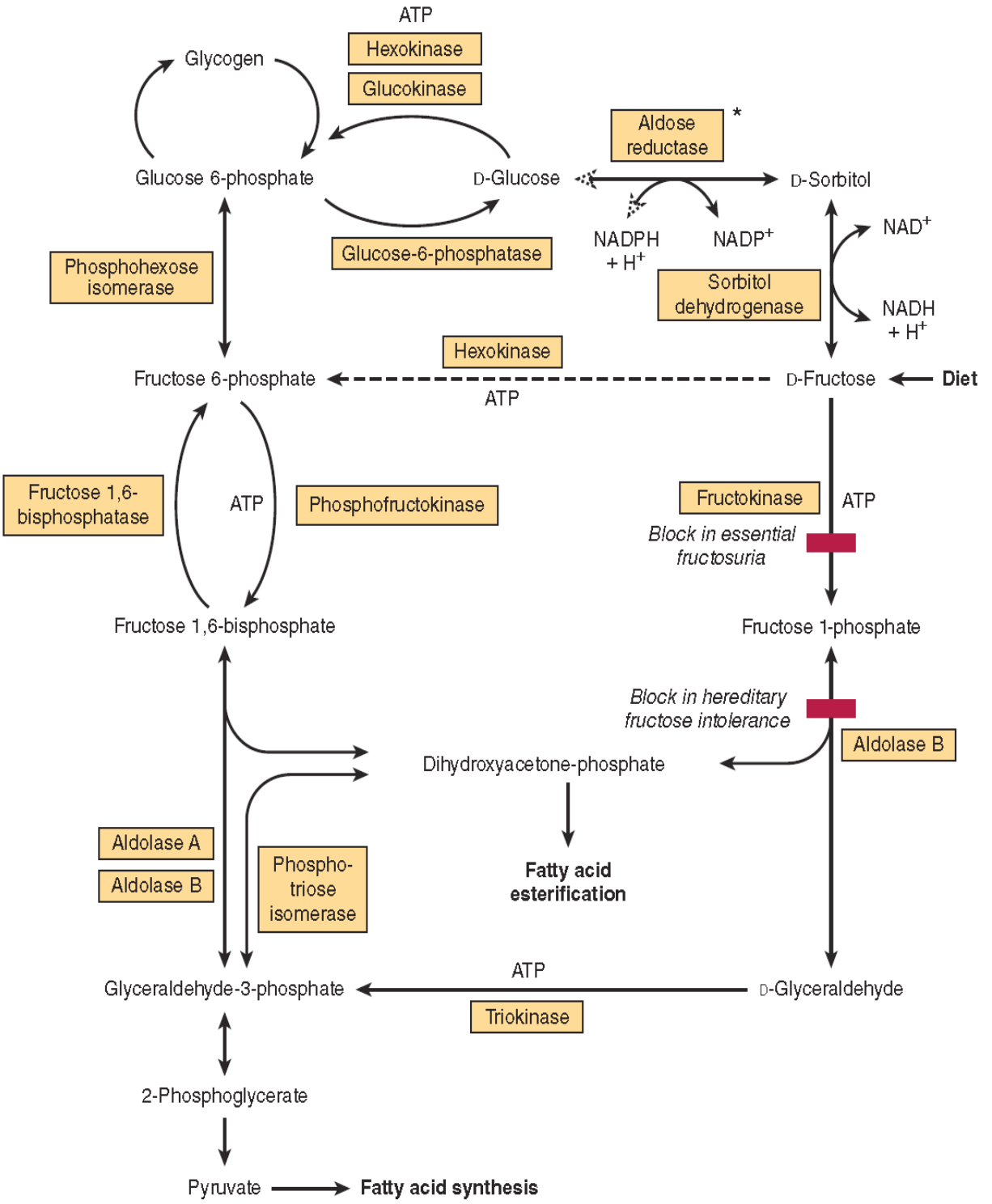
A: Acidosis

D: Increase in 2,3 BPG

E: Exercise

T: Increase in Temp

METABOLISM OF FRUCTOSE & POLYOL PATHWAY OF GLUCOSE



→ GOLDEN MCQ POINTS

FRUCTOLYSIS is seen ONLY in LIVER CYTOSOL due to presence of
ALDOLASE B only in liver.
→ Inborn Errors in Fructose Metabolism
1. Fructokinase deficiency leads to Essential
Fructosuria
2. Aldolase B deficiency leads to Hereditary
Fructose Intolerance that requires attention.
MC Age of presentation →

→ Fructose is more atherogenic than glucose as overconsumption of fruits leads to Hypercholesterolemia, hypertriglyceridemia & Hyperuricemia
EXPLANATION:

→ Net gain of ATP in fructolysis is always ATP

- Sorbitol Pathway – ABSENT in Liver

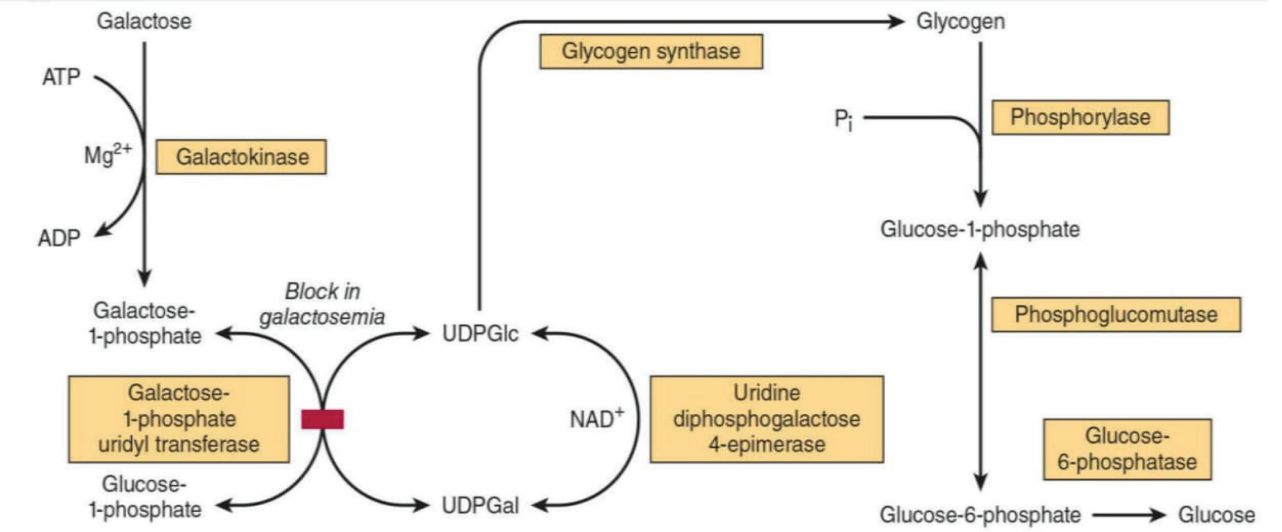
- ONLY Possible in → BRAIN & LENS due to presence of enzyme ALDOSE REDUCTASE in these tissues

- Sorbitol deposition in lens of eyes – development of SNOWFLAKE cataract in DM

Snow Flake Cataract in Diabetes mellitus due to Sorbitol deposits

✹ GALACTOSE METABOLISM

GALACTOSE IS NEEDED FOR SYNTHESIS OF LACTOSE, GLYCOLIPIDS, PROTEOGLYCANS & GLYCOPROTEINS



GALACTOSEMIA

- Deficiency is Galactose-1-phosphate uridyl transferase (GALI PUT) > Galactokinase leads to Galactosemia

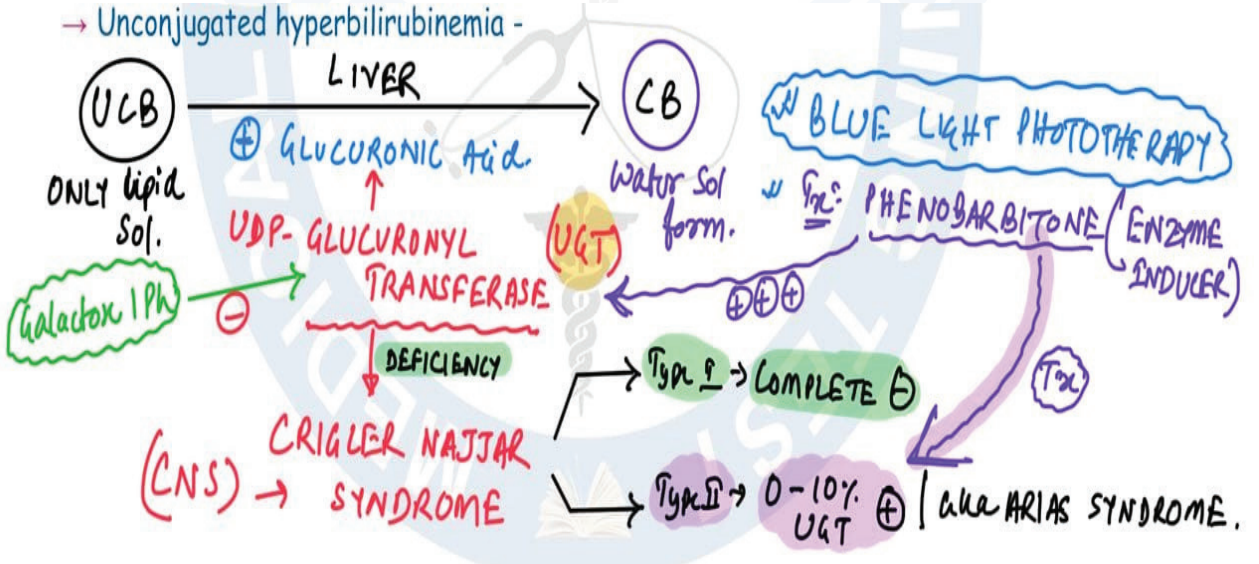
- OIL DROP Cataract is seen in galactosemia d/t deposition of galactitol/dulcitol in lens

❑ Galactosemia C/F: Presents usually on Day 3 or 4 after birth

→ Classical Presentation - child REFUSES TO SUCK BREAST MILK but ACCEPT GLUCOSE WATER,

→ Breast Feeding is CONTRAINDICATED,

So Tx is --



✹ GLUCURONIC ACID PATHWAY OF GLUCOSE

→ Alternative oxidation pathway for glucose is Glucuronic acid pathway which generates UDP Glucuronic acid that is needed for Conjugation of bilirubin,

steroids, various drugs and Synthesis of glycosaminoglycans (GAG) & Proteoglycans

→ Essential Pentosuria is d/t deficiency of Xylulose Reductase or Xylulose Dehydrogenase

→ Disorders included in Garrod's tetrad are Alkaptonuria, Albinism, Cystinuria and Essential Pentosuria

→ Human being can't produce vitamin C d/t deficiency of L-gulonolactone oxidase

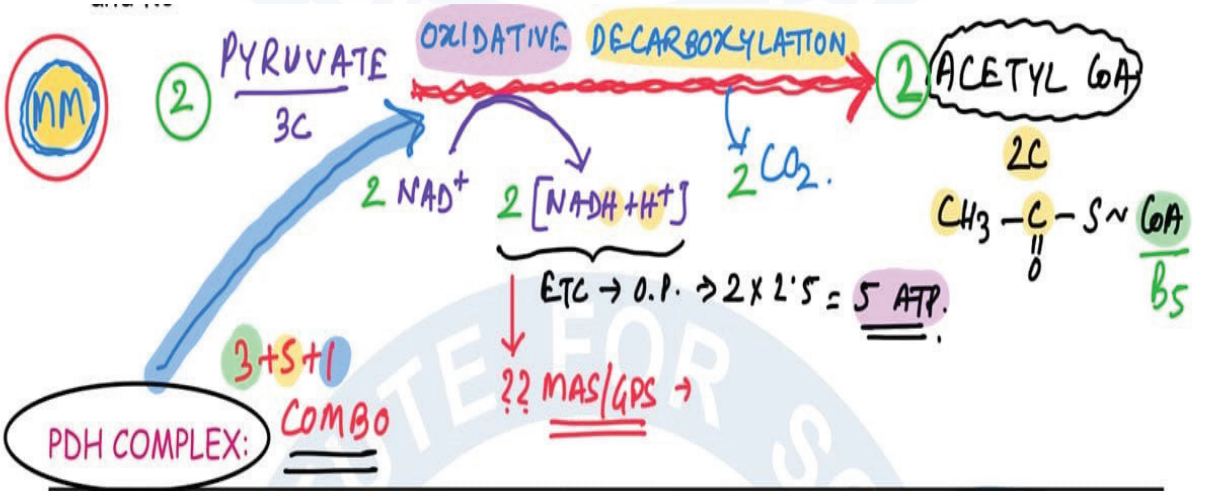
FATE OF PYRUVATE

Formation of

1. Acetyl CoA by oxidative decarboxylation (LINK REACTION) that finally enters into TCA cycle	3. Alanine (Transamination by enzyme ALT / SGPT)
2. Latic acid (Anaerobic glycolysis) by reduction,	4. Glucose (Gluconeogenesis)

Role of Pyruvate Translocase :

- Conversion of pyruvate to acetyl CoA: → Link Reaction between aerobic glycolysis and KC



PDH COMPLEX:

3 ENZYMES	5 COENZYMES	1 COFACTOR
Pyruvate Dehydrogenase	TPP (vit B1)	Mg2+
Dihydrolipoyl transacetylase	FAD (vit B2)	
Dihydrolipoyl dehydrogenase	NAD (vit B3)	
	CoA (vit B5)	
	Lipoic acid	

- Enzyme NOT related to PDH complex – Pyruvate Decarboxylase

PYRUVATE ==========→ ACETALDEHYDE

Extra Edge:

ALCOHOL

METABOLISM

- Why does alcohol consumption lead to Lactic Acidosis?

→ PDH complex deficiency

- Can be treated by ……supplementation

- Diet Prescribed for PDH Complex deficiency:

- Normal Lactate to Pyruvate ratio **(L/P ratio) is 6 – 10.** The ratio serves as a marker of tissue hypoxia in circulatory and septic shock

→ GOLDEN MCQ POINTS of KREB"S CYCLE PATHWAY

- IRREVERSIBLE reactions of TCA cycle
- CLASSICAL CONDENSATION reaction
- ONLY 5C compound of KC
- Rate limiting enzymes in Kreb's cycle
- Vitamins required for Kreb's cycle are
- Only Substrate level phosphorylation in TCA cycle
- Only NON-Matrix Enzyme in KC
- Net gain of ATP in Kreb's cycle

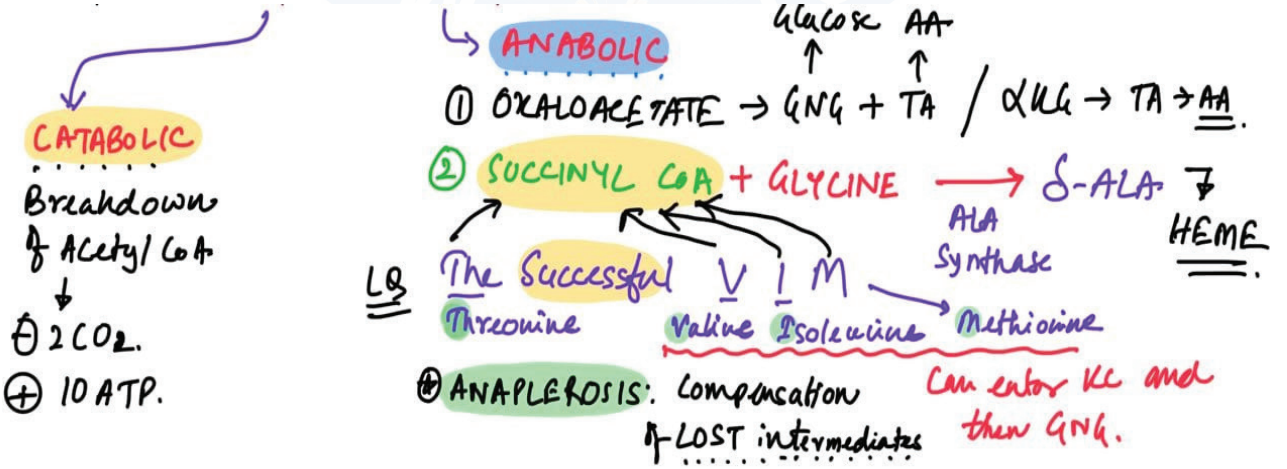
Extra Edge:

→ Complete oxidation of one molecule of glucose under aerobic condition produces

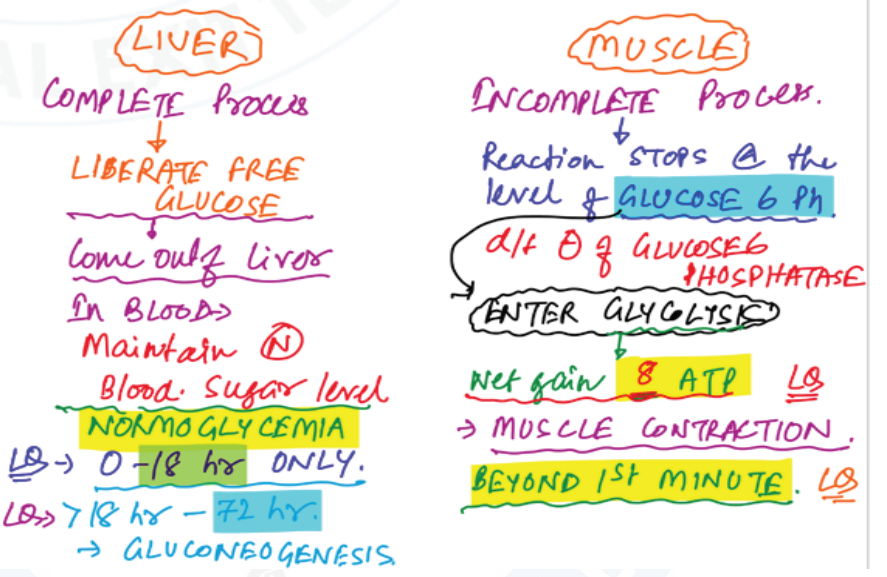
→ Complete oxidation of one molecule of pyruvate under aerobic condition produces

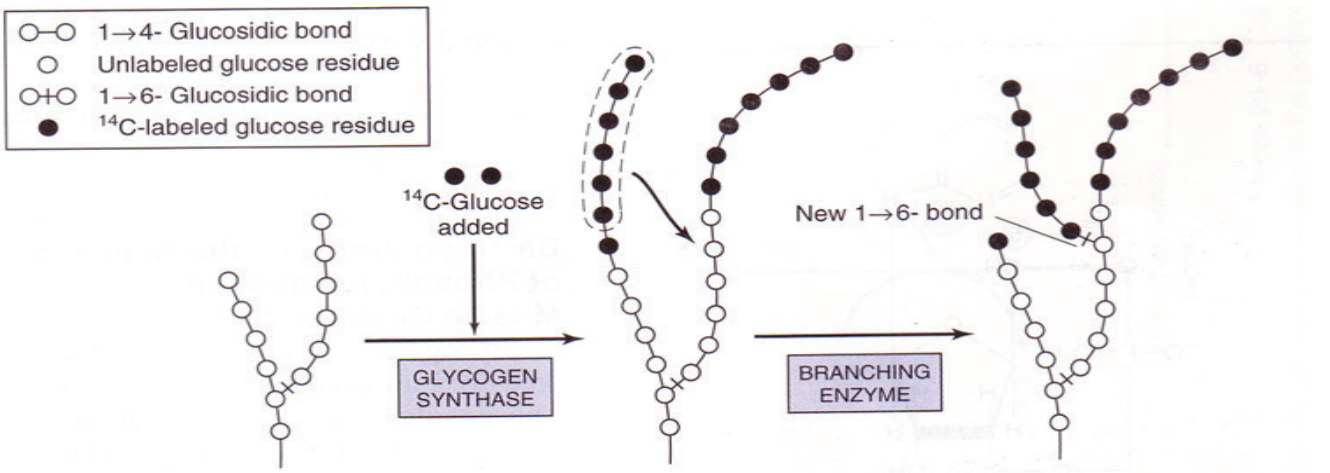
- Inhibitors of TCA cycle:
  - F
  - A
  - M
- Can we convert Acetyl CoA back to Glucose?

- Why KC is an Amphibolic Pathway?



GLYCOGEN METABOLISM

GLYCOGENESIS	vs	GLYCOGENOLYSIS
■ Seen in Well fed state ■ INSULIN dependent ■ Muscle stores glycogen more than liver due to greater muscle mass. ■ However, density or concentration wise liver stores more glycogen) ■ In glycogen glucose are linked to linear linkage by alpha 1,4 and at branching point alpha 1,6 bonds		■ Seen in Fasting and STARVATION ■ GLUCAGON & EPINEPHRINE dependent  ■ Creatine Phosphate > Glycogen is source of energy in 100-meter sprinter. Hence, Long distance runners do not compete with sprinters!!



GOLDEN MCQs POINTS of GLYCOGENESIS & GLYCOGENOLYSIS

- Alpha 1:4 BOND Formation: GLYCOGEN SYNTHASE → RLE.
  ⊕ ↑ (L)+(M)
  INSULIN ⊕ HK-IV/GK. (LIVER).
- ALPHA 1:6 BOND Formation: (BE) → GLUCOSYL 4,6 TRANSFERASE.
- Alpha 1:4 BOND Breakdown: → GLYCOGEN PHOSPHORYLASE → RLE (B6/PLP)
  GLUCAN TRANSFERASE
  ⊕ CAMP (2nd m) ⊕ GLUCAGON (1st m) Epinephrine
- ALPHA 1:6 BOND Breakdown: (DBE) → α 1:6 GLUCOSIDASE → free GLUCOSE
- Enzyme common to both glycogenesis and glycogenolysis- Phosphoglucomutase
  n n ⌐ n GLUCONEOGENESIS ⊻ Liver n → GLUCOSE 6 PHOSPHATASE

GLYCOGEN STORAGE DISEASES

TYPE	Glycogen Storage Disease	Deficient Enzyme
Type 0	LEWIS'	Hepatic Glycogen Synthase
Type Ia	VON GIERKE'S	Hepatic Glucose-6-phosphatase
Type II	POMPE'S	Acid maltase / Alpha 1,4-glucosidase
Type III	CORI-FORBE'S/Limit Dextrinosis	Debranching enzyme / Alpha 1,6 glucosidase
Type IV	ANDERSEN'S / Amylopectinosis	Branching enzyme / Glucosyl 4,6 Transferase
Type V	McARDLE'S	Muscle Glycogen phosphorylase
Type VI	HER'S	Hepatic Glycogen phosphorylase
TYPE VII	TARUI's disease	Muscle Phosphofructokinase

✹ GOLDEN MCQ POINTS regarding GSDs:

- ✓ MC GSD in Children & Overall →
- ✓ MC GSD in Adults →
- ✓ Most fatal or lethal GSD →

- ✓ MC GSD treatable with ERT →

- ✓ Exercise induced MYOGLOBINURIA →

- ✓ Muscle involvements are NOT seen in →
- ✓ Abnormal type glycogen deposition in →
- ✓ MC GSD predisposing to cirrhosis →

❖ VON GIERKE's Disease:

(0–28 d) EARLY (NEONATAL)

① FASTING HYPOGLYCEMIA in b/w feeds.

② LACTIC ACIDOSIS → ↑ GLYCOLYSIS.

③ HYPERLIPIDEMIA → ↑ MOBILIZATION of STORED FATS

④ KETOSIS → d/t ↑ KB Production

⑤ HYPERURICEMIA ↑ UA in Blood.

↑ GLUCOSE 6 PHOSPHATE

LIVER

Excess → ENTER → HMP SHUNT.

↑ PRODUCTION of (PRPP) PHOSPHO RIBOSYL PYRO PHOSPHATE.

↑ PURINE PRODUCTION.

↑ PURINE BREAKDOWN

↑ URIC ACID

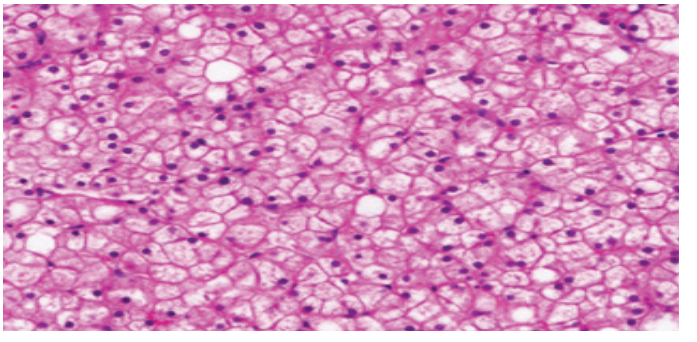
LATE after 3-4 MONTHS.

* MASSIVE HEPATOMEGALY
PROTUBERANT ABDOMEN
'POT BELLY'
* KIDNEY ENLARGEMENT.
* XANTHOMA FORMATION.
MC → BUTTOCKS.

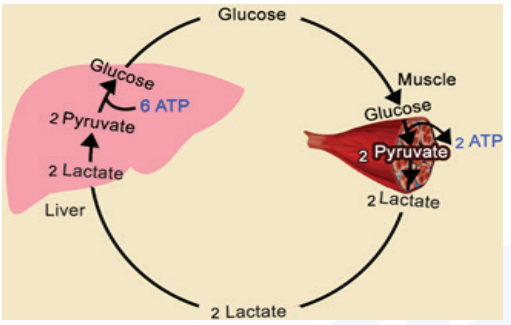
CHUBBY CHEEKS
'DOLL LIKE FACIES'.

* (N) LIVER ENZ
ALT / AST.

GOUT > 6 months.



✹ CORI CYCLE



- Cori Cycle is occurring in Cytosol
- concerned with transport of muscle lactate to liver for gluconeogenesis during strenuous exercise
- Its equally important for RBC for disposing excess Lactate

✹ GLUT - GLUCOSE LINKED UPTAKE TRANSPORTERS

BI DIRECTIONAL TRANSPORTERS

Rani	Laxmi	Bai	Made	India	Successful
GLUT 1	GLUT 2	GLUT 3	GLUT 4	GLUT 5	GLUT 6
RBC	Liver	Brain	Sk. Muscle	Intestine	Spleen
Others Brain, Kidney, Placenta	Others GB, Pancreatic β cell Kidney *NON-INSULIN DEPENDENT	Others Placenta	Others Cardiac muscle, Adipose Tissue, LMG *INSULIN DEPENDENT	*Exclusive for fructose absorption	

UNI DIRECTIONAL TRANSPORTERS (Sodium Glucose Linked Transporters)

a) SGLT 1 – Absorption of glucose in intestine by secondary active transport.

Co-transporter of glucose in intestine is – Sodium

b) SGLT 2 – Reabsorption of glucose in kidney tubules (90% in PCT itself)

- Gene mutation of SGLT 2 leads to Renal glucosuria where Renal threshold of glucose is brought down to 155 mg/dl.

✹ HEXOSE MONOPHOSPHATE SHUNT (HMP SHUNT)

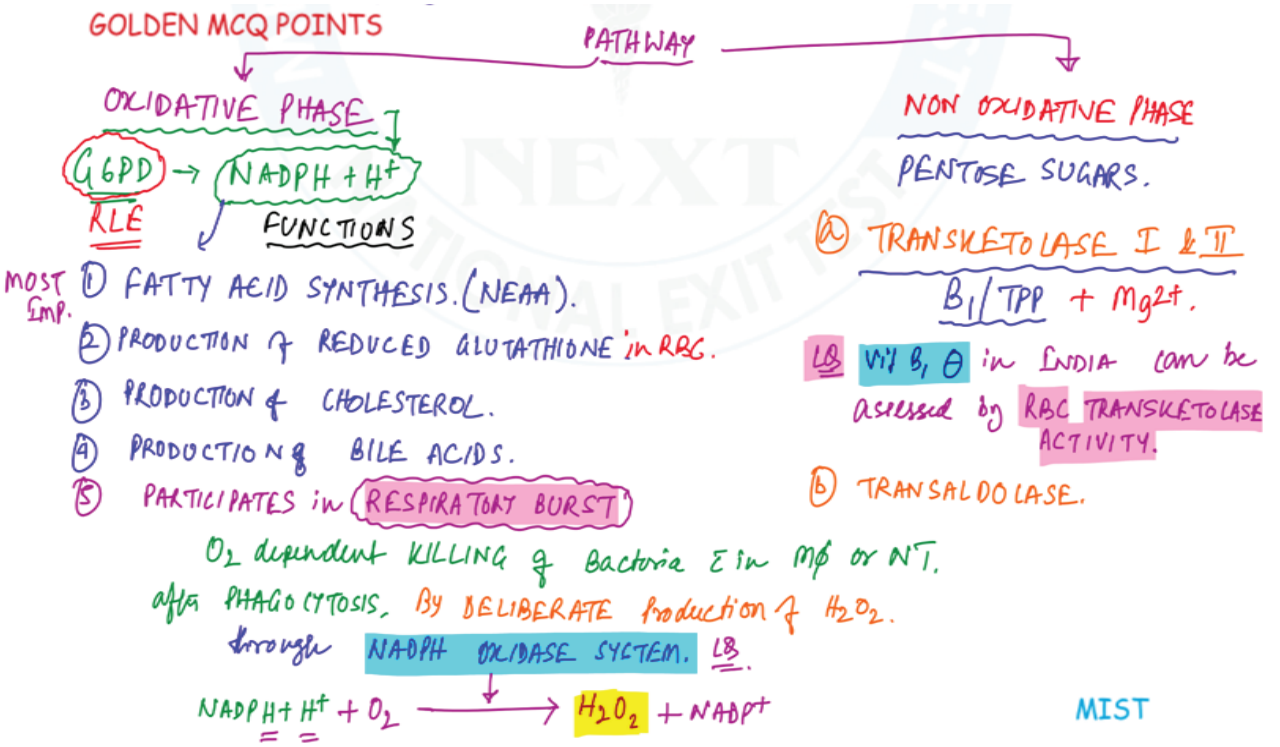
What is it?

ther names: PENTOSE PHOSPHATE PATHWAY (PPP), DIRECT OXIDATIVE PATHWAY, Phosphogluconate Oxidation PATHWAY, Warburg, Lipmann, Dickens SHUNT (WLD PATHWAY)

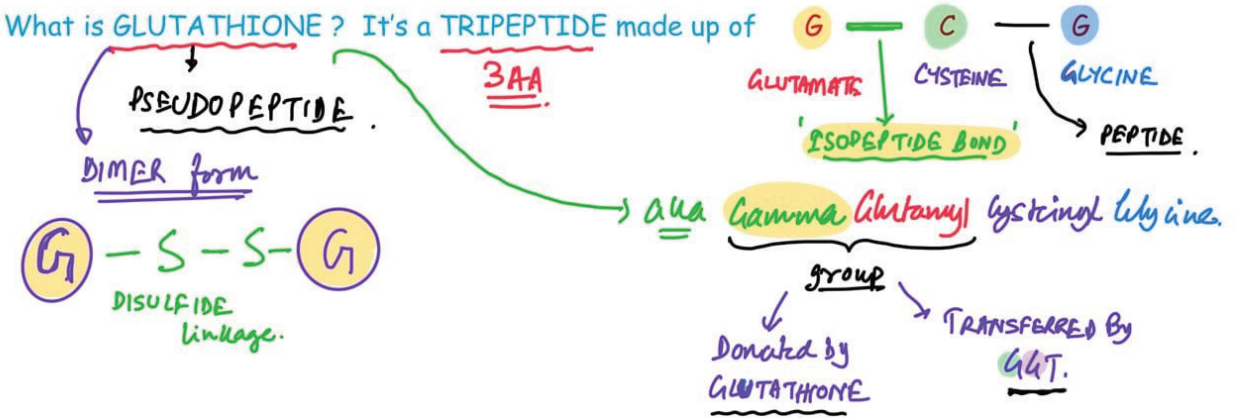
SITE::

MAXIMUM NADPH Quantity is seen in →

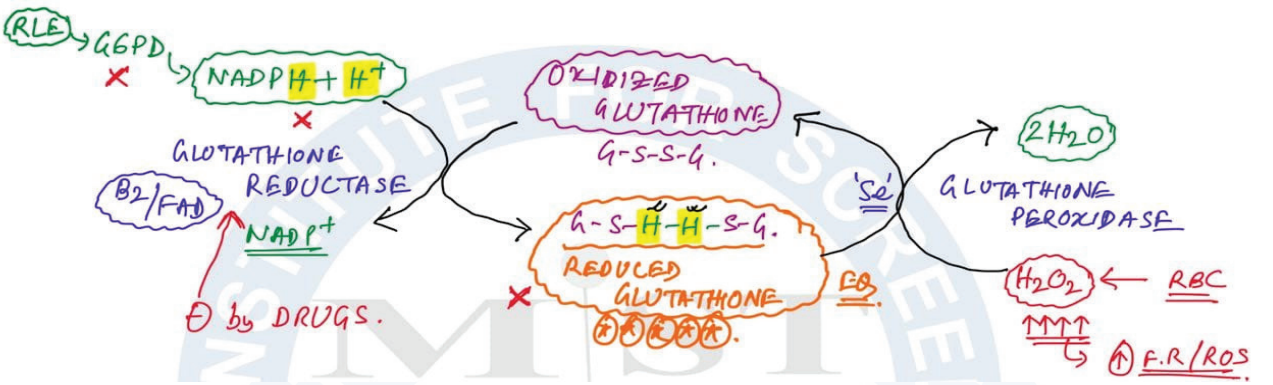
MAXIMUM NADPH concentration is seen in →



Formation of reduced glutathione (GSH) →



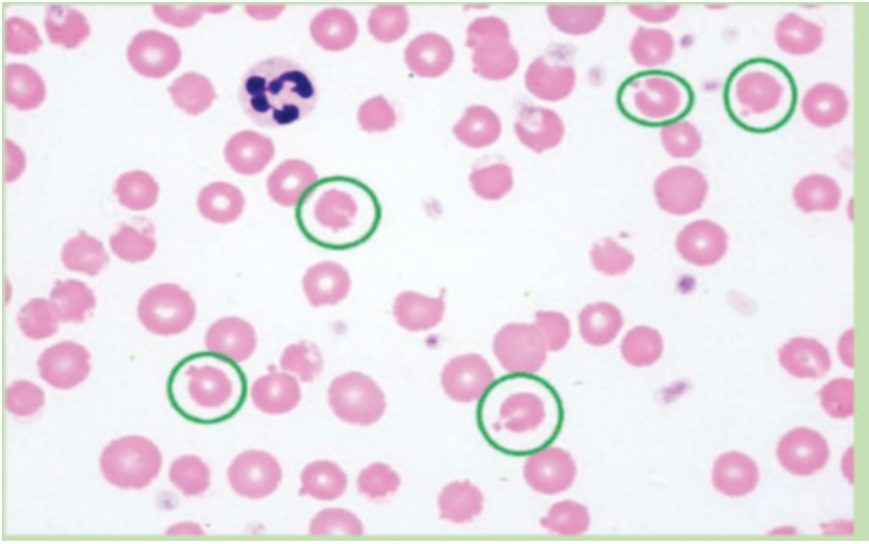
SPECIAL PATHWAY in RBC



- Vitamin B2 deficiency can be assessed by -

DISORDERS Associated with HMP:

- HEMOLYTIC ANEMIA – MCC of Hemolytic anemia in India is G6PD deficiency followed by pyruvate kinase deficiency

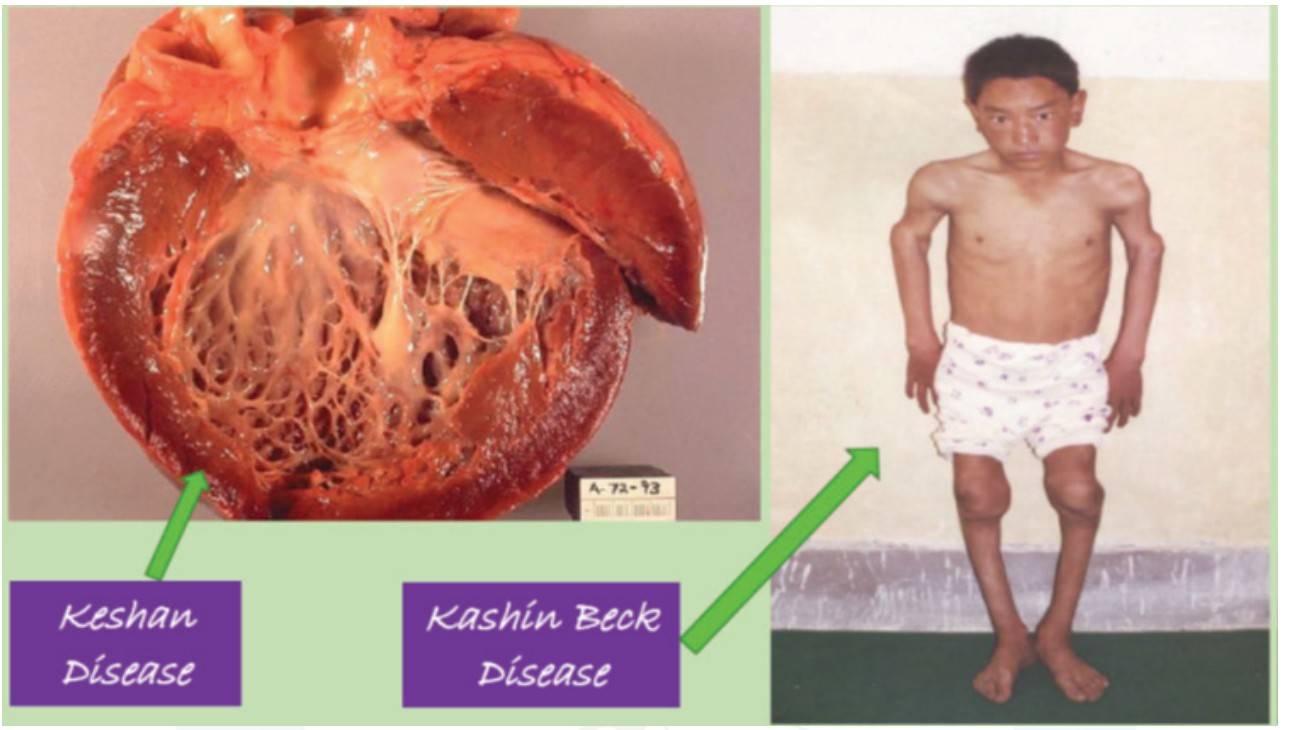


Heinz Bodies d/t condensation of Hb in Hemolytic Anemia due to G6PD Deficiency

- Antimalarial drugs, antipyretic drugs & sulfa group of antibiotics (SULFAMETHOXAZOLE) can worsen a case of G6PD deficiency leading to hemolysis. Hence, they are CONTRAINDIACATED.
- G6PD deficiency confers some protection against plasmodium falciparum malaria

- WERNICKE's KORSAKOFF SYNDROME

SELENIUM DEFICIENCY DISORDERS:



PRACTICE MCQS:

Most important rate limiting enzyme of glycolysis is

a. PFK-1

b. Pyruvate kinase

c. Glucose 1,6- bisphosphatase

d. Hexokinase

PFK-I is allosterically activated by

a. Fructose 1, 6 bisphosphate

b. Fructose 2, 6 bisphosphate

c. Phosphoenolpyruvate

d. Pyruvate

Insulin affects all of the following enzymes in glycolysis EXCEPT:

a. Phosphofructokinase

b. Pyruvate kinase

c. Glucokinase

d. Hexokinase

Which one of the following statements concerning glycolysis is correct?

a. The conversion of glucose to lactate requires the presence of oxygen

b. Hexokinase is important in hepatic glucose metabolism only in the absorptive period following consumption of a carbohydrate – containing meal.

c. Fructose 2, 6 – bisphosphate is a potent inhibitor of phosphofructokinase

d. The regulated reactions are also the irreversible reactions

Which of the following is the end product of glycolysis in an RBC

a. Lactic acid

b. Acetyl Co A

c. Enters Krebs cycle

d. Ethanol

Inhibition of thiamine deficiency affects conversion of

a. Pyruvate to acetyl Co-A

b. Succinyl CoA to fumarate

c. Fumarate to succinyl CoA

d. Acetyl CoA to pyruvate

E coli sepsis seen in

a. Urea disorder

b. Galactosemia

c. Glycogen storage

d. Lysosome disorder

An infant had frequent attacks of vomiting and tremors on weaning off from milk and addition of fruit juices. Urine reducing sugar was positive but glucose was negative. The infant is suffering from:

a. Fructokinase deficiency

b. Galactokinase deficiency

c. Glucose-6-phosphatase deficiency

d. Aldolase –B deficiency

Substrate level phosphorylation is seen in reaction catalysed by

a. Succinate dehydrogenase

b. Malate dehydrogenase

c. Succinate thiokinase

d. Isocitrate dehydrogenase

Which step in TCA cycle is irreversible

a. Succinate thiokinase

b. Alpha Ketoglutarate dehydrogenase

c. Iso Citrate Dehydrogenase

d. Aconitase

Succinyl CoA can be formed by

a. Histidine

b. Leucine

c. Valine

d. Lysine

Skeletal muscle is deficient in

a. Glucose 6 phosphatase

b. Hexokinase

c. Isomerase

d. Phosphofructokinase

In McArdle's disease, Muscle cannot make use of glycogen for energy due to the absence of

a. Glucose 6 phosphatase

b. Glucokinase

c. Phosphorylase

d. Phosphoglucomutase

Workbook Sheet

CHEMISTRY OF AMINO ACIDS & PROTEINS

- Proteins has only a minor role in energy supply but they are the structural and functional units of the body – BUILDING BLOCK
- Protein metabolism is important -- growth, repair and reconstruction of tissue.
- All proteins have same 20 AA, but they vary in NUMBER, COMPOSITION and SEQUENCE (THINK About HbA vs HbS)
- If the availability of AA is more than required, they are converted to ketoacids in liver, which further enters into kreb's cycle through Acetyl CoA and used for energy.
- Urinary nitrogen is the chief end product of protein metabolism.
- Urea represents 95% of urinary nitrogen in human beings while remaining 5% is contributed by Uric acid & Creatinine
- Human has only L-a-Amino acids
- New Concept: only the assembly of polypeptide chains in 3D structure i.e. Tertiary or Quaternary structures are proteins. Rest are called polypeptides.

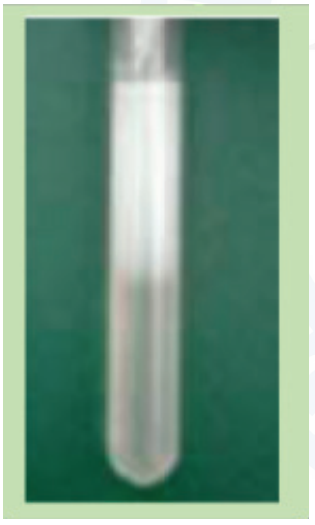
Basic Structure of AA:

- Branched chain AA are Valine, Isoleucine, Leucine (VIL) that plays role in energy production.
- Sulfur containing AA are Methionine Cysteine and Cystine (2 mol of cysteine together) while Hydroxyl group containing AA are Tyrosine, Serine, Threonine (Mn: Try Three Sari).
- Basic amino acids are Histidine, Arginine and lysine (Mn: His Basics Are Loose) & Arginine is the most basic amino acid.
- Aromatic amino acids are Histidine, Tyrosine, Tryptophan & Phenylalanine (Mn: HTTP) and they canabsorb UV light (Max-Tryptophan).

- Proline is imino acid and contains pyrrolidine ring. Prevents Proper Protein Folding Process
- Leucine is most non polar while uncharged polar AA are Glycine, Serine, Asparagine, Tyrosine, Threonine, Glutamine & Cysteine (Mn: Good Souls Are Truly The God's Children).
- Leucine is cardioprotective
- Purely/Only/100% ketogenic AA is Leucine while partially glucogenic and partially ketogenic are Lysine, Tyrosine, Tryptophan, Phenylalanine, Isoleucine (Mn: Le Tai Tai Ph Is) and most glucogenic is Alanine.
- Essential amino acids are (Mn: TV TILL 8 PM) threonine, valine, tryptophan, isoleucine, leucine, lysine phenylalanine and methionine while Histidine and Arginine are semi-essential amino acids.
- Tyrosine and Cysteine are both non-essential amino acids as they are produced in liver from phenylalanine & methionine respectively.

Extra Edge:

- → HISTIDINE: contains imidazole ring that can PROTONATE or DE-PROTONATE at Physiological pH for which it acts as blood buffer and ensures Hb also to act as buffer in PLASMA at physiological pH
- → Tyrosine contains phenol ring while Tryptophan contains indole ring
- → Arginine contains guanidinium/ guanidino group and it can be detected by Sakaguchi test
- → Ninhydrin test detects a AA and generates Ruhemann's purple color



HEAT & ACETIC ACID COAGULATION TEST -

Detects PROTEINURIA, most likely ALBUMINURIA - REMEMBER Presence of Albumin in Urine is ALWAYS PATHOLOGICAL

However, the Normal Protein that's passed in urine during micturition →

Max Up to →

✹ MICROALBUMINURIA

- Urinary Albumin Creatinine Ratio (UACR): SPOT test
- Males - < 17 mg of Albumin/gm of Creatinine
- Females - < 25 mg of Albumin/gm of Creatinine
- Moderately Increased Albuminuria (KDIGO)

✹ SOME IMPORTANT POINTS in STRUCTURES OF PROTEIN

✹ Various bond formation:

Covalent: PEPTIDE and DISULFIDE

Non covalent: H Bond, Hydrophobic, Electrostatic/Ionic/Salt Bridge, Van der Waal's forces

• Overall strongest bond in protein structure is Peptide bond f/b disulfide bond while van der waal's forces are weakest • Peptide bond is rigid and planar with partial double bond character where Distance between two α carbons is 0.36nm or 3.6 A° • H bonding GUIDES a protein into folding process	• MC AA found in a helix is alanine and best source of β pleated sheet is silk fibroin • Tertiary structure is a complete 3D structure that is stabilized by multiple weak interactions (Hydrophobic bonds, Electrostatic bonds, Van der waal's forces) and denotes Vibrational Spectrum of Proteins • Although VWF is the weakest of all, BUT it contributes MAXIMUM towards the STABILITY of the 3° of proteins. While MINIMUM contribution is shown by HYDROPHOBIC BONDS.
• Proline destabilizes both α helix & β pleated sheet by introducing kinks or bends while proper folding in a polypeptide chain is favoured by glycine	• Single PP chain always ends with tertiary structure viz. albumin, myoglobin • Skin keratin is softer than nail keratin due to less number of disulfide linkages • Best technique to detect 3D structure is X-Ray Diffraction Crystallography

Extra Edge:

PRION DISEASES: Prions are proteinaceous infective particles that have normal primary structure but abnormal tertiary structure due to improper β pleated sheet in their secondary structure

i. MC form of Human prion disease → Creutzfeldt Jacob Disease AKA TRANSMISSIBLE SPONGIFORM ENCEPHALOPATHY where cerebral cortex looks like sponge)

ii. Fatal familial insomnia,

iii. GSSD

iv. KURU

✹ ALZHEIMER's Disease

1. HYPER PHOSPHORYLATED Tau Proteins	2. β Amyloid Precursor Proteins (β-APP or AβP)
↓	↓
Neurofibrillary Tangles	DEPOSITED in BRAIN
↓	↓
Loss of MICROTUBULES in BRAIN	SENILE DEMENTIA
↓	
Decrease in Acetylcholine Production	
↓	
MEMORY LOSS	

✹ PARKINSON's disease: a Synuclein + Parkin

❑ REMEMBER Alzheimer's Disease and Parkinson's Disease both are reported due to abnormal folding of proteins but are NOT prion disease

✹ COLLAGEN

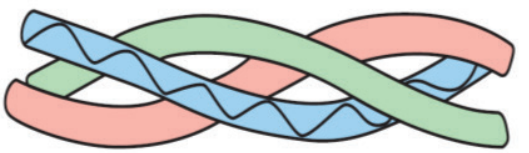
→ Most abundant protein in body

→ Most abundant plasma protein

→ Most abundant amino acid found in collagen

COLLAGEN Structure:

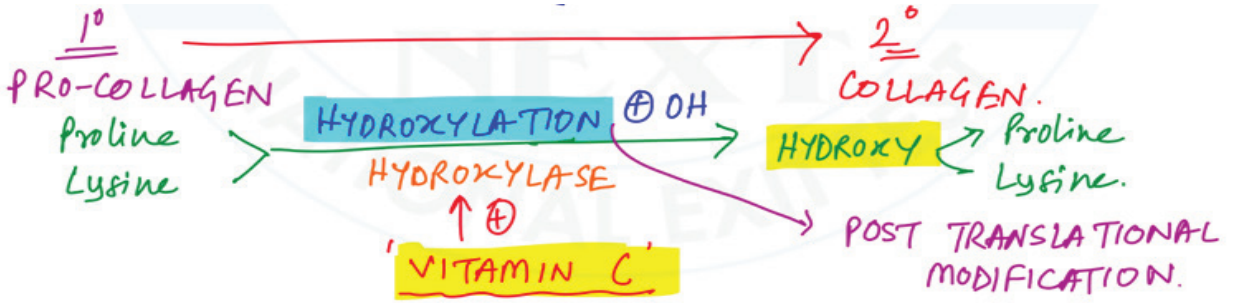
Triple helix



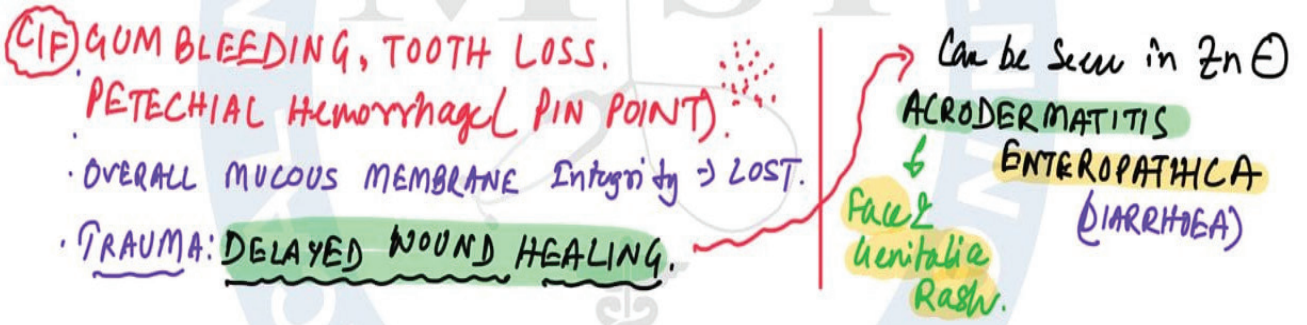
✹ Collagen has 3 individual a chains that finally attains a triple helical structure (Tropocollagen) with every 3rd AA is each a chain as Glycine (Min 33% of AA)

Amino acid sequence –Gly – X – Y – Gly – X – Y – Gly – X – Y –

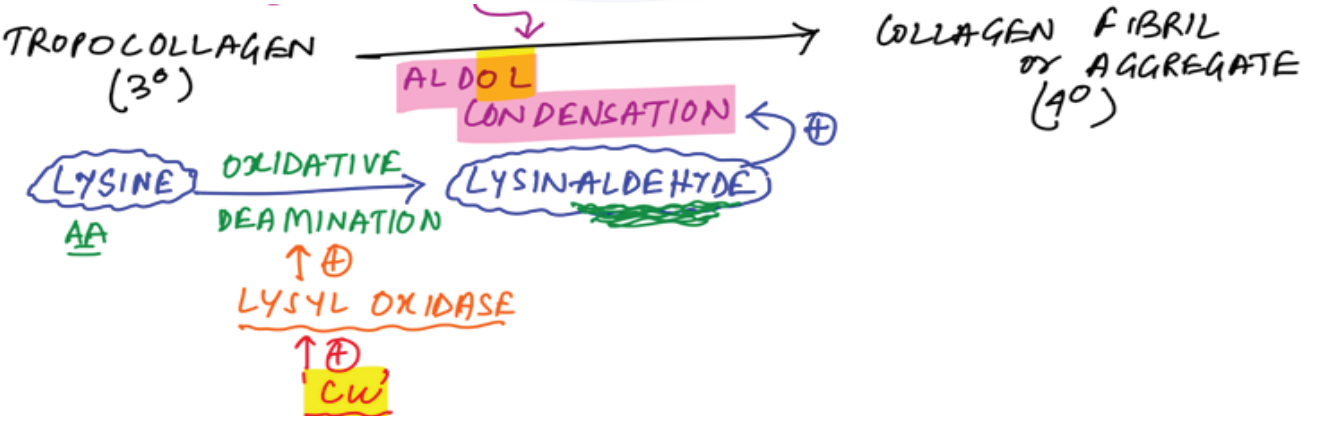
✹ COLLAGEN MATURATION: (1° → 2°)



→ Applied Aspect: SCURVY



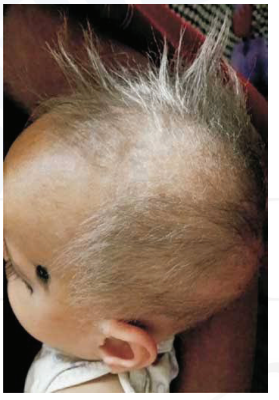
✹ COLLAGEN CROSS LINKING (3° → 4°)

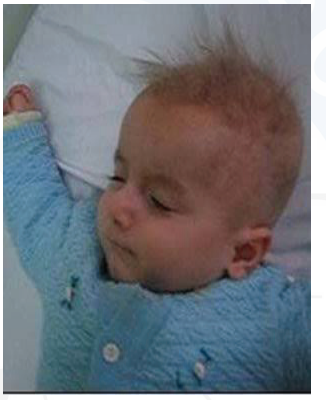


→ Applied Aspect: MENKE's KINKY HAIR SYNDROME:

ATP7A gene
↓
Cu BINDING ATPase
(INTESTINE).
↳ Cu Transporter for ABSORPTION.

↓ MALABSORPTION of Cu → ↓↓ Cu.
3° → 4°
Weak Collagen.
Excessive Curly/Kinky Hair



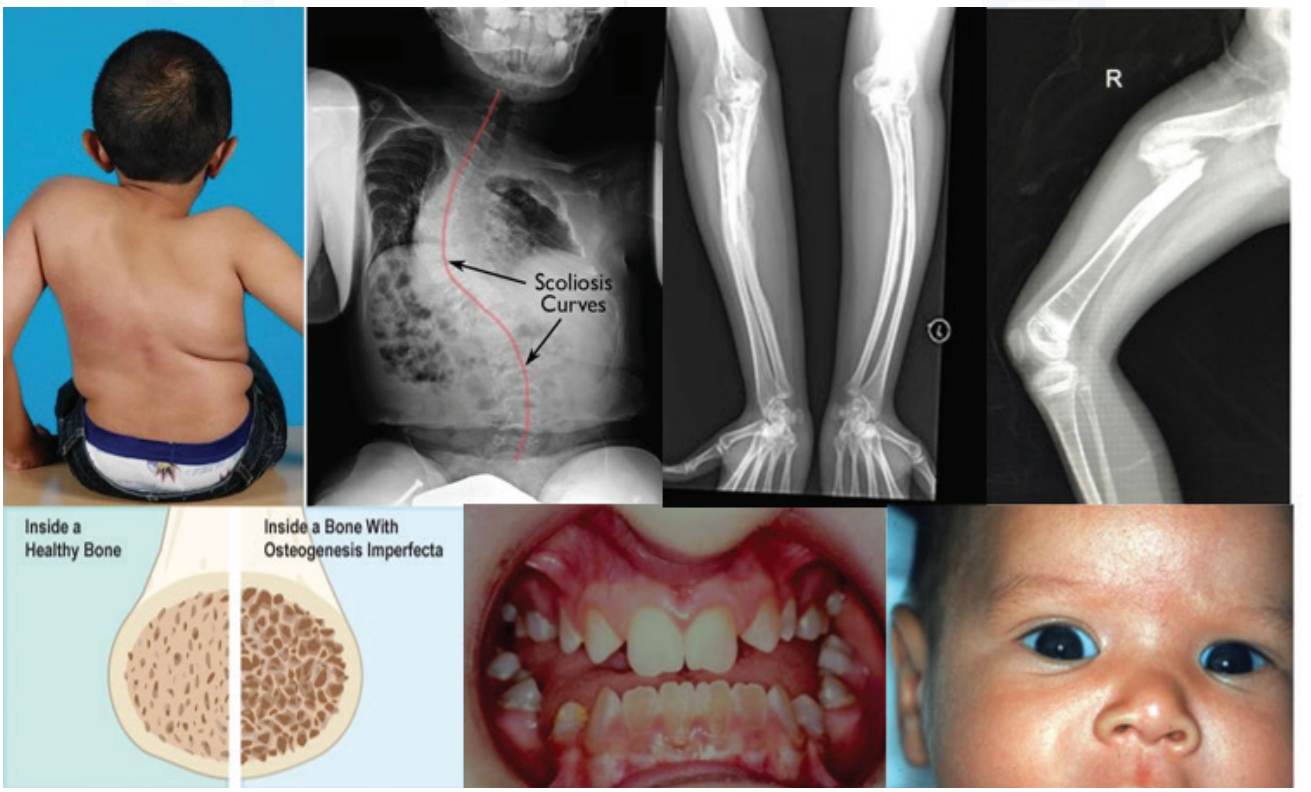


→ Types of Collagen: Depending on the amino acid variations, there are 27 types of collagens.

Type I	Skin, Bone, Tendon, Ligament, Perichondrium	S
Type II	Cartilage, Vitreous humor of eye, Joint capsule	C
Type III	Arteries, Veins, Capillaries, Lung, Skin	A
Type IV	Basement membrane, Lens, Cochlea (inner ear)	B

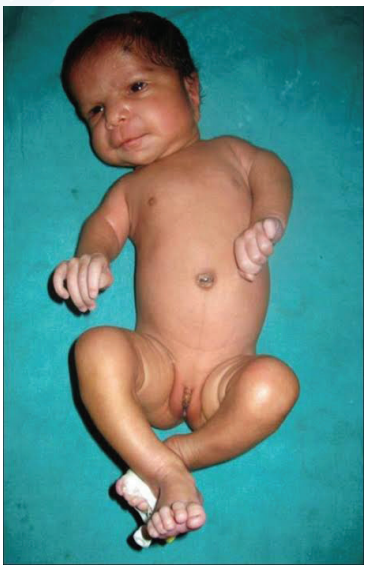
Collagen Vascular Disorders Aka Collagen Associated Disorders:

A) Type I Collagen Defect:



RICKETS:

B) Type II Collagen Defect:

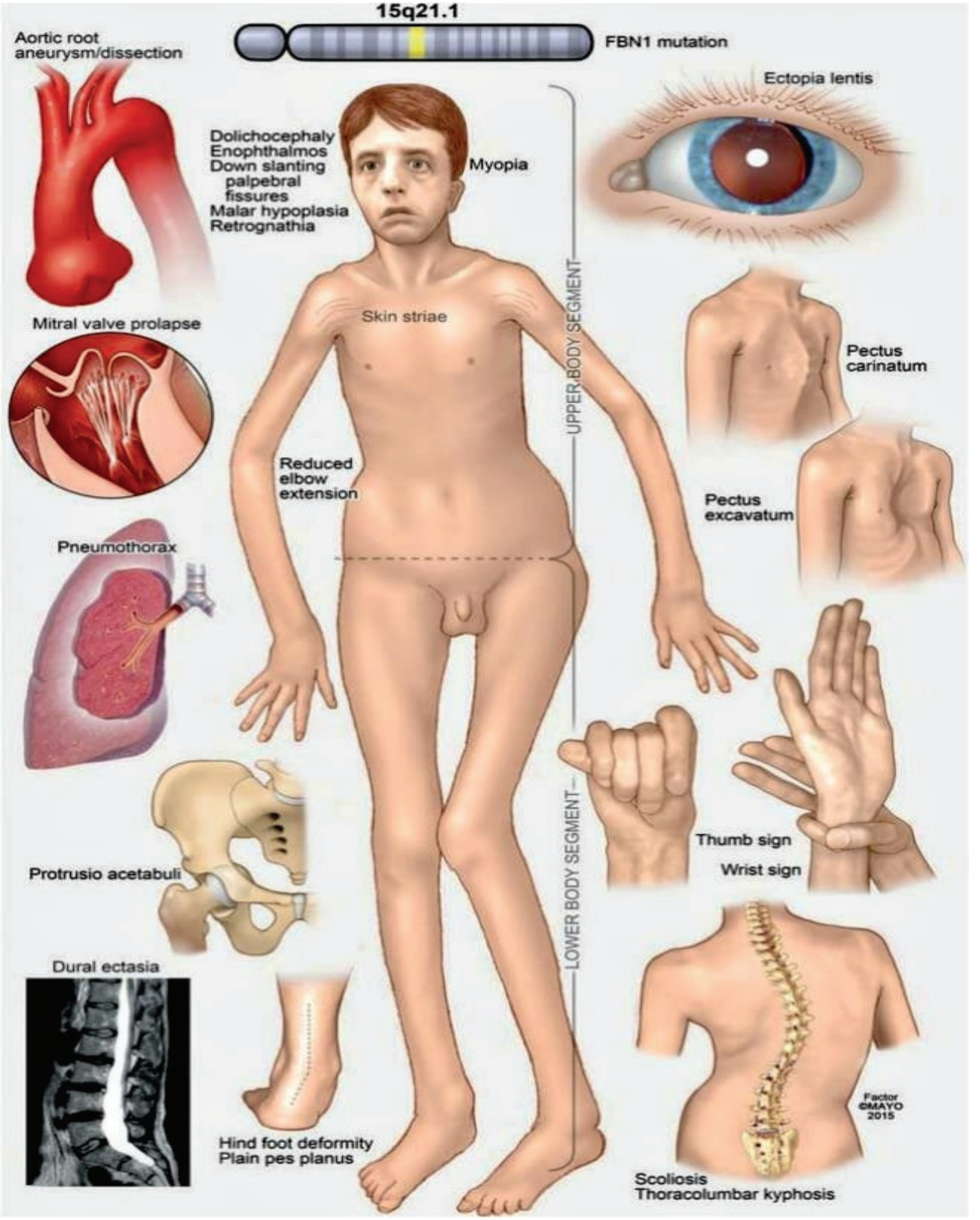


C) Type III & V Collagen Defect:

D) Type IV Collagen Defect:

E) Type VI & VII Collagen Defect:

Are we missing out on an important Collagen Vascular Disease??


15q21.1
FBN1 mutation
Aortic root aneurysm/dissection
Ectopia lentis
Dolichocephaly
Enophthalmos
Down slanting palpebral fissures
Malar hypoplasia
Retrognathia
Myopia
Skin striae
Mitral valve prolapse
Pectus carinatum
Pectus excavatum
Reduced elbow extension
UPPER BODY SEGMENT
Pneumothorax
Thumb sign
Wrist sign
Protrusio acetabuli
LOWER BODY SEGMENT
Dural ectasia
Hind foot deformity
Plain pes planus
Scoliosis
Thoracolumbar kyphosis
Factor ©MAYO 2015

- CHAPERONES: Protein folding is assisted by chaperones that are also known as 'Heat Shock Proteins'.
  - Acts as QUALITY CONTROL or EDITING MECHANISM during protein synthesis.
  - Chaperone action is considered as a type of post-translational modification
  - Examples of them are HSP – 70 (MC type), HSP – 60 (chaperonin), Protein disulfide isomerase, Peptide prolyl cis-trans isomerase etc.
- Denaturation: During heat denaturation except primary structure all other structural organization in proteins can be denatured because primary structure is stabilized by strongest peptide bond & Disulfide bond both.

→ Best technique to estimate Serum Total Protein:

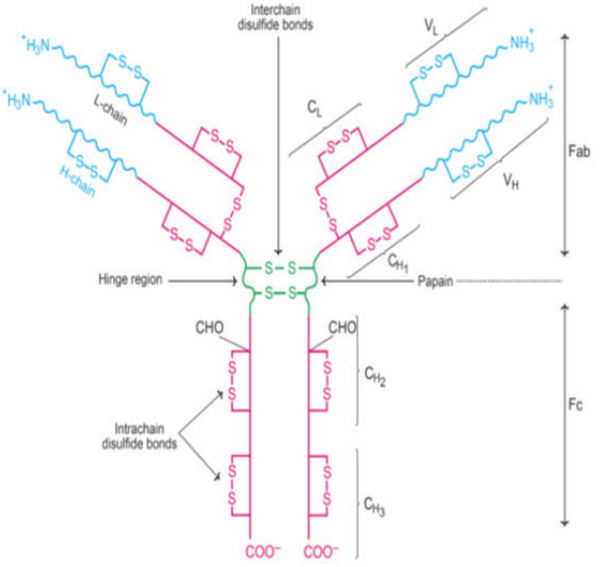
→ Best technique to estimate Serum Albumin:

→ Best technique to detect protein Purity

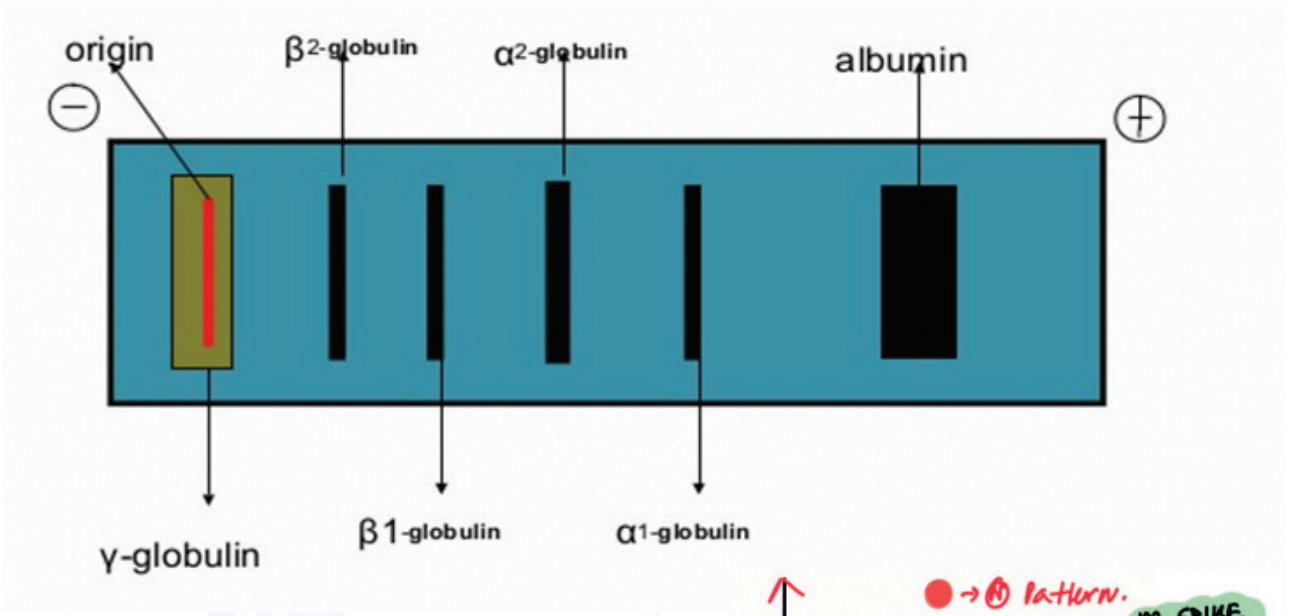
→ Best technique for molecular weight determination of proteins

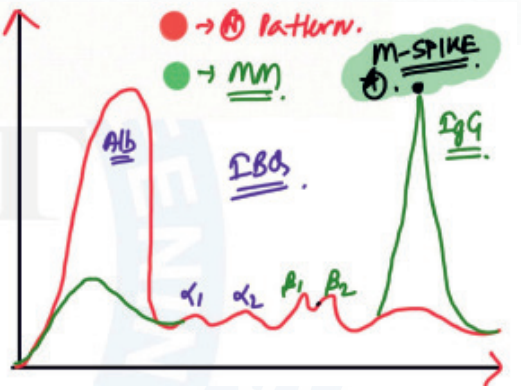
- IMMUNOGLOBULINS:

IMMUNOGLOBULINS:



Type	IMPORTANT ROLES & MCQ POINTS
IgG	major antibody, antibody seen in secondary immune response -triggers foreign cell destruction mediated by complement system. can cross placental barrier, and protects the newborn child from infections. Also explains the Rh isoimmunization - fetal hemolysis, neonatal jaundice, and in severe cases, neonatal death or miscarriage
IgA	Role in MUCOSAL barrier, Secretory antibodies seen in seromucous secretions of colostrum, gastrointestinal tract, nasopharyngeal tract, urogenital tract, tears, saliva, sweat.
IgM	Macroglobulins, Natural antibodies, Five subunits are joined together by a J-chain polypeptide. Antibody in primary response - most effective against invading microorgnisms.
IgD	May function as B-cell receptor
IgE	Mediate allergy, Immediate type hypersensitivity and anaphylaxis


origin
β2-globulin
α2-globulin
albumin
⊖
⊕
γ-globulin
β1-globulin
α1-globulin


Pattern.
MM.
M-SPIKE
IgG
Alb
IBG
α1
α2
β1
β2

Extra Edge

→ MULTIPLE MYELOMA: NEOPLASM of B-Cell origin./Plasma Cell origin.
DISPROPORTIONATE/MISMATCH Production of Heavy & Light chains.
OVERPRODUCTION of LIGHT Chains.
Released in Urine
κ
λ
GOLD STANDARD for Dx of MM
BENCE JONES PROTEIN (BJP).
↑IgG. overproduction.
M-BAND b/w β2 & β2. in SPE.
MYELOMA or MONOCLONAL BAND.
CRAB
C
R
A
B
↑Ca2+
Renal Stones
↑U.O.
ANEMIA
LYTIC BONE Lesion
(Low Back Pain)
Bradshaw Test
White Ring
JACOBSON & MILNER TEST aka HEAT Test.
BJP PRECIPITATED ONLY b/w 45°C-58°C
*

PRACTICE MCQS:

Major collagen in basement membrane

a. Type 1

b. Type 2

c. Type 3

d. Type 4

Which of the following enzyme required for Transformation of pro-collagen to collagen:

a. Zinc

b. Iron

c. Selenium

d. Copper

Menke kinky hair disease is linked to

a. Cu transporter

b. Fe transporter

c. Zn transporter

d. Mg transporter

Strongest bond from the following:

a. Hydrophobic

b. Vander waal's

c. Electrostatic

d. Hydrogen

Which of the following amino acid protonate and deprotonate at neutral pH and provides buffering action to blood at physiological pH

a. Histidine

b. Arginine

c. Lysine

d. Glycine

Separation between protein by their mass is performed by

a. SDS-PAGE

b. Electrophoresis

c. Salting out

d. Ion exchange chromatography

Amino acid with hydroxyl group

a. Leucine

b. Tyrosine

c. Cysteine

d. Valine

One of the following proteins does not have a quaternary structure

a. Myoglobin

b. Hemoglobin

c. Lactate dehydrogenase

d. Immunoglobulin G

Proteins can be precipitated by the following methods, EXCEPT

a. Adding alcohol and acetone

b. Saturating with ammonium sulphate

c. Using salts of heavy metal

d. Shifting the pH away from the isoelectric point

Selenocysteine is part of

a. NADP reductase

b. NADPH dehydrogenase

c. Thioredoxin reductase

d. PDH Complex

Which of the following forces is the most unfavorable for protein folding?

a. Conformational entropy

b. Hydrophobic Interactions

c. Van der Waals interactions

d. Hydrogen bonds

The purely ketogenic amino acids

a. Leucine

b. Lysine

c. Glycine

d. All of the above

Antibody mostly found in mucous membrane

a. IgG

b. IgM

C.IgA

d. IgD

Which of the following immunoglobulin elevated in multiple myeloma is the cause of M-band in serum protein electrophoresis

a. IgG

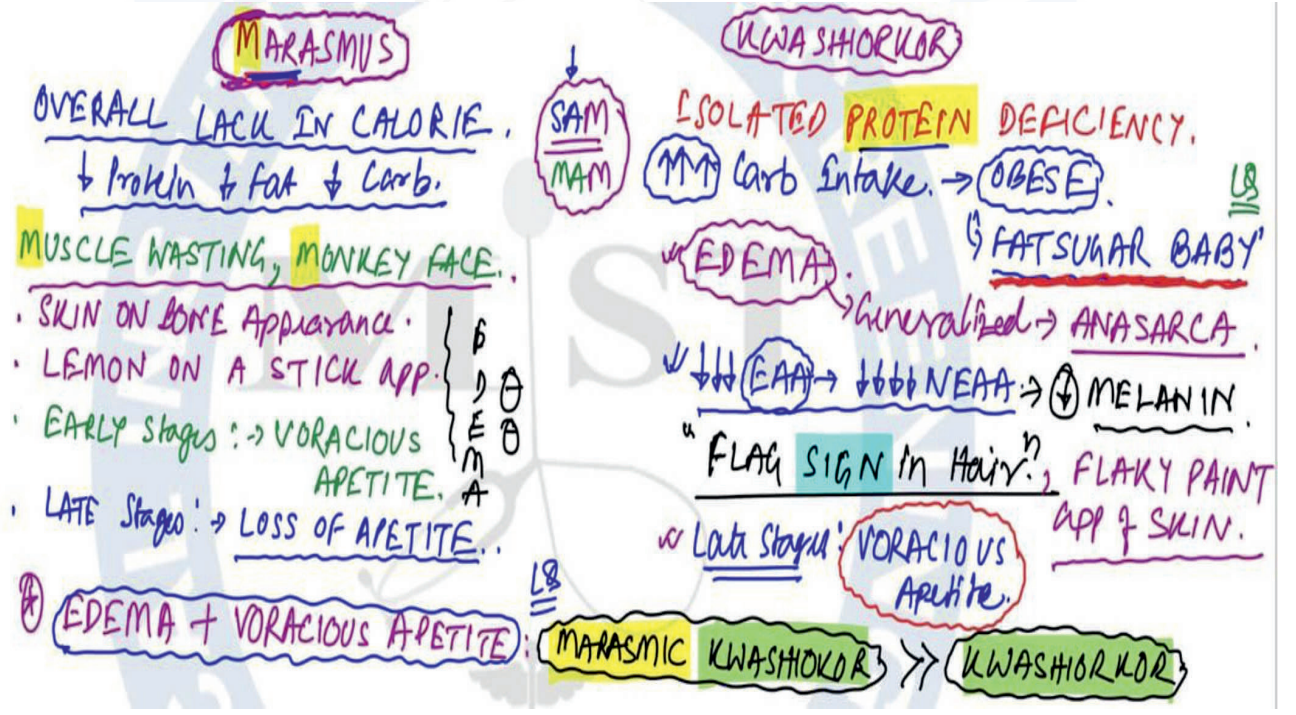
b. IgM

C.IgA

d. IgD

GENERAL AMINO ACID & PROTEIN METABOLISM

- NITROGEN BALANCE
- Normal nitrogen balance: intake=excretion |
- Positive nitrogen balance: intake>excretion |
- Negative nitrogen balance: intake<excretion |



✹ Digestion of dietary proteins

The major sites for protein digestion are the stomach and intestines.

✹ Absorption of amino acids

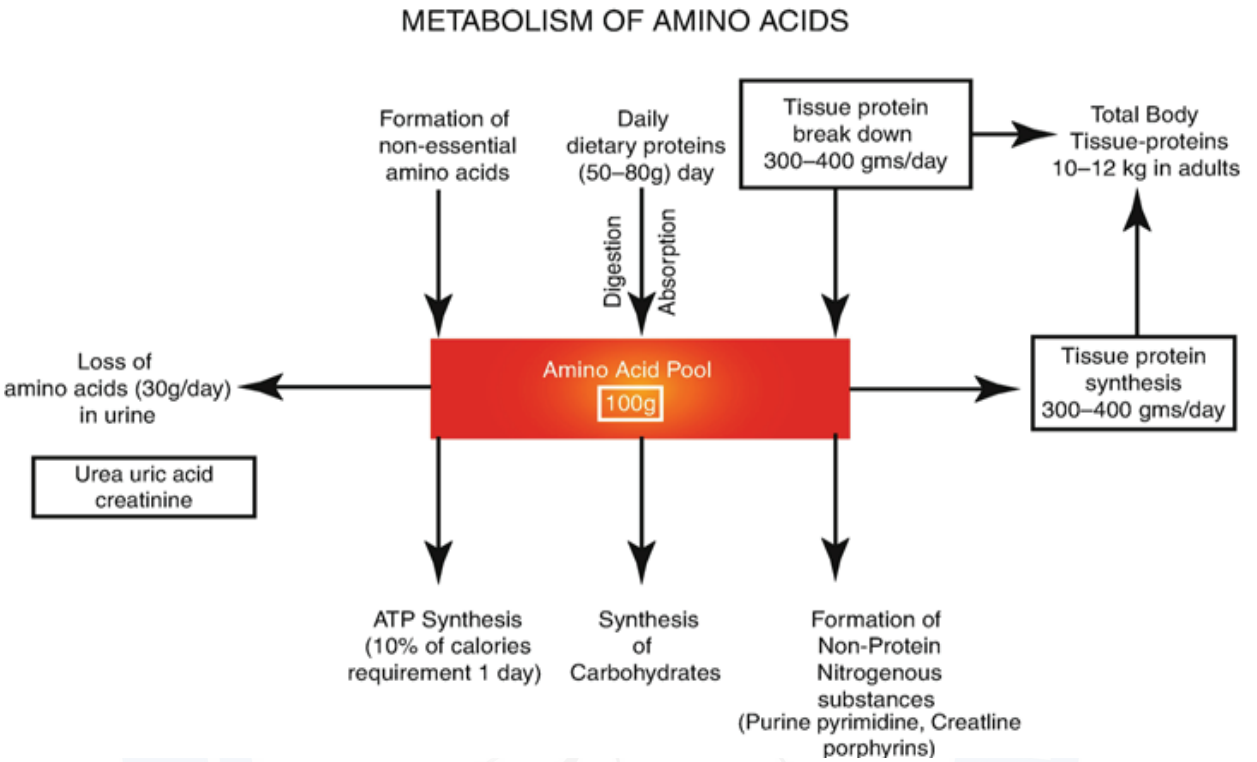
Amino acids and dipeptides are absorbed into the body by an energy-spending mechanism called "active transport".

ATP is required to CROSS the →

Extra Edge:

→ MEISTER Cycle:

→ AMINO ACID POOL



✹ INTRACELLULAR PROTEIN DEGRADATION

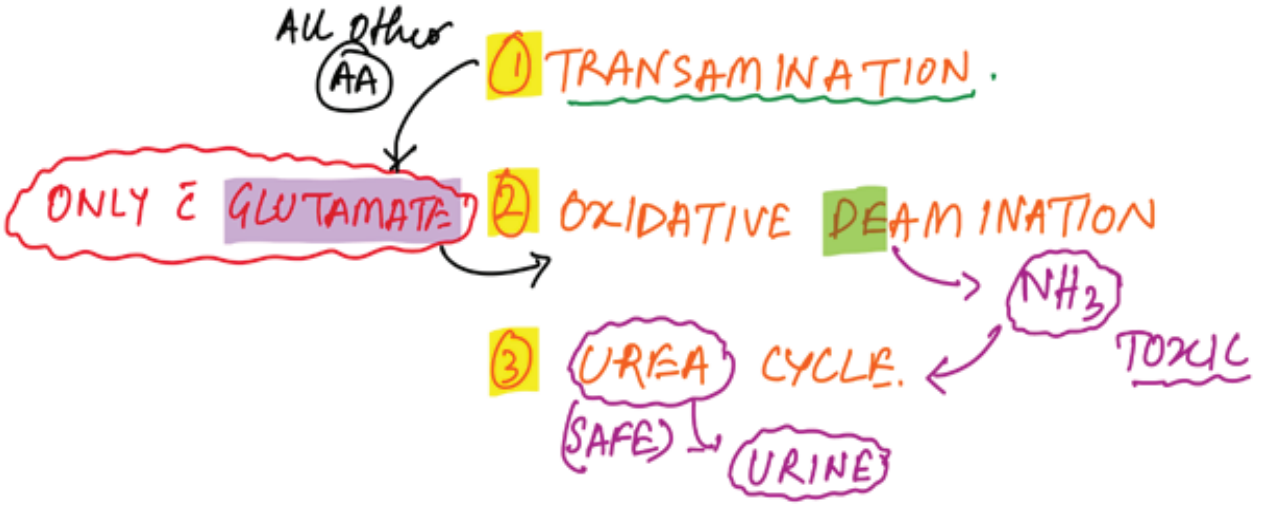
LYSOSOMAL DEGRADATION.
L&S
SUICIDAL BAG

CONTROLLED PROTEOLYSIS
SER: ERAD → Endo. Ret. Associated Degradation.
UBIQUITIN
S (SELECTION/TAG.
MARK OF DEATH
PROTEASOME
PROTEOLYSIS

✹ AMINO ACID DEGRADATION

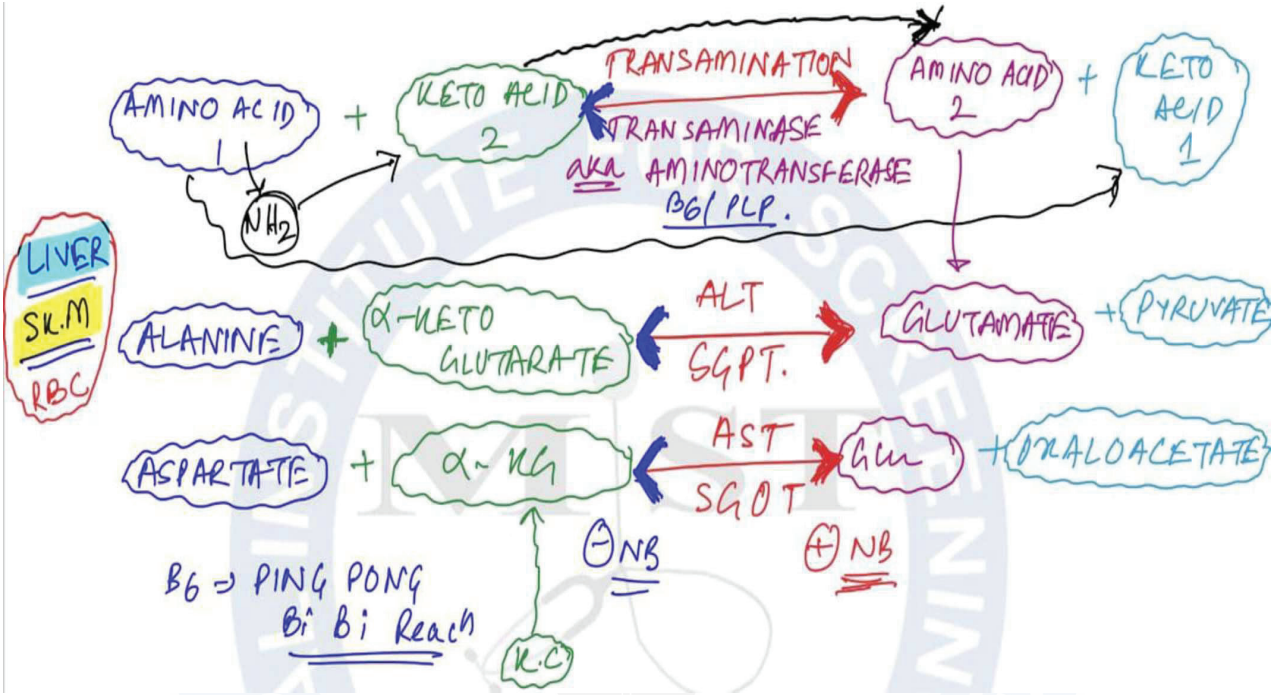
1. Any amino acids generated by protein catabolism that are not needed as building blocks for new bimolecular synthetic reactions are degraded to carbon skeletons in the liver which can be utilized in Gluconeogenesis.

2. The first step in amino acid degradation is the removal of nitrogen.

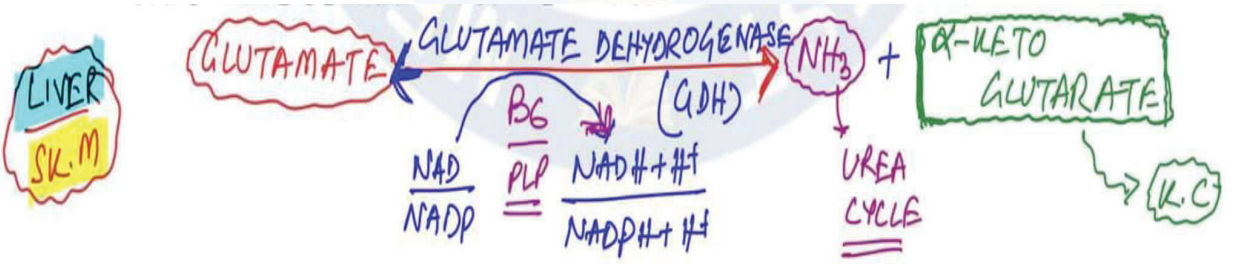


✹ TRANSAMINATION REACTION (Braunstein & Kritzmann Reaction)

→ Transfer of Amino group (NH2) from one amino acid to a keto acid with the generation of a new amino acid and a new keto acid



✹ OXIDATIVE DEAMINATION REACTION:

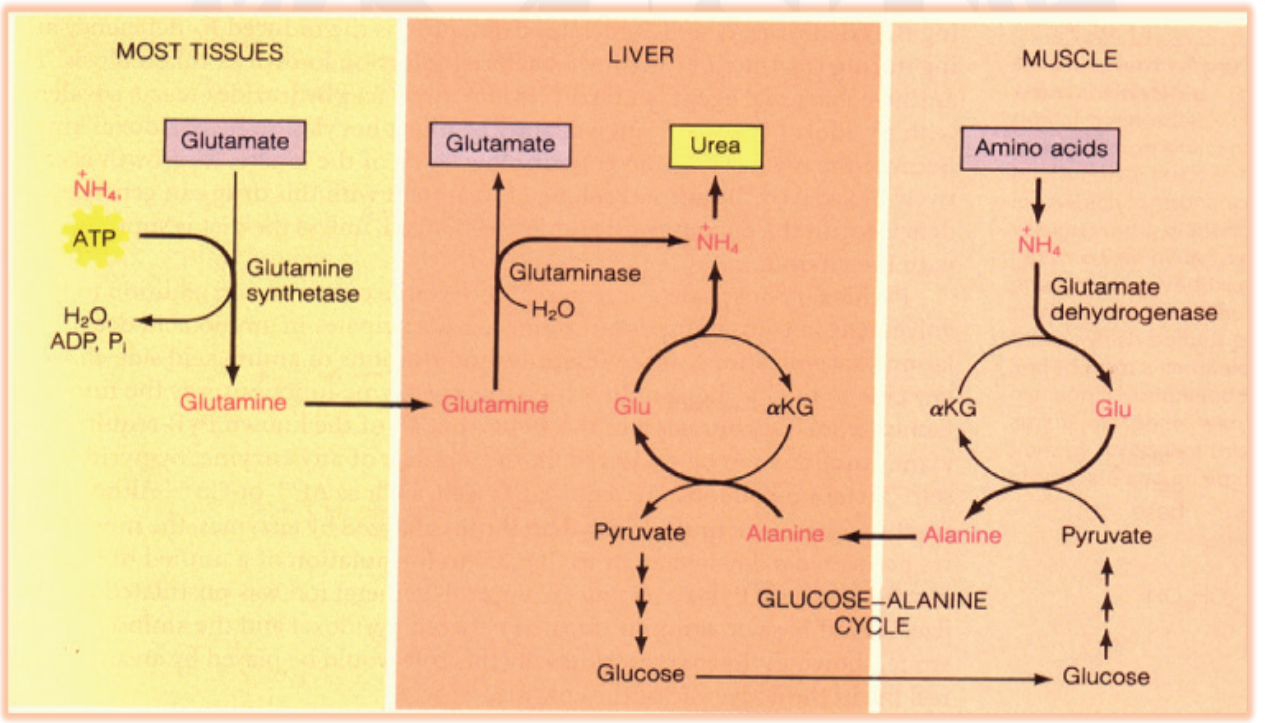


GDH gene Mutation : HIHA SYNDROME

✹ Fate of Ammonia & Ammonia toxicity → HYPERAMMONEMIA

→ Clinical Features: TREMOR, SLURRED SPEECH. BLURRED VISION, COMA

✹ Ammonia Transport



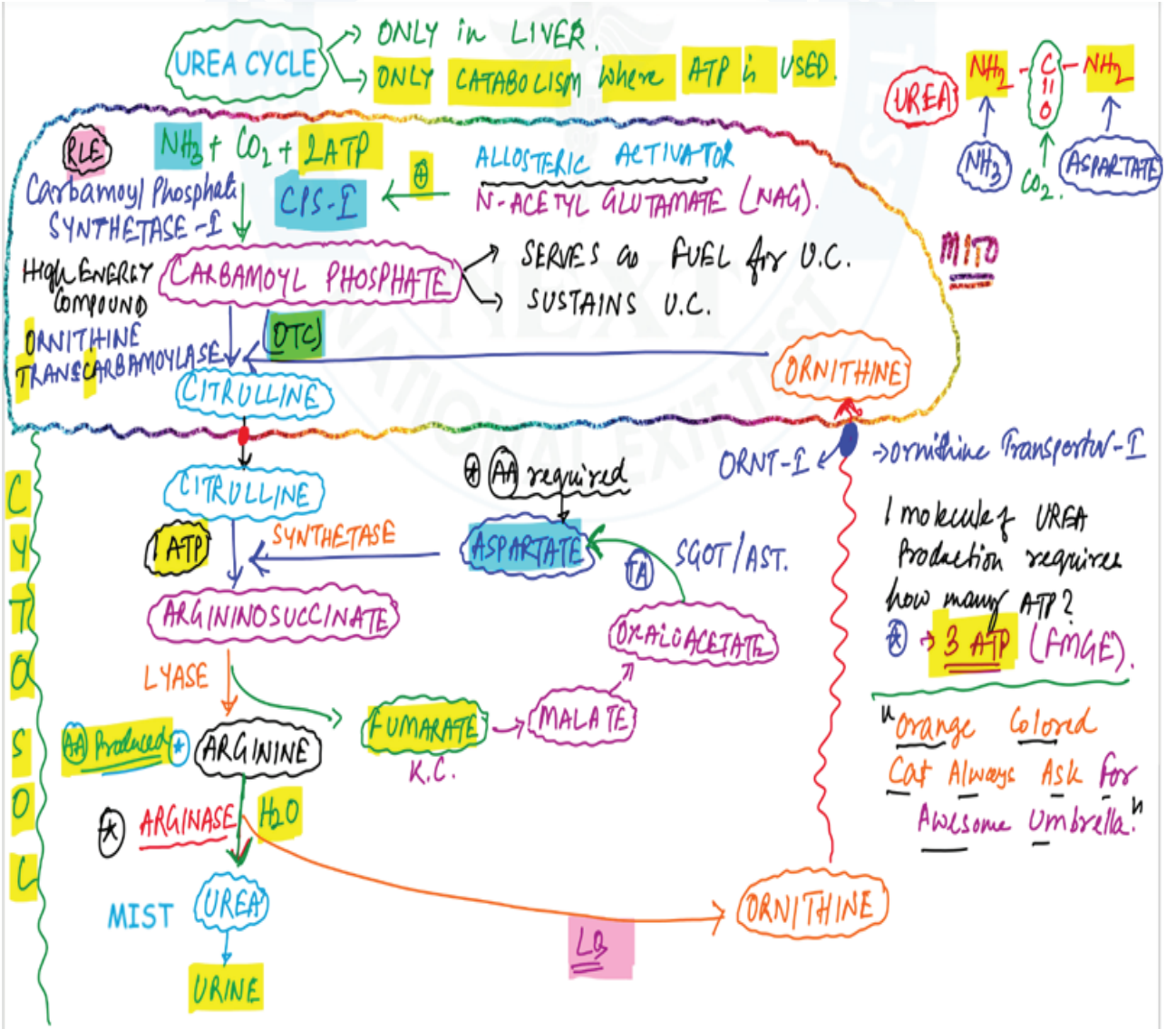
Transport form from BRAIN →

Transport form from MUSCLE →

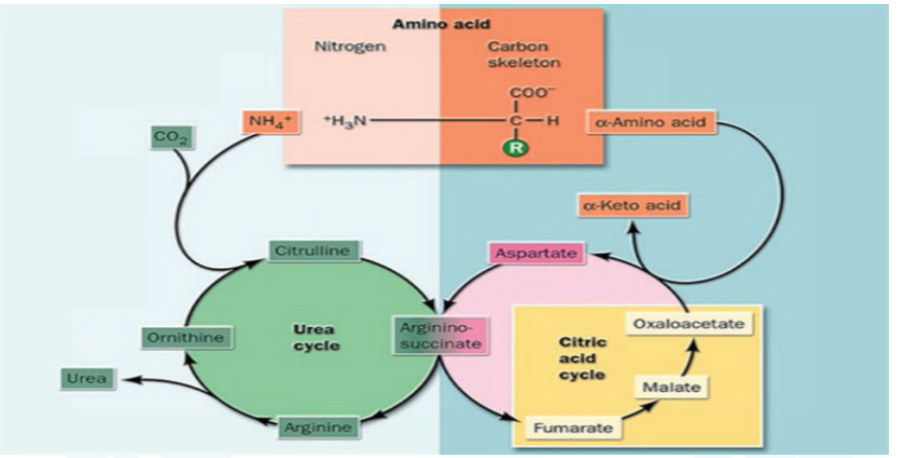
Extra Edge:

→ Increased level of Serum Alanine after fasting suggest

UREA CYCLE:



Extra Edge → "urea bicycle".



→

→

✹ Urea cycle disorders (TOTAL 6 types)

TYPE I HYPERAMMONEMIA : ⊖ CPS-I. ∴ NH_3 ↑↑↑↑↑↑↑↑ in BLOOD.
MOST FATAL :
· Compensatory Mechⁿ → FAIL
· AR, MR.

TYPE II HYPERAMMONEMIA : ⊖ OTC → NH_3 ↑↑↑ in Blood.
LB MC in INDIA :
Compensatory Mechⁿ → ACTIVE
↑ Carb Phos → Excess can enter into Pyrimidine Synthesis.
↑ GLUTAMINE in Blood
↑ URACIL in Blood + OROTIC ACIDURIA.

TYPE III HYPERAMMONEMIA (MILD)
ORNT-I DEFECT.
LB
· HYPERORNITHINEMIA
· HYPER-AMMONEMIA
· HOMOCITRULLINURIA
HHH SYNDROME

PRACTICE MCQS:

Ubiquitin is involved in

a. Intracellular proteolysis

b. ETC

c. Transport of ATP

d. Protein folding

Trans-amination of pyruvate & glutamic acid leads to the formation of

a. Oxaloacetate

b. Alpha Ketoglutarate

c. Aspartate

d. Malate

Which of the following vitamin acts as coenzyme in transamination

a. Cyanocobalamin

b. Pyridoxine

c. Folate

d. All of the above

Which product of citric acid cycle is used in detoxification of ammonia in brain

a. Oxaloacetate

b. Alpha ketoglutarate

c. Succinate

d. Citrate

The amino acid that serves as a carrier of ammonia from skeletal muscle to liver is

a. Alanine

b. Methionine

c. Arginine

d. Glutamine

Hydrolysis occurs in urea cycle during

a. Formation of fumarate

b. Formation of citrulline

c. Formation of arginine

d. Formation of ornithine

Urea cycle is linked to TCA cycle by

a. Arginine

b. Ornithine

c. Oxaloacetate

d. Fumarate

Carbamoyl phosphate synthetase-I is

a. Cytosolic enzyme

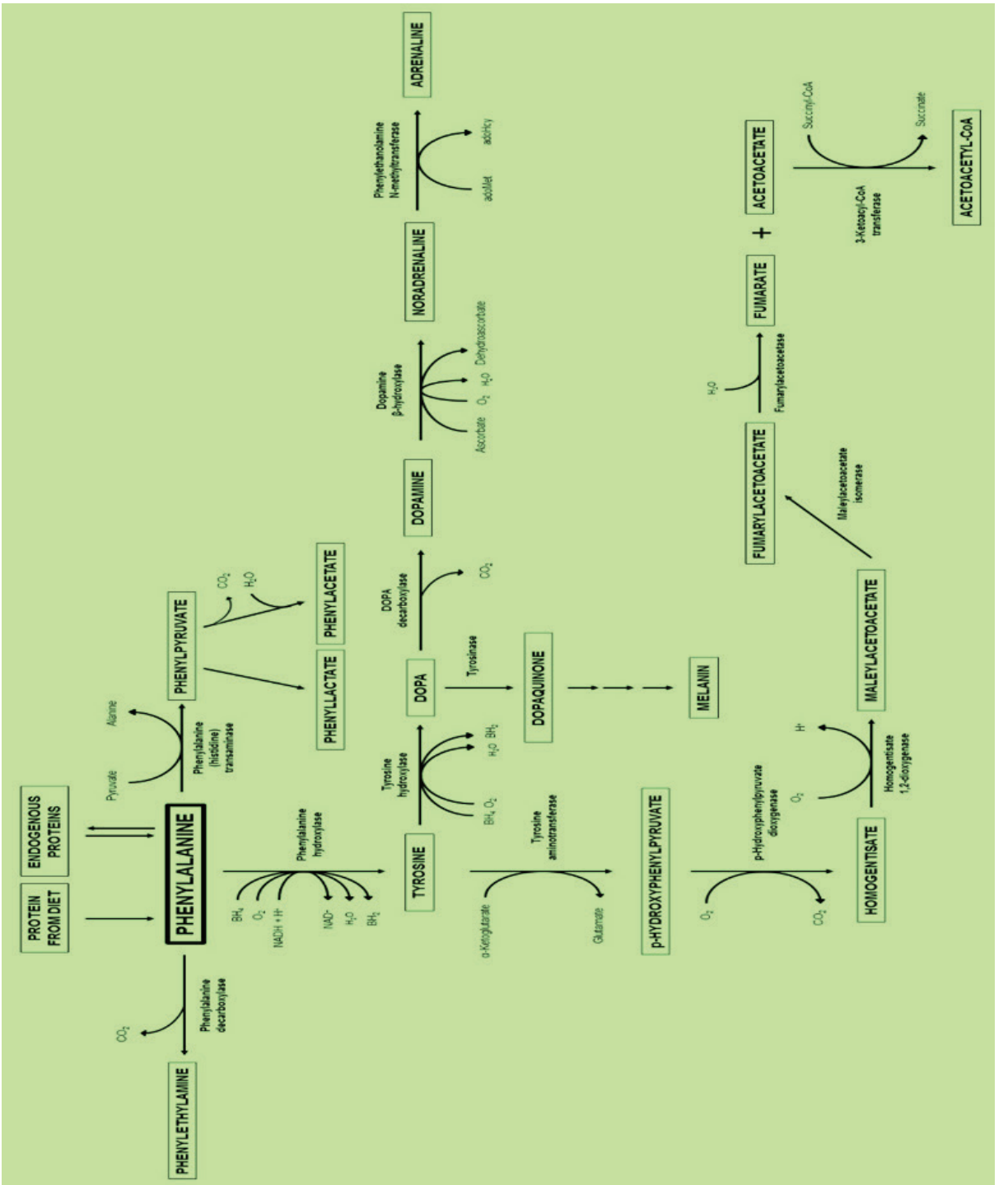
b. Lysosomal enzyme

c. Mitochondrial enzyme

d. All of the above

INDIVIDUAL AMINO ACID METABOLISM & ASSOCIATED DISORDERS

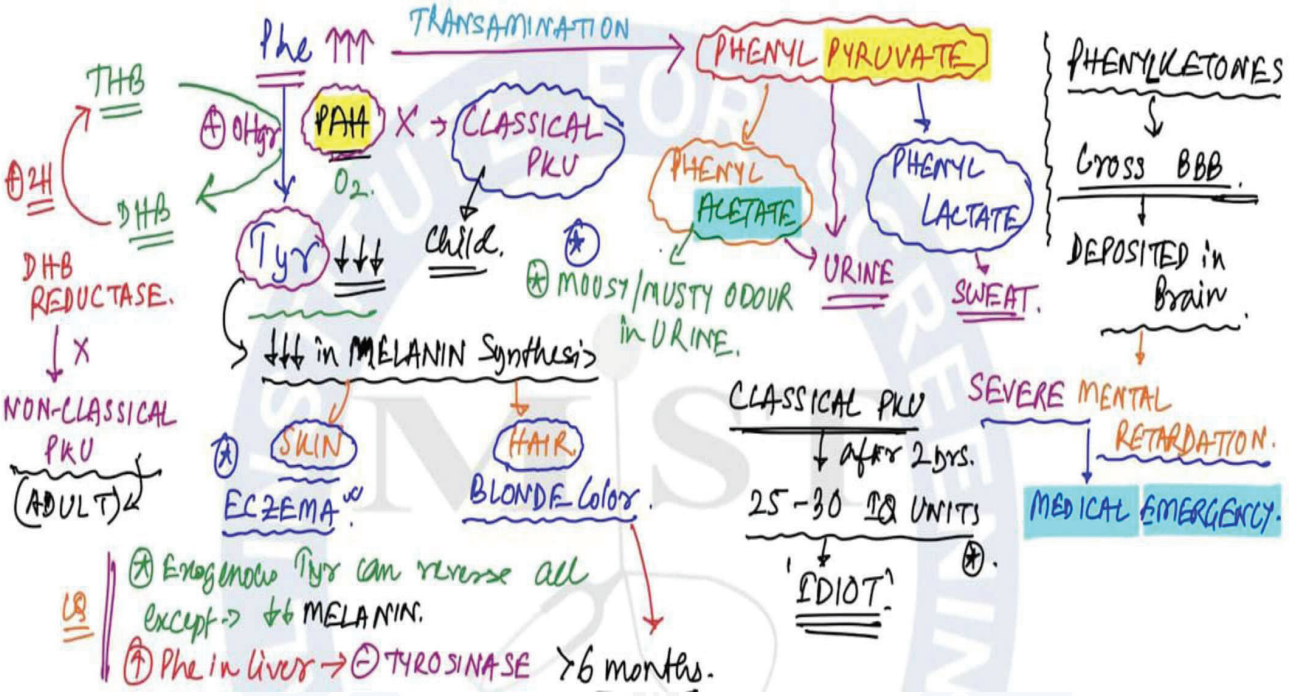
✹ PHENYLALANINE & TYROSINE METABOLISM & ASSOCIATED DISORDERS



✹ PHENYLKETONURIA(PKU)

- Most Common Metabolic disorder in Amino Acid metabolism and also considering all metabolism.

Pathogenesis & Important Clinical Features of PKU



Laboratory Diagnosis of PKU

Treatment of PKU

✹ TYROSINEMIAS:

Type 1	Type 2	Type 3
- AKA TYROSINOSIS / HEPATORENAL TYROSINEMIA - Fumarylacetoacetase OR Fumaryl acetoacetate hydrolase deficiency - CABBAGE like or MUSHROOM like or Rancid butter body odour	- AKA Richner Hanhart Syndrome / OCULOCUTANEOUS TYROSINEMIA - deficiency of tyrosine transaminase in liver - C/F Resemble a lot with PKU	- AKA Neonatal tyrosinemia or BENIGN TRANSIENT HYPERTYROSINEMIA of NEWBORN - PHPP dioxygenase deficiency - Vitamin C supplementation can be tried - Swimming pool odour in urine is seen in HAWKINSINURIA (variant of type 3)

✹ ALKAPTONURIA AKA BLACK URINE DISEASE

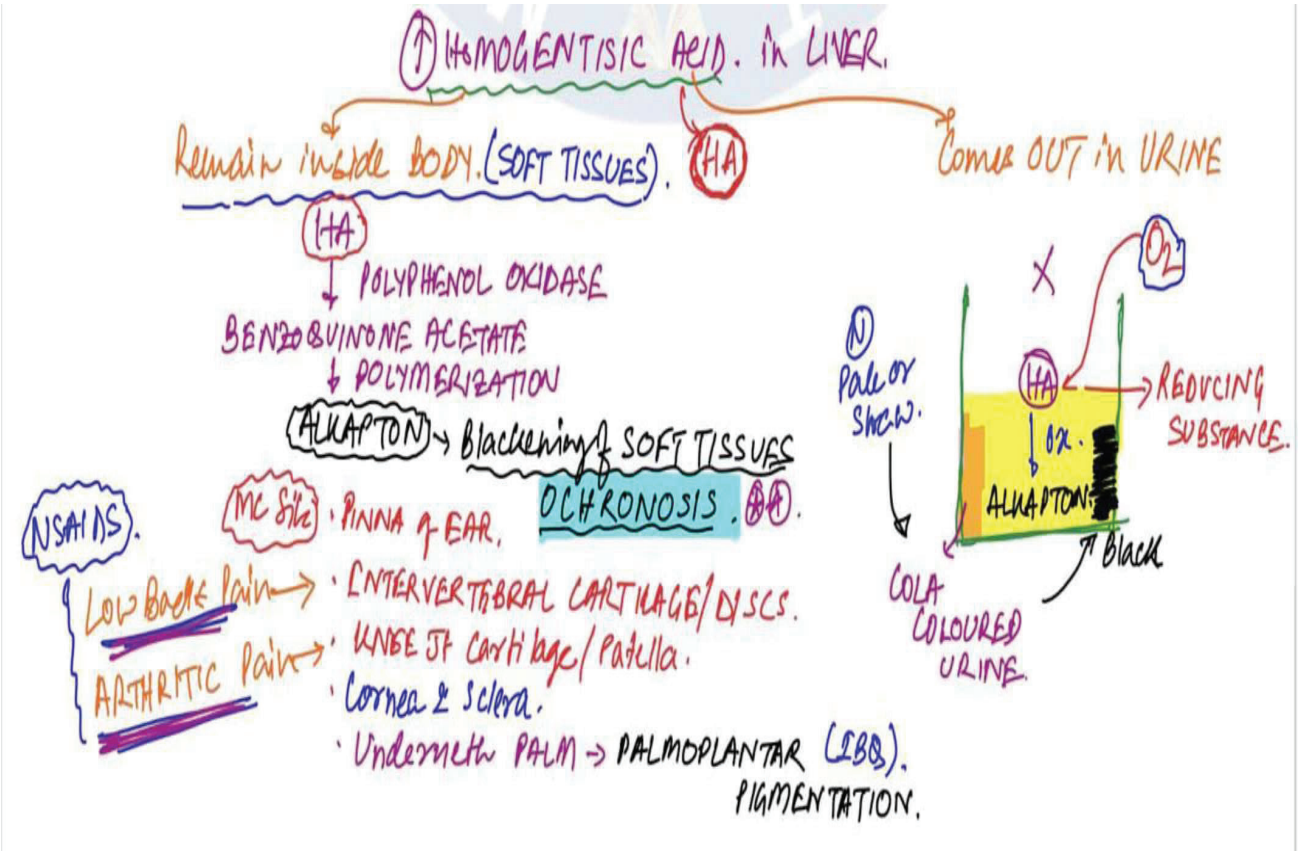
→ 1st IEM identified

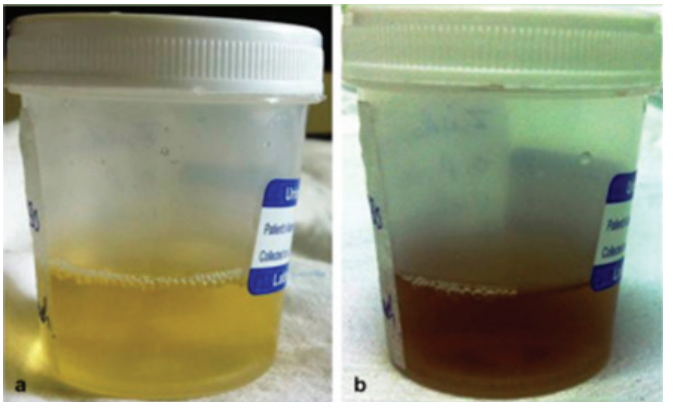
→ Part of GARROD's TETRAD

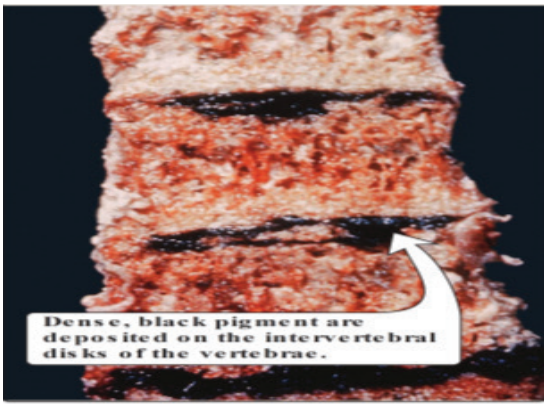
→ Deficiency of HOMOGENTISIC ACID OXIDASE or HOMOGENTISATE OXIDASE

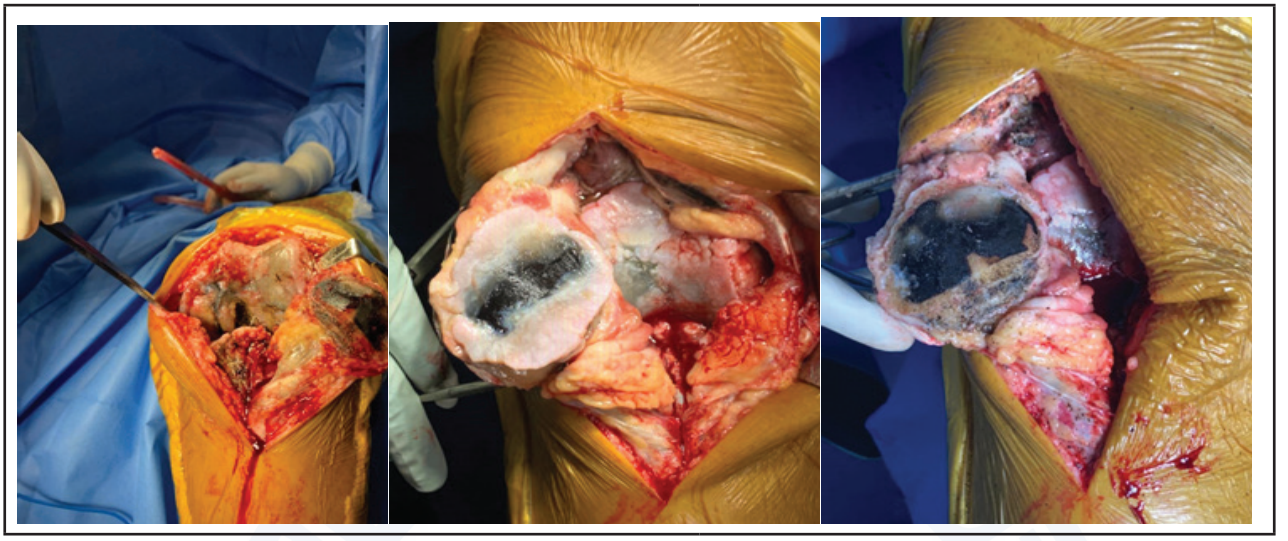
→ MC Age of presentation is 3rd or 4th decade of life

Pathogenesis & Important Clinical Features of Alkaptonuria









Laboratory Diagnosis of Alkaptonuria:

a)

b)

c)

Treatment of Alkaptonuria:

ALBINISM

→

→

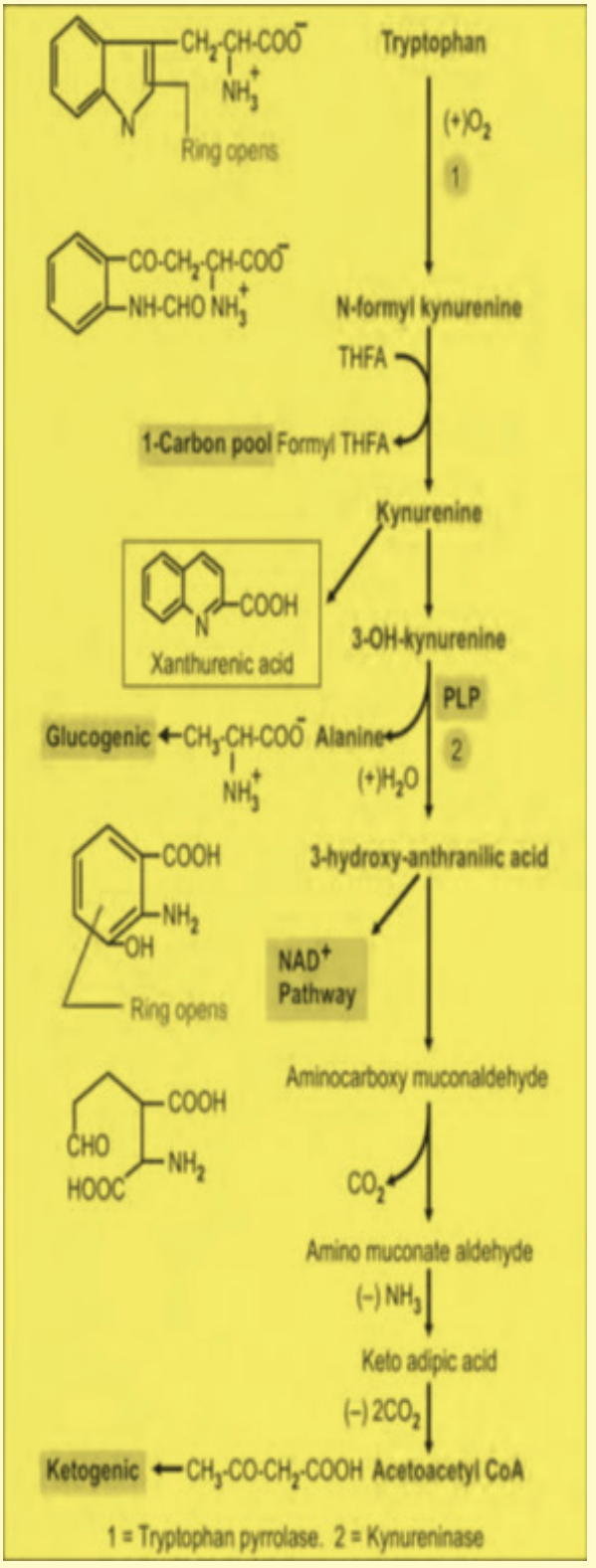
✹ TRYPTOPHAN METABOLISM - PATHWAYS & GOLDEN POINTS

→ Formation of Kynurenine – anthranilate

→ Formation of Serotonin & Melatonin

→ Formation of Nicotinic acid

→ Formation of Tryptamine

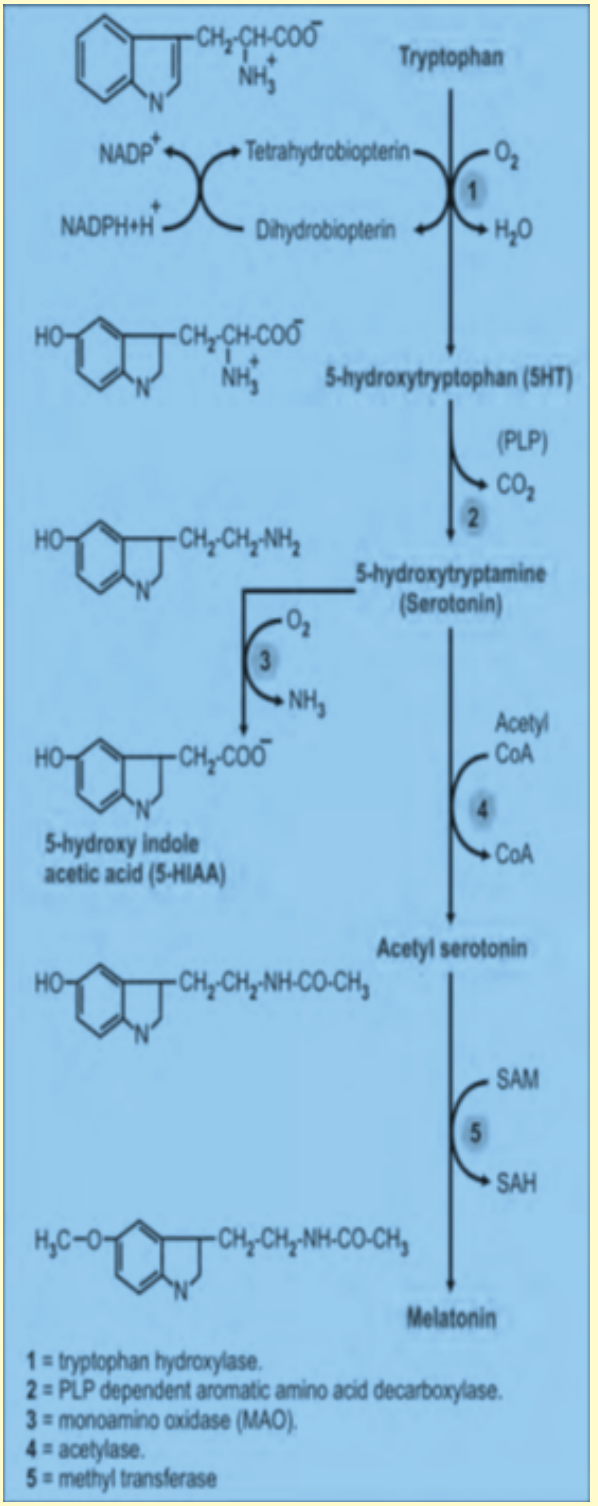


- B6 deficiency can also cause →

- As Kynureninase is B6 or PLP dependent

- B6 deficiency can be assessed by measuring →

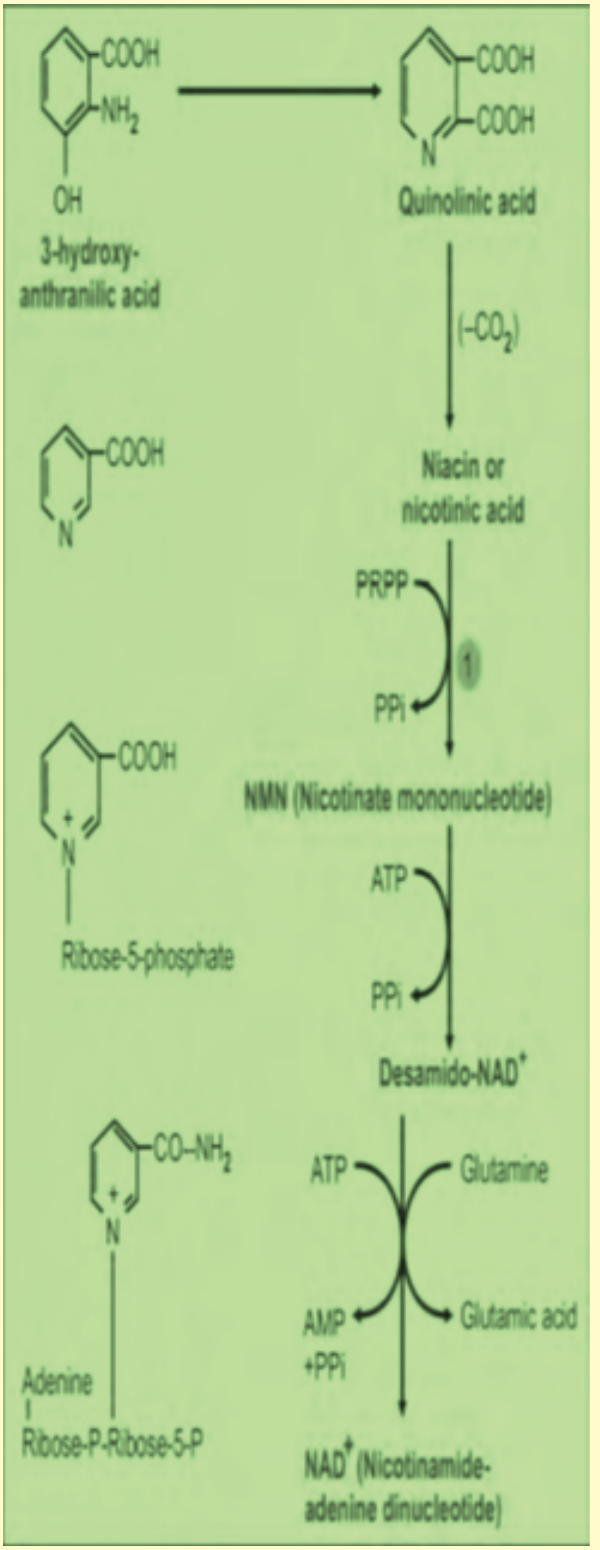
- B6 deficiency can also lead to



- ONLY 1 % of dietary Tryptophan enters for SEROTONIN synthesis
- Serotonin production is dependent on Biopterin and B6 (PLP)
- Melatonin production in pineal gland is dependent on Acetyl CoA & SAM
- Argentaffinoma / Malignant carcinoid syndrome:
- Here 60% of Tryptophan is channelized to produce serotonin
- MC originating from GIT

  I R A

  (Ileum > Rectum > Appendix)
- Presents with intermittent diarrhoea, Hot flush, sweating and pellagra like symptoms
- Diagnosis is done by 24-hour urinary estimation of 5-HIAA (Hydroxy Indole Acetic Acid)



- Metabolically only 3% of dietary Tryptophan goes for B3 production

- 60 mg of dietary Tryptophan forms 1 mg of niacin

- RLE of Niacin synthesis is QUINOLINATE Phospho Ribosyl Transferase (QPRT)

- 4D's of pellagra (dermatitis, diarrhoea, dementia, and death.

- Dermatitis is the first sign and it classically starts around neck – CASAL's Necklace

- MCC of PELLAGRA

◊ Hartnup's Disease:

PELLAGRA
(A)(B)(C).

TRYPTOPHAN TRANSPORTER DEFECT

X GIT → MALABSORPTION of Trp.

Trp → Intestinal Bacteria → PUTREFACTION of Trp.

→ INDOLE / SKATOLE → FOUL SMELLING STOOL

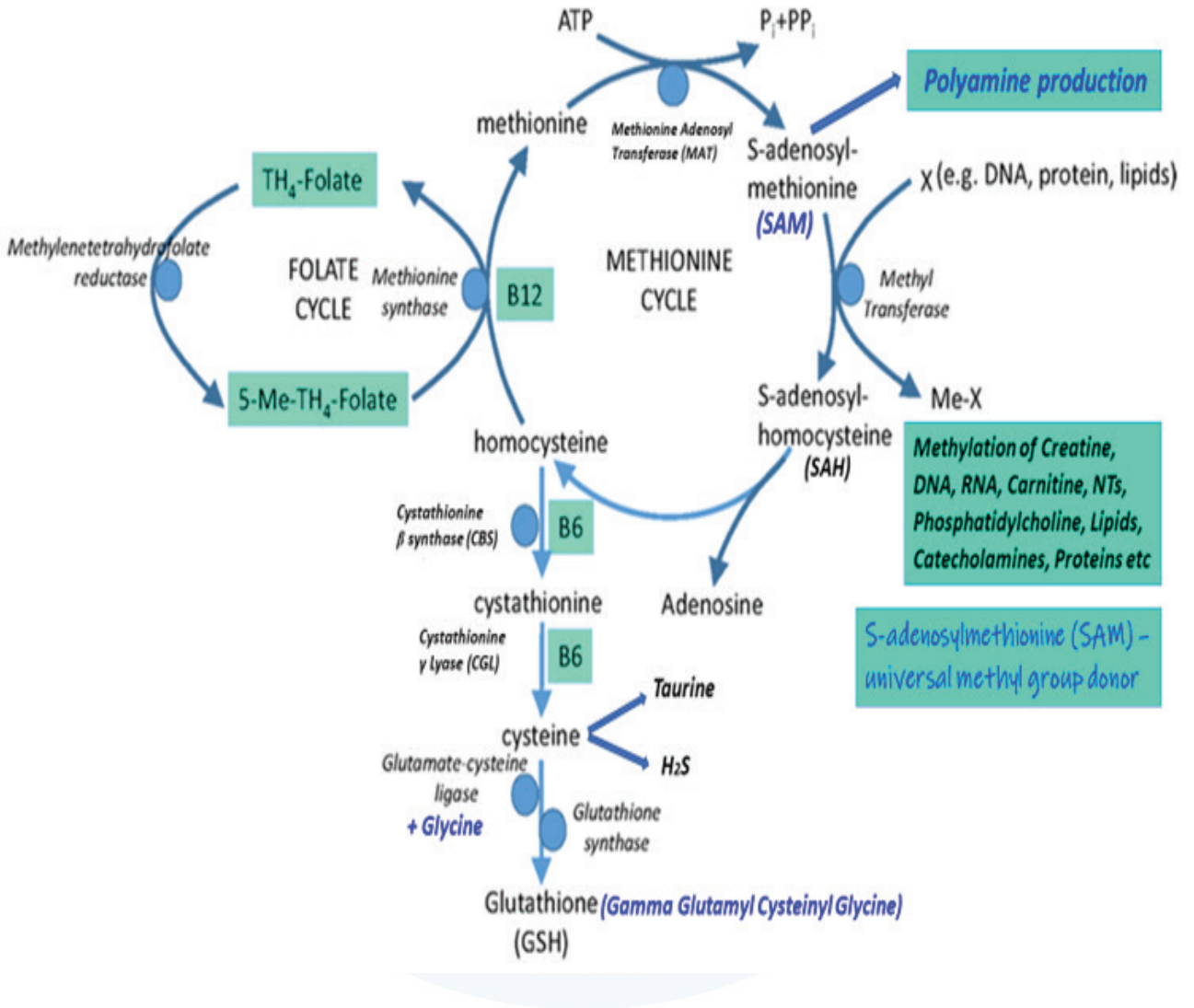
X Kidney → -> NO Reabsorption of Trp. in kidney Tubules.

Trp → INDICAN → URINE

Dx → INDICANURIA (light blue color)

- Diagnosis of Hartnup's disease: OBERMEYER Test

✹ SULPHUR CONTAINING AMINO ACID METABOLISM – PATHWAY & GOLDEN POINTS

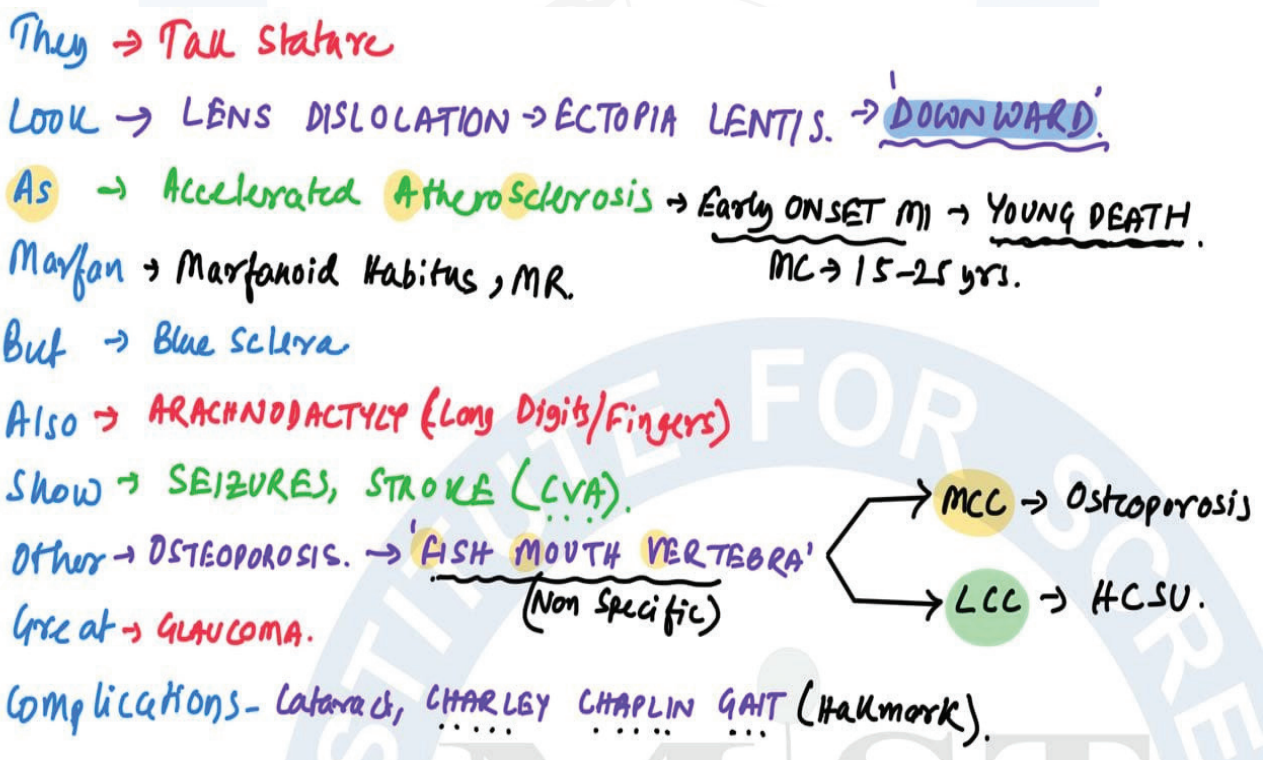


Extra Edge:

FOLATE TRAP →

◊ Homocystinuria:

Pathogenesis & Important Clinical Features →



Diagnosis

Treatment

◊ CYSTINURIA

◊ Histidine Metabolism & Associated Disorders (IEM)

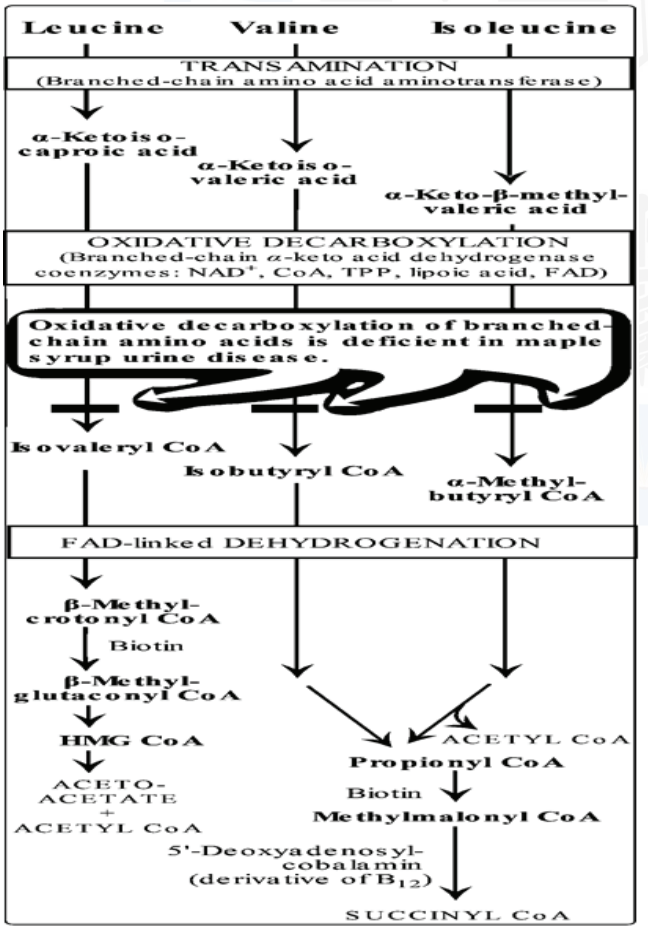
a)Urocanic acid pathway

FIGLU EXCRETION TEST is done to detect →

AKA

b) Histamine Production:

◊ Branched chain amino acid Metabolism & Associated Disorders (IEM):



BCAA metabolism:

◊ Maple Syrup Urine Disease (MSUD):

◊ ISOVALERIC ACIDEMIA -

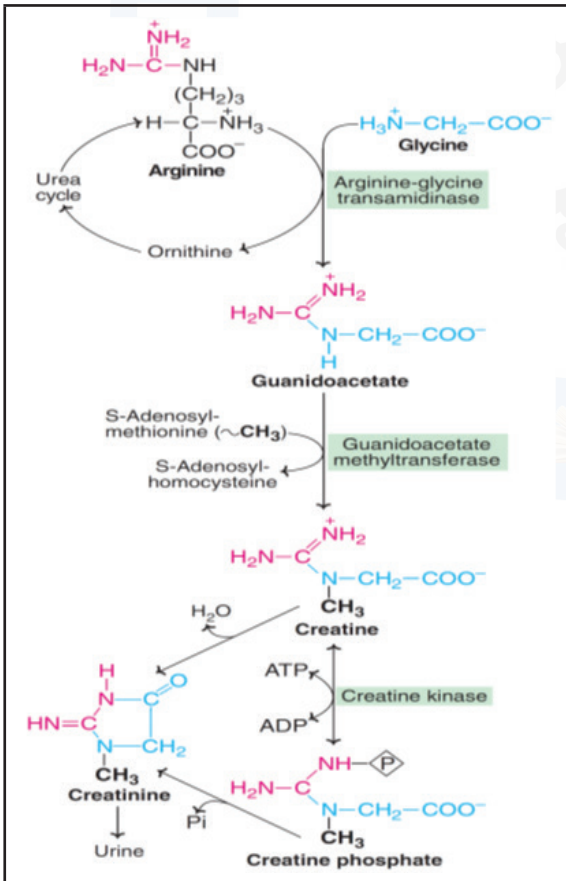
- d/t deficiency of FAD linked Isovaleryl CoA dehydrogenase that leads to Sweaty feet odour in body fluids.

◊ Glycine Metabolism & Associated disorders (IEM) with specialized products obtained from Glycine:

GOLDEN MCQ POINTS on Glycine

A → Alpha Carbon of Glycine has 2H atoms (R→H) K.A.C.A.
Heart → HEME SYNTHESIS.
Full → FLEXIBILITY OF FOLDING is FAVOURED.
Of → OPTICALLY INACTIVE
Careless → COLLAGEN, CREATINE SYNTHESIS.
Sentiments → SMALLEST + SIMPLEST.
Usually → PURINE, GLUTATHIONE, HIPPURIC Acid Synthesis.
FAils → Folic Acid + B6 → Required for Glycine Synthesis
BAdly → Bile Acid Conjugation.

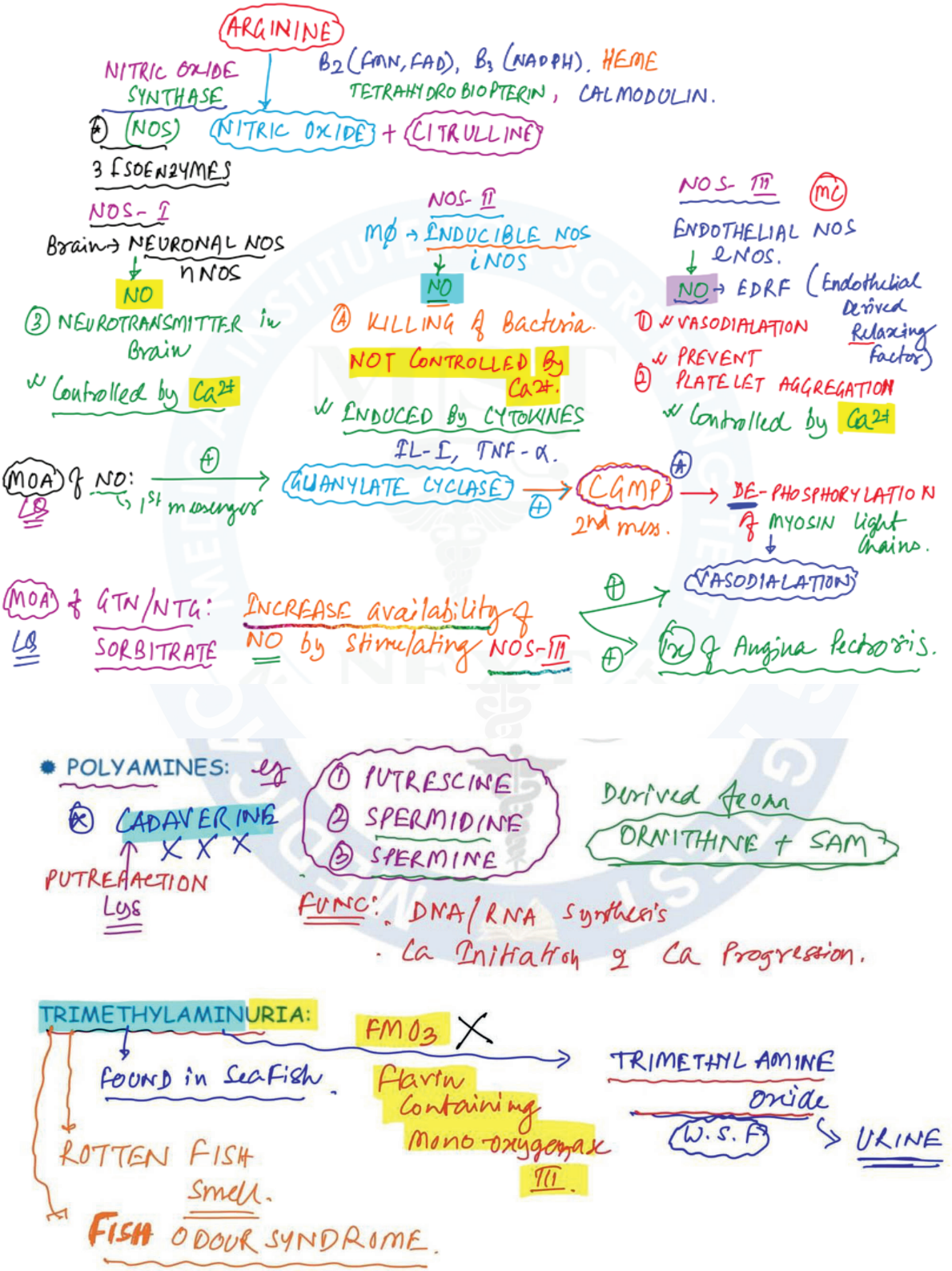
✹ BIOSYNTHESIS OF CREATINE AND PHOSPHOCREATINE



- Creatine is a tripeptide derived from (GAM) Glycine, Arginine and Methionine

- Creatine phosphate is energy store of muscle and participates in Lohmann's Reaction during ATP lack in muscle

ARGININE METABOLISM



QUICK RECAP On INBORN ERRORS OF AMINO ACID METABOLISM ASSOSIATED WITH ABNORMAL ODOUR

Inborn Error of Metabolism	Urine/Body fluid Odour
Gultaric Acidemia Type II	**Acrid, Sweaty feet**
Isovalericacidemia	**Cheesy odour in breath &Sweaty feet odour in body fluid**
Maple Syrup urine disease	**Burnt Sugar/Maple syrup**
Tyrosinemia/ Hypermethioninemia	**Boiled or Rotten cabbage/ Mushroom like/Rancid butter**
Phenylketonuria	**Mousy or musty**
Trimethylaminuria	**Rotten fish**
Oasthouseurine disease	**Hops-like**
Multiple Carboxylase deficiency	**Tomcat urine**
Hawkinsinuria	**Swimming Pool**
3-Hydroxy 3-methylglutaric aciduria	**Cat Urine**

PRACTICE MCQS:

Phenylalanine is the precursor of all the following except

a. Tyrosine

b. Epinephrine

c. Thyroxine

d. Melatonin

Norepinephrine to epinephrine conversion requires which enzyme:

a. Phenyl ethanolamine-N-methyltransferase

b. Transaldolase

c. Tyrosine hydroxylase

d. Alpha ketoglutarate

Rancid smelling butter urine is a feature of

a. Phenylketonuria

b. Tyrosinemia

c. Isovaleric acidemia

d. Multiple carboxylase deficiency

Blackening of urine on exposure to atmosphere is seen in

a. Phenylketonuria

b. Maple syrup urine disease

c. Hartnup disease

d. Alkaptonuria

Synthesis of melatonin is from

a. Tryptophan

b. Serotonin

c. Histidine

d. Phenylalanine

Amino acid secreted in cystinuria are all EXCEPT

a. Ornithine

b. Arginine

c. Histidine

d. Lysine

Disease due to deficiency of branched chain alpha keto-dehydrogenase complex enzyme:

a. Hartnup' s disease

b. Maple syrup urine disease

c. Alkaptonuria

d. Alport syndrome

Fact about selenocysteine except

a. Considered 21st amino acid

b. Coded by UGA

c. Occurs in glutathione peroxidase

d. Made of cysteine and methionine

The most direct precursor of taurine is:

a. Glycine

b. Cysteine

c. Methionine

d. Glutathione

MEDICAL INSTITUTE FOR SCREENING TEST MIST

LIPID CHEMISTRY & METABOLISM

Digestion & Absorption of LIPIDS:

Major dietary lipids are triacylglycerol, cholesterol and phospholipids.

Lingual lipase:

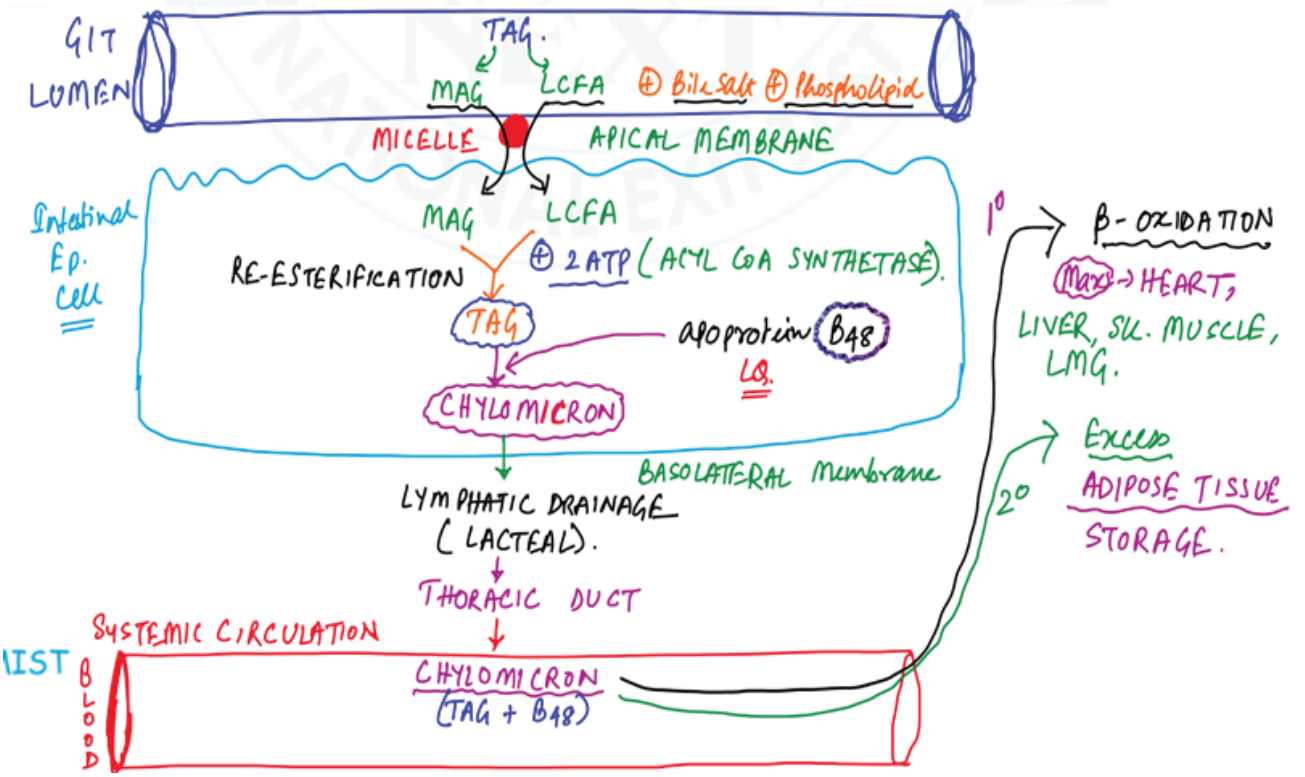
Optimum pH of 2.5 – 5, More significant in the newborn infants, breaks SCTs present in milk, butter and ghee.

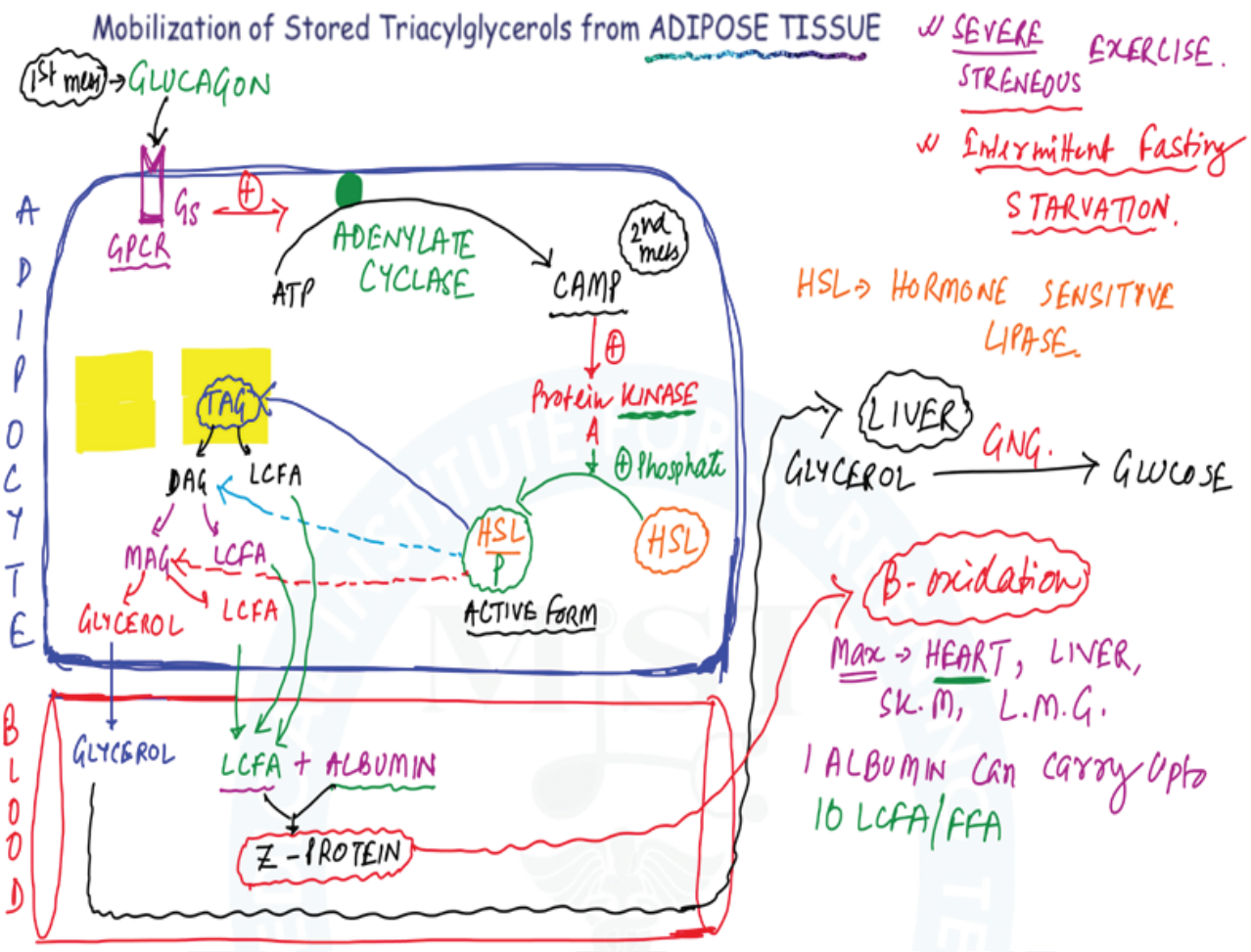
Gastric lipase: Optimum pH about 5.4, 30% digestion of TAG.

Emulsification is a pre-requisite for digestion of lipids in Intestine

1. Bile salts (detergent action)
2. Peristalsis (mechanical mixing)
3. Phospholipids.

✹ ABSORPTION OF LIPIDS:



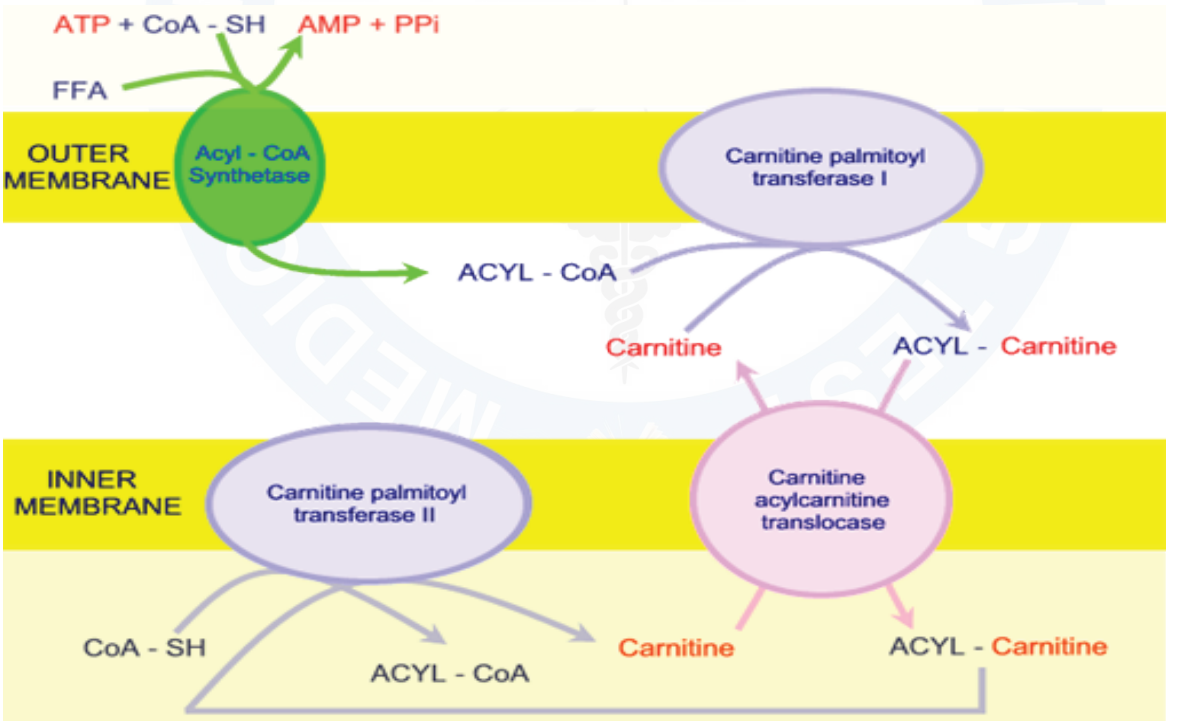

Mobilization of Stored Triacylglycerols from ADIPOSE TISSUE
1st mes → GLUCAGON
GPCR
Gs
⊕
ADENYLATE CYCLASE
ATP
CAMP
2nd mes
⊕
Protein KINASE A
⊕ Phosphate
HSL
HSL P
ACTIVE FORM
TAG
DAG
LCFA
MAG
LCFA
GLYCEROL
LCFA
ADIPOCYTE
BLOOD
GLYCEROL
LCFA + ALBUMIN
Z-PROTEIN
LIVER
GLYCEROL
GNG.
GLUCOSE
β-oxidation
Max → HEART, LIVER, SK. M, L.M.G.
1 ALBUMIN Can Carry Upto 10 LCFA/FFA
SEVERE STRENEOUS EXERCISE.
Intermittent fasting
STARVATION.
HSL → HORMONE SENSITIVE LIPASE.

✹ BETA OXIDATION OF FATTY ACIDS

3 stages:

Stage 1: Activation of Fatty Acids

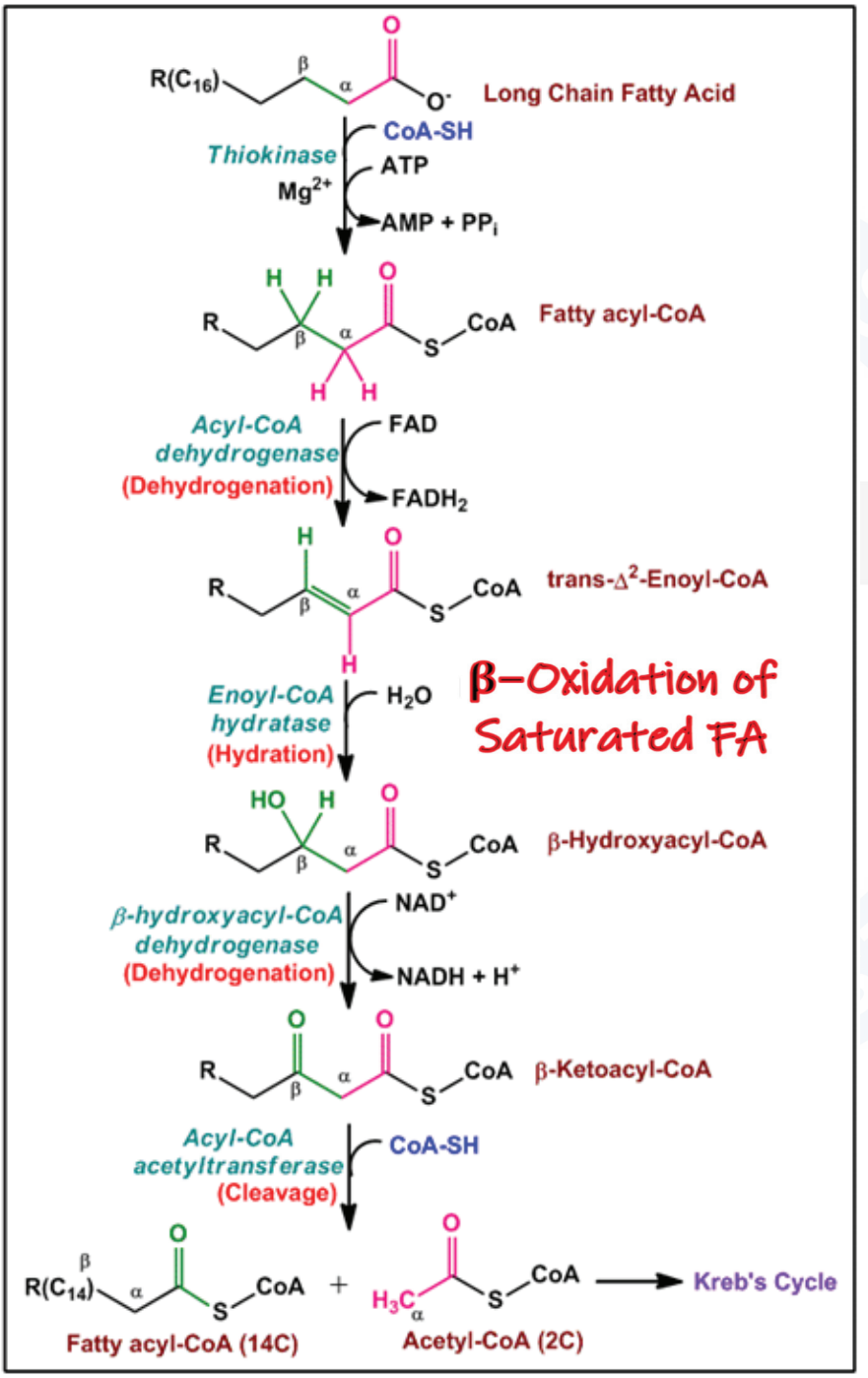
Stage 2: CARNITINE SHUTTLE



CARNITINE is obtained from Vit C dependent synthesis from

…… and …… in liver & kidney.

Stage 3: BETA OXIDATION PROPER



*Fat is burned under the fire of carbohydrates

OMEGA OXIDATION	ALPHA OXIDATION	PEROXISOMAL BETA OXIDATION
- Site: SER of liver and kidney. -Minor pathway. -Active when β Oxidation is defective. - Preferred substrates are MCFA (10 or 12 carbon FA) found mainly in Breast Milk. - double-ended fatty acids AKA dicarboxylic acids are formed.	- Site: SER & Peroxisomes Used for fatty acids that have a methyl group at the β -carbon *Does not generate energy as releases propionyl-CoA, not acetyl-CoA. - Seen in Phytanic acid which is a methylated fatty acid derived from GLV and ruminant animal fats, fish viz. tuna, cod, and haddock	- Appropriate for VLCFA Oxidation (C24-C36). - Molecular O2 is used - Functions to shorten the chain length to a point so that β-oxidation can complete in Mitochondria. - Produces H2O2

ORGANIC ACIDURIAs:

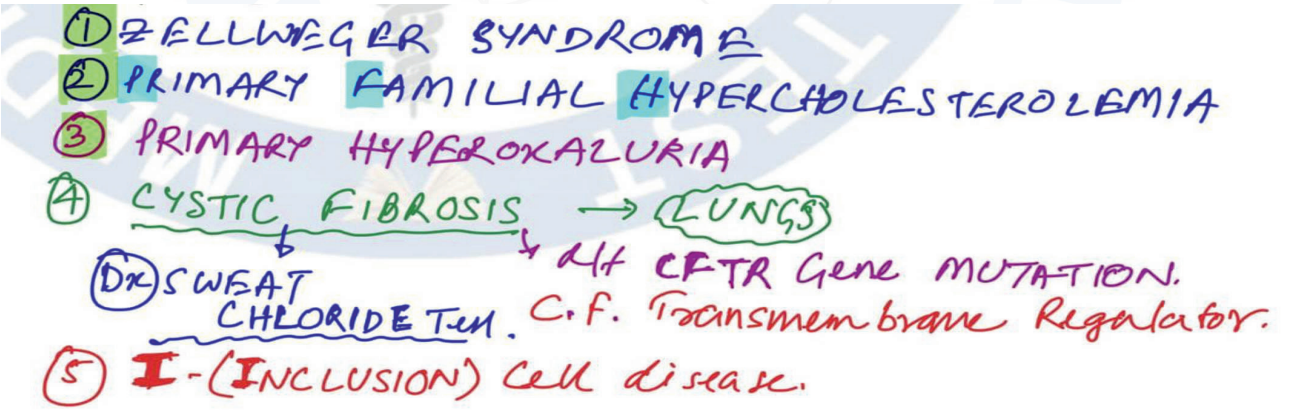
- Medium Chain Acyl CoA Dehydrogenase (MCAD) deficiency:
  - 1 in 2,500 live births., AR, ACADM gene mutation.
  - 2nd MC inborn error of metabolism
  - MC inborn error of FA metabolism. MC in INDIA →
  - Almost No ATP is produced → Sudden infant death syndrome (SIDS)-baby dies overnight due to not feeding

- Jamaican vomiting sickness:
  - Caused by a toxin (HYPOGLYCIN A & B) from the unripe fruit of the ackee tree from Africa.
  - Methylene Cyclo Propyl Acetic acid (MCPA) the metabolite of hypoglycin & a compound found in lychee seeds inhibits MCAD

Zellweger's (cerebrohepatorenal) syndrome:

- Defects in peroxisomal β-oxidation; AR; PEX 1, 2, 3 gene mutation
- Complete loss of peroxisomes in all tissues i.e. Peroxisome biogenesis disorder
- Progressive degeneration of brain. liver, kidney.
- Accumulation of C26-C38 polyenoic acids in severe neurological symptoms
- HYPOMYELINATION d/t PUFA accumulation. Death in 1st yr of life.

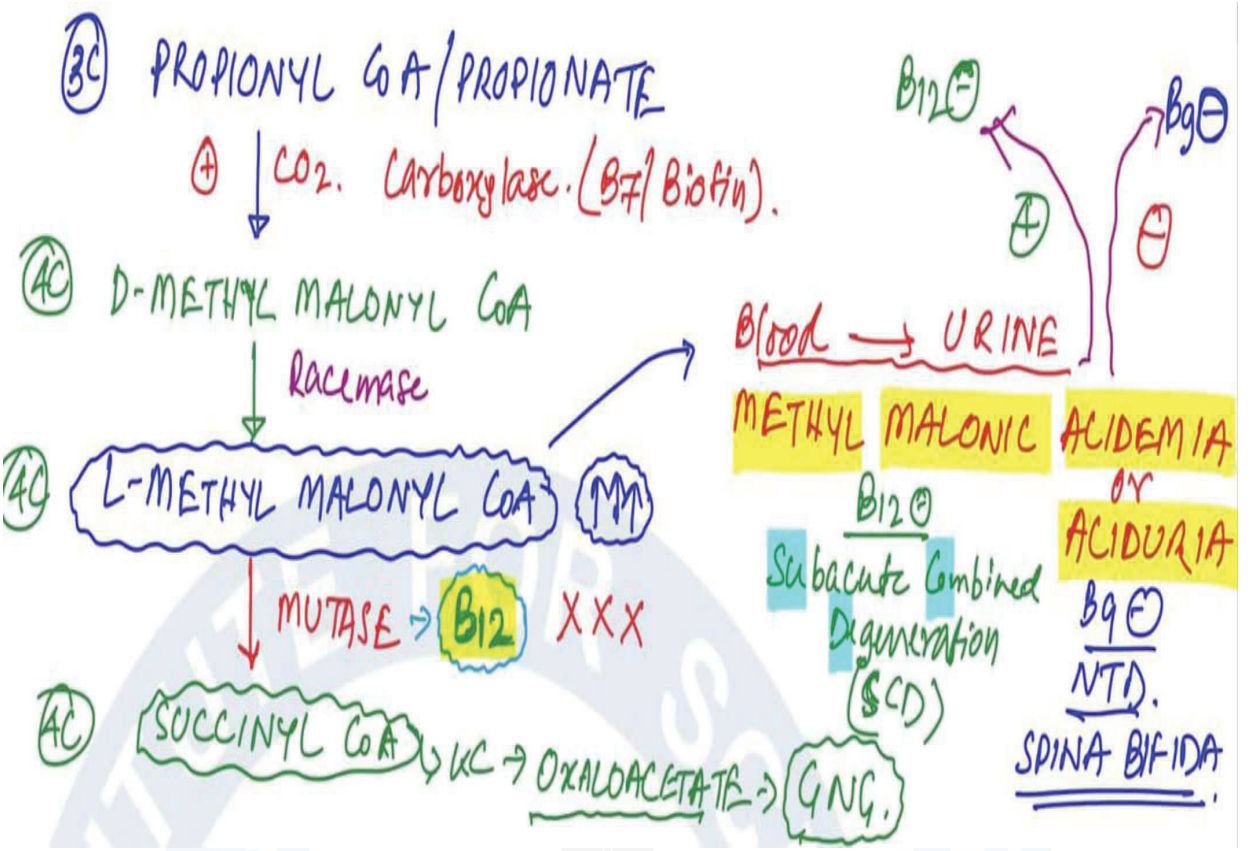
Extra Edge: Protein Targeting Defects:



Refsum disease: AKA heredopathia atactica polyneuritiformis

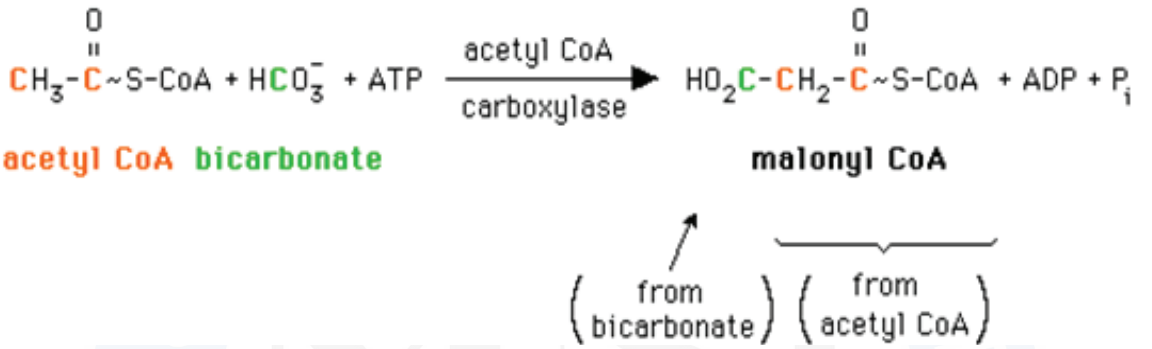
- Deficiency of Phytanic acid a-hydroxylase aka phytanic acid a-oxidase aka Phytanoyl CoA a-Hydroxylase or peroxin-7 activity.
- Phytanic acid accumulation in tissues - phytanic acid storage disease
-

✹ Inborn Errors of Propionate Metabolism:



✹ DE NOVO SYNTHESIS OF FATTY ACIDS

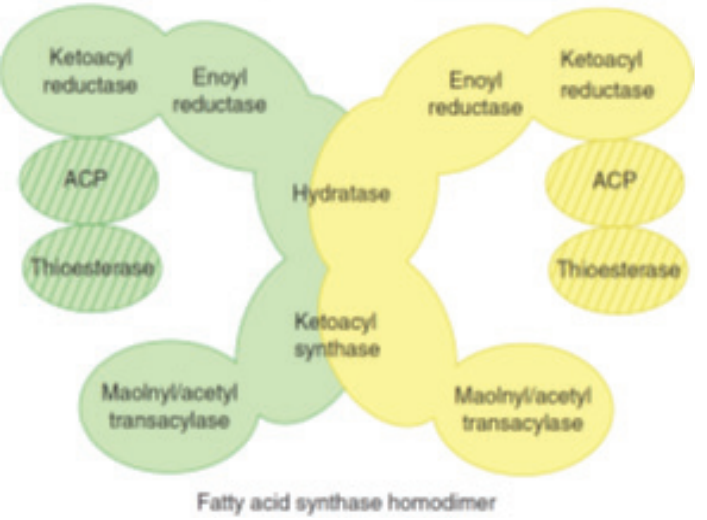
- Site: Cytosol of liver, adipose tissue, kidney, brain, and lactating mammary glands.
- Transport of Acetyl CoA to Cytoplasm by citrate malate shuttle system and subsequent ACTIVATION.



RLE:

Hydrogen donor:

FAS I COMPLEX:



Fatty acid synthase homodimer

Hormonal Regulation:

Coordinated Regulation between FAS & Beta Oxidation

Difference in Beta-oxidation & Fatty acid synthesis

	Beta-oxidation of FA	Fatty acid synthesis
Site	Mitochondrial Matrix	Cytoplasm
Enzymes	Present as independent units	Multi-enzyme complex
Sequential units	2C units split off as 2C acetyl CoA	2C units are added as 3C malonyl CoA
Co-enzymes	NAD+ and FAD are re-duced	NADPH used as reducing power
Carrier Mechanism	Carnitine Shuttle	Citrate – Malate Antiport
Regulation	Insulin ↓ & Glucagon ↑	Insulin ↑ & Glucagon ↓

ESSENTIAL FATTY ACIDS:

- Linoleic Acid (w6): w6 FA → Lower plasma cholesterol + LDL + HDL

  Dietary Source: Salicornia oil (75%) > Safflower oil (74.62%) > Sunflower oil (65.7%)

- a-Linolenic Acid (w3): w3 FA→ Do NOT lower LDL & HDL but they are CARDIOPROTECTIVE.

  Dietary Source: Flaxseed/Linseed oil > Soyabean oil > Canola oil > Mustard Oil > Rice Bran oil

- Arachidonic acid (w6): 20C - 4 double bonds

  Dietary Source: meat products including chicken, beef, pork, and fishes, egg, milk.

- Eicosapentaenoic acid (EPA) (w3): 20C – 5 DB

  Dietary Source: fish oil / fat

- Docosahexaenoic acid (DHA, w3): 22C – 6 DB

  AKA CERVONIC acid

  Dietary Source : Breast milk > Fish oil > algae oil

  → Best combination in diet: w6 FA & w3 FA (Ratio 4:1)

TRANS FATTY ACIDS	EICOSANOIDS
■ By-product of the saturation of unsaturated FA during partial hydrogenation, or hardening, of natural vegetable oils in the manufacture of margarine to improve shelf life & decrease volatility ■ i.e. stability at high temperature while deep frying. ■ But the undesirable effect of hardening is conversion of cis double bonds to trans double bonds TFA is structurally similar to saturated FA. ■ TFA compete with EFA and exacerbate EFA deficiency. Associated with increased risk CVD & DM. ■ Max permissible limit →	They are formed from C20 polyunsaturated fatty acids viz. ARACHIDONIC ACID PG2, TX2 series (prostanoids) - cyclooxygenase pathway LT4 and LX4 - lipoxygenase pathway

✹ Clinically Important Phospholipids and Glycolipids

■ Cardiolipin

- 2 Phosphatidic acid share 1 Glycerol i.e. Di Phosphatidyl Glycerol
- Phospholipid present in Inner membrane mitochondria → maintains ETC
- Triggers programmed cell death (apoptosis)
- Anti Cardiolipin Antibodies are increased in Syphilis, APLA syndrome, TB, Malaria, SLE.

■ Lecithin

- Phosphatidylcholine aka LECITHIN is the most abundant phospholipid of cell membrane & represents a large proportion of body's store of choline which is needed for Acetylcholine production.

- Di Palmitoyl-Phosphatidyl Choline (DPPC) decreases surface tension at the air-liquid interface and thus greatly reduces the work of breathing and major constituent of Surfactant i.e. mainly produced by Type II PNEUMOCYTES

- Lung maturity of fetus: DPPC to Sphingomyelin (L/S) ratio in amniotic fluid.

- ≥ 2 ratio indicates MATURITY & transition from S→L in Type II pneumocytes by 32 weeks of gestation.

- Deficiency of lung surfactant causes infant respiratory distress syndrome (IRDS).

■ Cephalin

- Involved in Blood Clotting

■ Phosphatidylserine

- Marker of apoptosis

■ Phosphatidylinositol

- is a precursor of second messenger that aid in signal transduction and membrane trafficking.

■ Plasmalogen

- Constitute 10% of PL in brain & heart muscle. Contains ETHER link instead of ester link

■ Platelet activating factor

- Contains ETHER link. mediates thrombotic events by platelet aggregation & degranulation.

- It also binds to Surface receptors & triggers potent acute inflammatory reactions, mediates hypersensitivity & anaphylactic reactions. Activates neutrophils & alveolar macrophages to generate superoxide radicals that kill bacteria.

■ Sphingomyelin

- Constituent of myelin & lipid rafts in plasma membrane. Plays role in cell signaling & apoptosis.

■ Gangliosides – 3 types GM1, GM2, GM3 – They are GLYCOLIPIDS

- GM1 Serve as cell surface receptor for cholera toxins & tetanus toxin in INTESTINE

Extra Edge:

■ Phospholipases - Degrades & Remodels Phospholipids

- PLA2 → found in venom of viper snakes that liberates Lysolecithin - hemolytic agent.

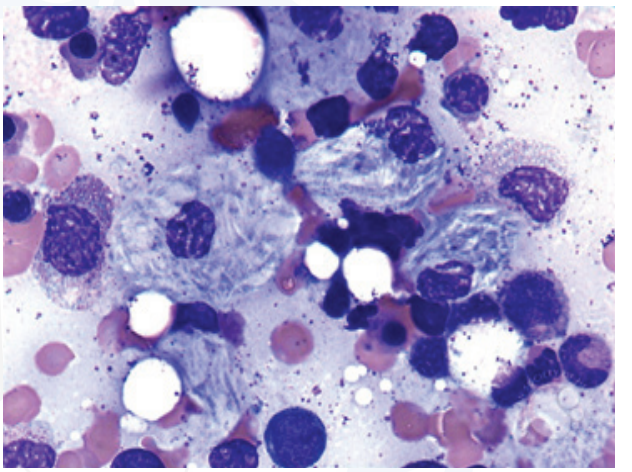
■ Blood group A can be differentiated from B due to presence of an additional N-Acetyl group in only in gr A

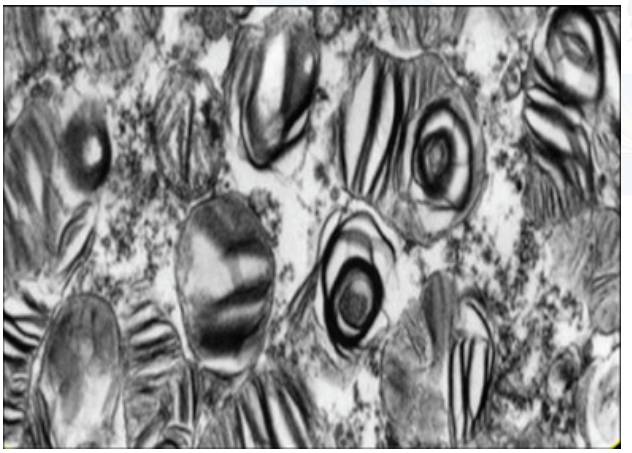
✹ LYSOSOMAL / LIPID STORAGE DISORDERS:

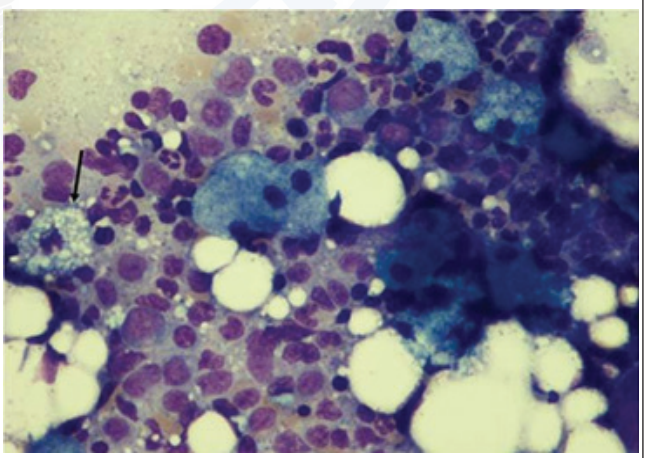
Disease	Enzyme defect	Disease	Enzyme defect
Gaucher's disease MC type of LSD	Beta glucosidase / Glucocerebro-sidase	Farber disease	Ceramidase
Niemann-Pick Disease (A+B)	Sphingomyelin-ase	Tay Sach's disease	Hexosaminidase A
Krabbe's disease	Beta galactosi-dase	Sandhoff's disease	Hexosaminidase A + B
Metachromatic leukodystrophy	Arylsulfatase A	Fabry's disease ONLY X -L -R	Alpha galactosi-dase A

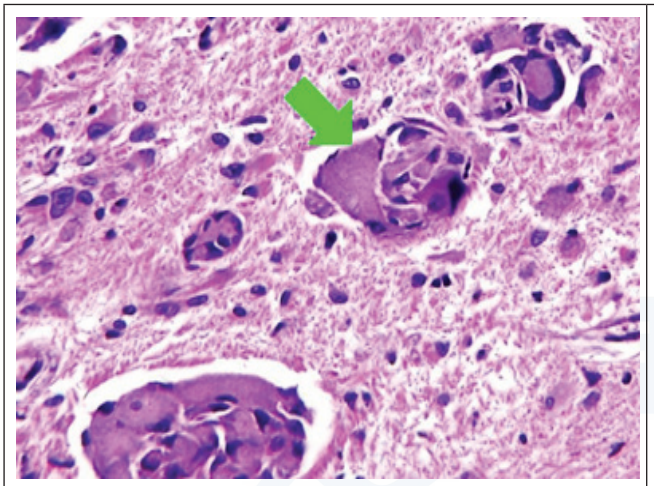
What is NPC?

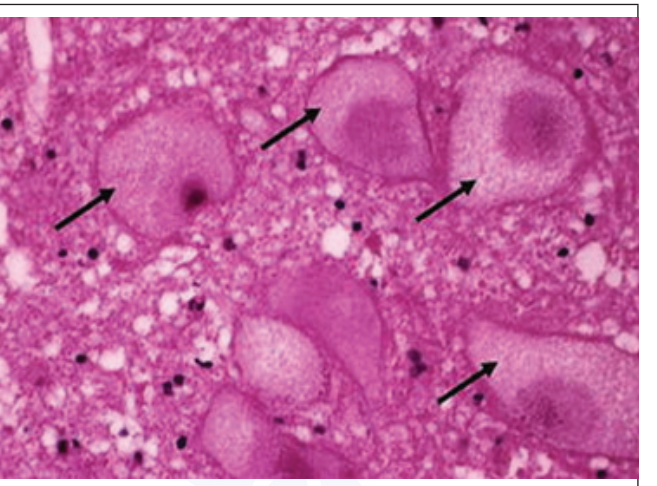
Cherry Tree Never Grow Tall in Sands











KETONE BODIES

KETOGENESIS	KETOLYSIS
• Ketone bodies are ABA (Acetoacetate, β-hydroxybutyrate and acetone). • Ketone bodies are synthesized from acetyl CoA in Liver • Acetoacetate is first Ketone body to be formed in liver mitochondria • β-hydroxybutyrate a pseudoketone as it lacks a keto (-C=O) group and its the storage form of ketone body in liver • Acetone is volatilized in lungs causing acetone breath – SWEET smell in breath in DKA & SKA • RLE of Ketogenesis: HMG CoA SYNTHASE	• Breakdown during starvation is done in BRAIN, skeletal muscles, heart and kidney d/t presence of Thiophorase in these tissues. • This THIOPHORASE is ABSENT in LIVER *Ketosis: TRIAD of ketonemia + ketonuria + smell of acetone in breath. Diagnosis of Ketonuria: ROTHERA's test.

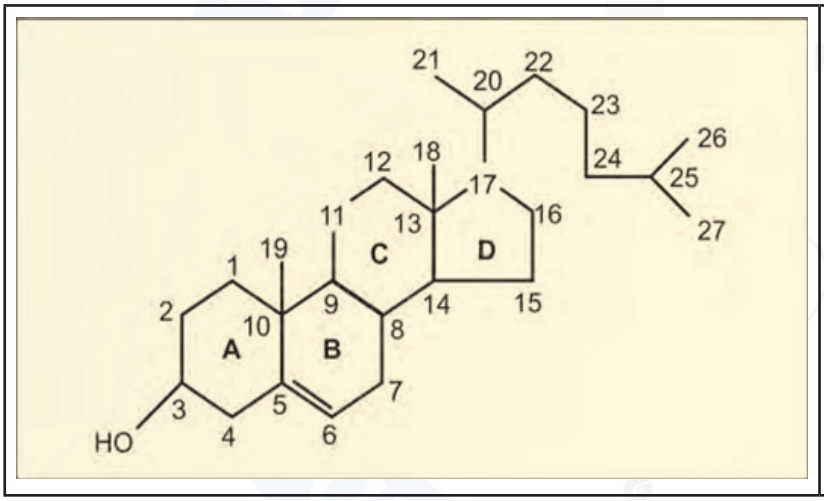
*Ketone Bodies Are Overproduced in Diabetes and during Starvation- But how will you differentiate DKA from SKA??

REMEMBER : In starvation there is hypoglycemia, lipolysis, ketoacidosis except hypercholesterolemia

✹ METABOLIC PATHWAYS DURING FASTING

- 0 - 18 hrs → GLYCOGENOLYSIS, GLYCOLYSIS. } minor β-ox.
- >18 hrs - 72 hrs → GLUCONEOGENESIS., GLYCOLYSIS.
- >3 days - 7 days → ↑ β-ox of FA, ↑ A.CoA → K.C. → ATP.
- >7 days - 2 wks → ↑↑↑ β-ox of FA → ↑↑↑ Acetyl CoA → KC } ↑ KB production
- >2 wks - 1 month → ↑ K.B. Breakdown., A.CoA → KC.
- >1 month → BODY PROTEIN LYSIS, DEATH.

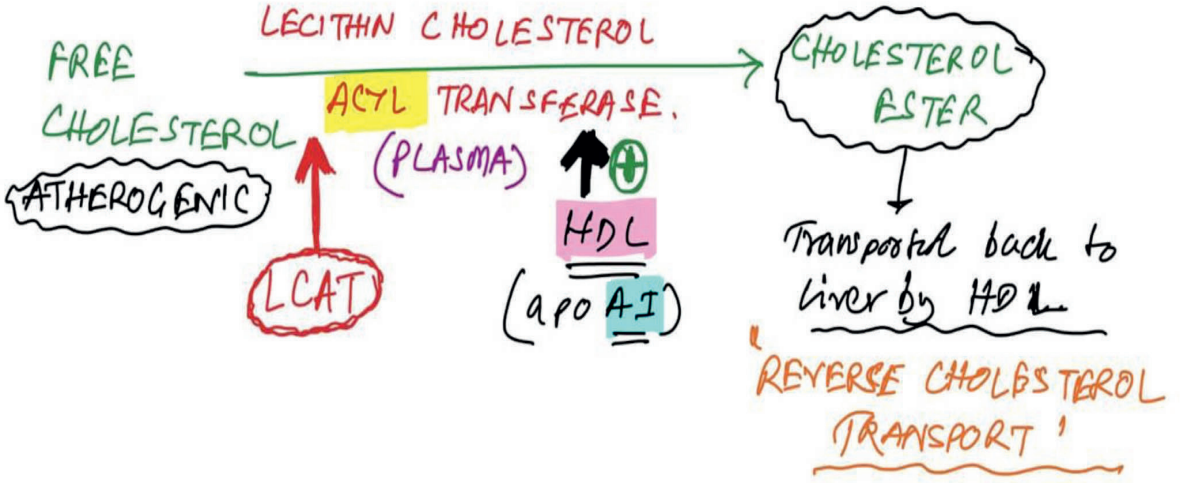
✹ Structure of Cholesterol:



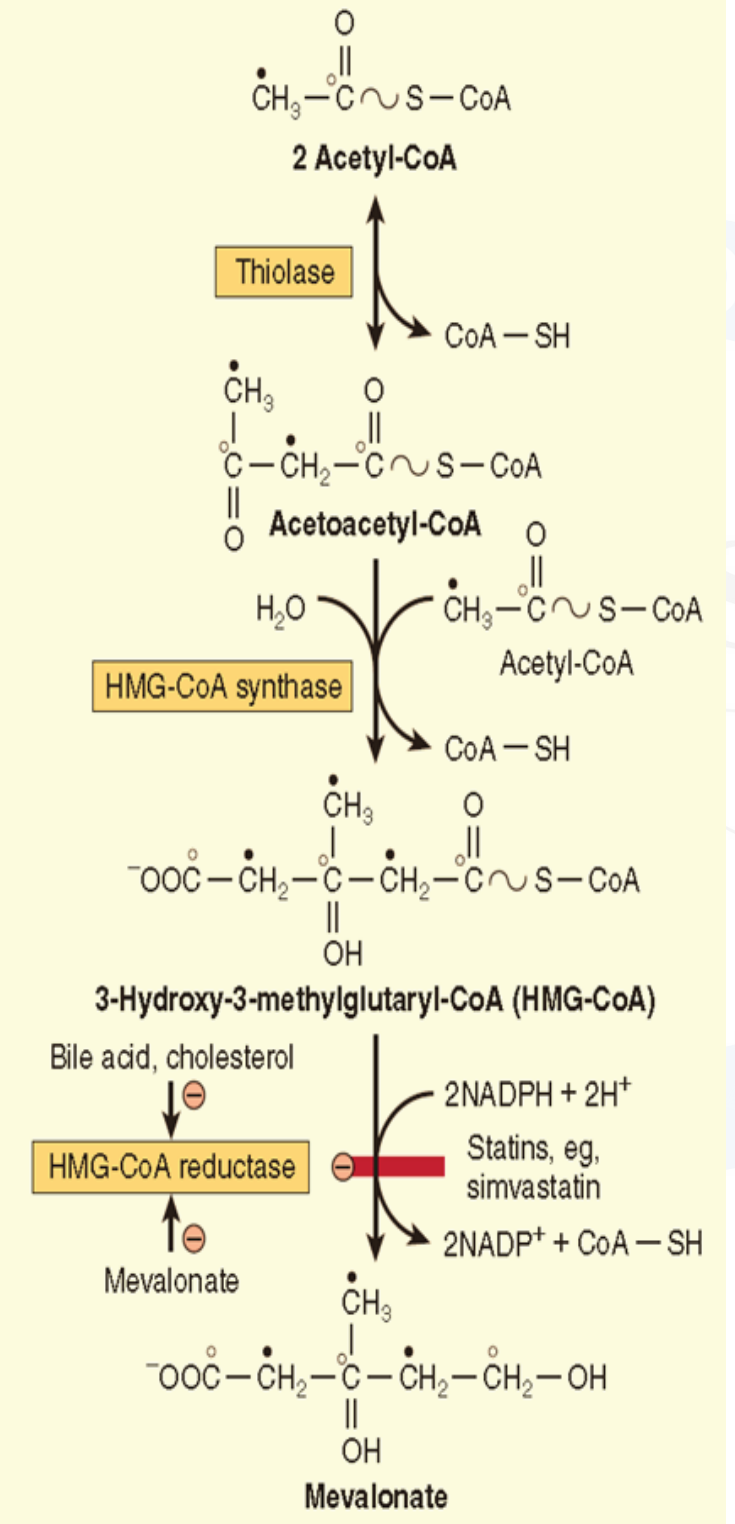
GOLDEN MCQ POINTS

Cholesterol is a steroid containing a 27 carbon cyclo pentano perhydro phenanthrene (CPPP) ring AKA STEROID RING

* Esterification:



Cholesterol Biosynthesis



GOLDEN MCQ POINTS

Source of All Carbon Atoms in Cholesterol →

major sites of synthesis of cholesterol : CYTOSOL of liver, adrenal cortex, testis, ovaries and intestine.

- squalene is 30 carbon precursor of cholesterol

→ Enzyme common to both cholesterol and ketone body synthesis is HMG CoA Synthase

→ Specialized Products Obtained from CHOLESTEROL

PRACTICE MCQS:

Stored TAG is hydrolysed by

a. Pancreatic lipase

b. Lipoprotein Lipase

c. Gastric lipase

d. Hormone sensitive lipase

True about medium chain fatty acids:

a. Require pancreatic lipase

b. Diffuse directly into portal circulation

c. Deposited in adipose tissue

d. All of the above

Which is involved in transport of long chain acyl-CoA in mitochondria

a. Ornithine

b. Xanthine

c. Carnitine

d. Albumin

Beta-oxidation of odd-chain fatty acids produces

a. Succinyl CoA

b. Propionyl CoA

c. Acetyl CoA

d. Malonyl CoA

Insulin causes lipogenesis by all except

a. Increasing acetyl- CoA carboxylase activity

b. Increases the transport of glucose into the cells

c. Inhibits pyruvate dehydrogenase

d. Decreases intracellular cAMP level

Omega 6 FA is

a. Cervonic acid

b. Alpha linolenic

c. Arachidonic

d. Timnodonic

Which is a glycolipid

a. Cerebroside

b. Plasmalogen

c. Sphingomyelin

d. Lecithin

Which of the following is considered a marker of apoptosis when present on the inner leaflet of cell membrane

a. Phosphatidylserine

b. Cardiolipin

c. Cephalin

d. Lecithin

Refsum's disease is due to accumulation of

a. Galactose

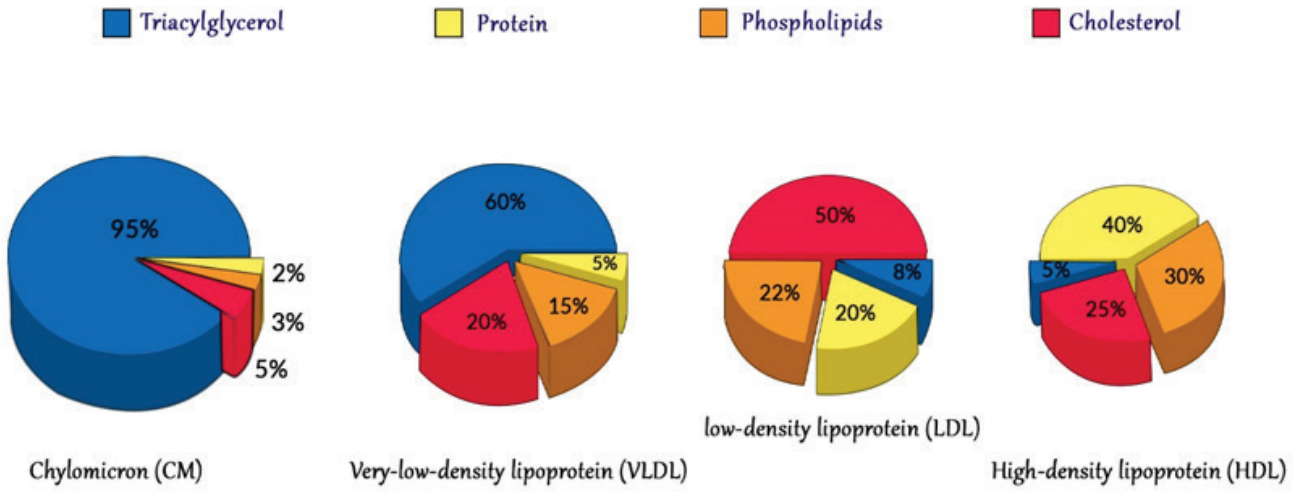
b. Sorbitol

c. Phytanic acid

d.Sphingomyelinase

Workbook Sheet

LIPOPROTEIN CHEMISTRY & METABOLISM



GOLDEN MCQ POINTS

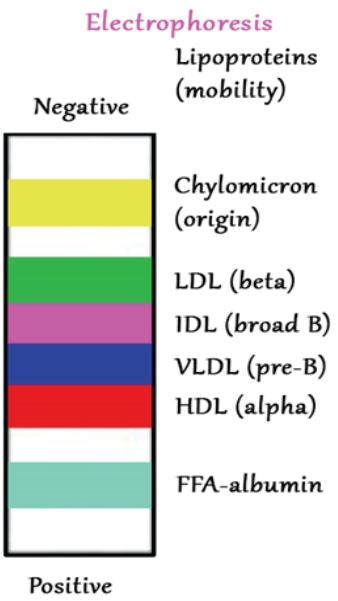
Chylomicrons possess largest size as they contain maximum triglyceride by weight - but least dense, - Assembled in intestinal mucosa - carry Exogenous TAG from Small intestine to peripheral tissue	- VLDL is produced in liver and carry Endogenous TAG from liver to peripheral tissue - Precursor of IDL & LDL	- LDL is a bad type cholesterol as it contains highest concentration of cholesterol - Carry cholesterol from Liver to Peripheral tissues - major culprit in causing atherosclerosis and CAD	- HDL is a good type cholesterol - Highest in density and richest in Phospholipids, protein and lipid - participates in Reverse Cholesterol transport and Cardioprotective - Scavenging action

BEST MARKER OF DYSLIPIDEMIA

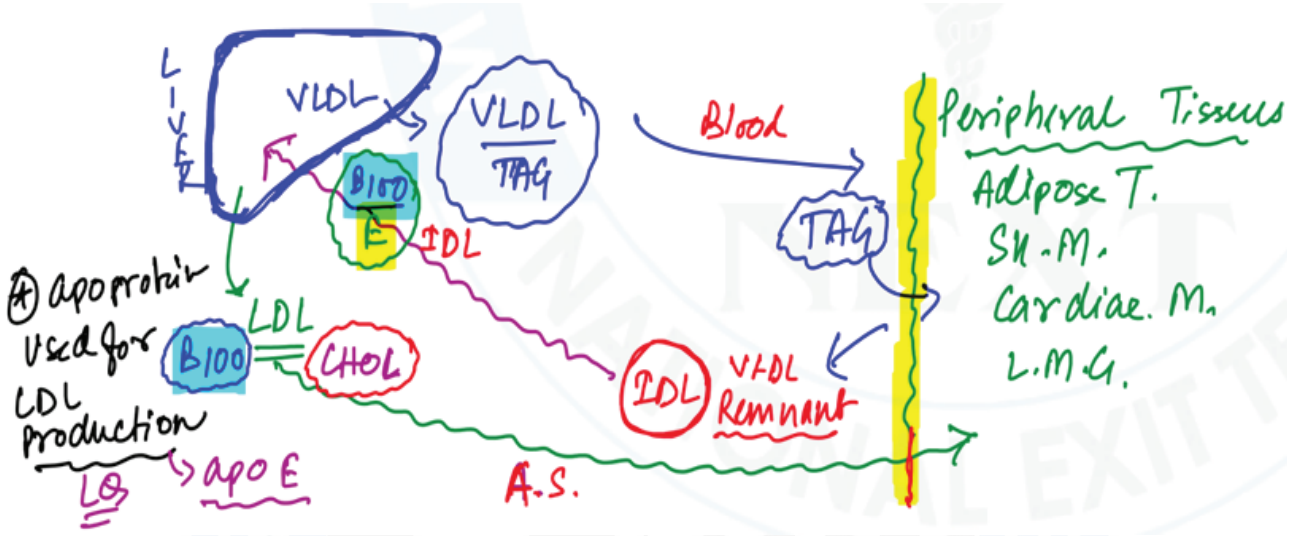
→

But Single Best Parameter -

Separation of PLASMA LIPOPROTEINS by ELECTROPHORESIS



✹ NON-HDL CHOLESTEROL METABOLISM:

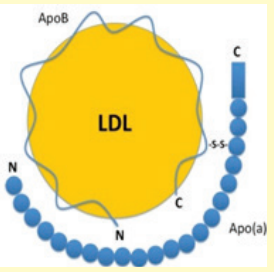


✹ Characteristics of different classes of lipoproteins:

Lipoprotein	Source	Corresponding Apoproteins
1. Chylomicron	Intestine	A-I, A-II, A-IV, B48, C-I, C-II, C-III, E
2. VLDL	Liver	B-100, C-I, C-II, C-III
3. IDL	VLDL	B-100, E
4. LDL	VLDL	B-100
5. HDL	Liver, Intestine	A-I, A-II, A-IV, C-I, C-II, C-III, D, E

REMEMBER: Though Apo B-48 & Apo B-100 are products of the same gene BUT they are differentiated due to RNA editing OR differential RNA splicing

Characteristics of apoproteins and their functions

Apoprotein	Component of	Functions
apo A-I	HDL, Chylomicron	Activation of LCAT; ligand for HDL receptor; Anti-atherogenic
apo A-II	HDL, Chylomicron	Inhibits LCAT; stimulates hepatic lipase but Inhibit Lipoprotein Lipase (LPL)
apo B-100	LDL; VLDL, IDL	Binds LDL receptor
apo B-48	Chylomicrons Only	48% size of B-100.
apo C-I	Chylomicron; VLDL, HDL	Activation of LCAT & LPL; anti atherogenic
apo C-II	Chylomicron; VLDL, HDL	Activates LPL
apo E	Chylomicron; IDL, HDL	Arginine rich protein that has 4 isotypes; ligand for hepatic uptake Apo E-III is most common and Apo E-IV is associated with Alzheimer's Disease & lipoprotein glomerulopathy
apo (a)	Lp(a) 	Attached to B-100 through a disulfide linkage; impairs fibrinolysis as it has Significant homology with plasminogen; highly atherogenic Treatment for high Lp(a) level:

Extra Edge:

LIPOPROTEIN LIPASE:

→ ABSENT in

→ PRESENT in

→ Highest conc. of LPL:

Heart LPL:

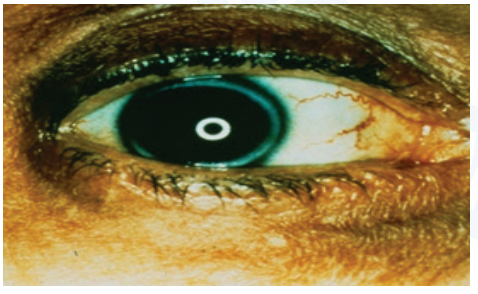
Adipose LPL:

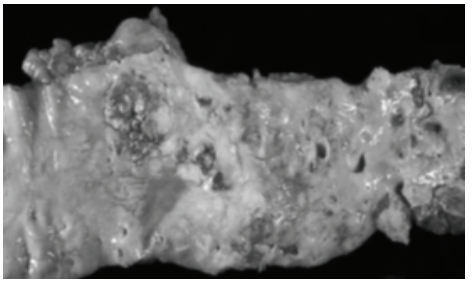
So, during Starvation → Most TAG is mobilized from Adipose tissue to Heart for β-oxidation and energy production

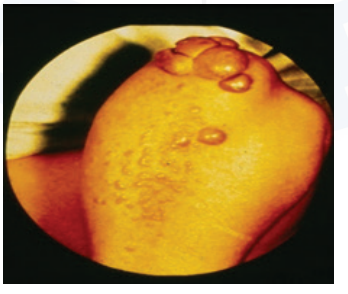
✹ Frederickson-Levy-Lees classification of Hyper-lipoproteinemias:

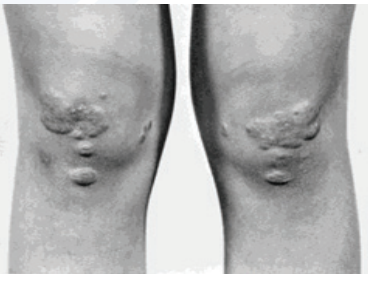
Type	Name	Lipoprotein fraction elevated	Cholesterol level	TAG level	Metabolic defect	Important features
Type I	Chylomi-cronemia or Chy-lomicron Syndrome					

Type	Name	Lipoprotein fraction elevated	Cholesterol level	TAG level	Metabolic defect	Important features
Type IIA	Primary Familial Hypercholesterolemia					- Most fatal - Autosomal dominant - Usually, death by 2nd or 3rd decade





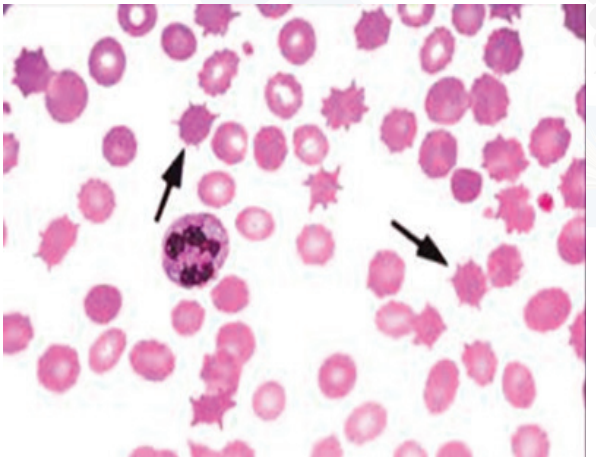


PALMAR XANTHOMA & TUBEROERUPTIVE XANTHOMAS → Type III

◊ Other causes of hyperlipoproteinemias

Familial LCAT deficiency: FISH EYE disease	Wolman's disease: Lysosomal acid lipase (LAL) or cholesterol ester hydrolase deficiency

◊ Hypolipoproteinemias

Abetalipoproteinaemia: Absence of B-apoprotein containing lipoproteins. So, Chylomicron – VLDL - LDL - IDL- 1st SIGN -	Hypo-alpha-lipoproteinemia aka Tangier Disease: ↓ in a LP Serum HDL is ALMOST close to NIL.
 ACANTHOCYTES	 ORA NGE YELLOW TONSILS

PRACTICE MCQS:

Defect in primary familial hypercholesterolemia in:

a. LDL-receptors
b. Apoprotein
c. Apoprotein B
d. VLDL

LCAT Activity is associated with which of the lipoprotein complex

a. HDL
b. VLDL
c. IDL
d. LDL

All of the following statement about apoproteins are true except

a. Apoprotein A-I activates LCAT
b. Apoprotein C-I activates lipoprotein lipase
c. Apoprotein C – II inhibits lipoprotein lipase
d. Apoprotein C- II activates lipoprotein lipase

Apo B is related to all except

a. HDL
b. LDL
c. VLDL
d. Chylomicrons

LDL receptor in liver can be detected by

a. Apo B 100 and apo E
b. Apo B 100 and apo A
c. Apo E
d. Apo E and apo A

Which one of these has the least density

a. Chylomicron

b. HDL

c. LDL

d. VLDL

Action of Lipoprotein Lipase

a. To form remnant

b. Promotes lipolysis

c. Form mature chylomicron

d. To form HDL

Which of the following lipoprotein transport triglyceride from liver to plasma

a. LDL

b. VLDL

c. Chylomicrons

d. HDL

Most important carrier of cholesterol is

a. VLDL

b. HDL

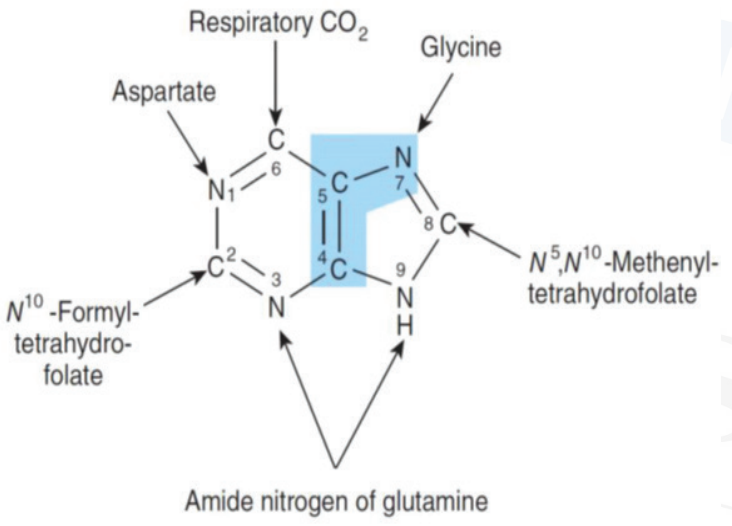
c. LDL

d. Chylomicrons

Workbook Sheet

NUCLEOTIDE CHEMISTRY AND METABOLISM & RELATED DISORDER

Contribution towards formation of purine ring:



* Nitrogen atoms are only contributed by amino acid

* The dietary purines and pyrimidines are neither converted to nucleotides nor incorporated into nucleic acids. They are directly catabolized.

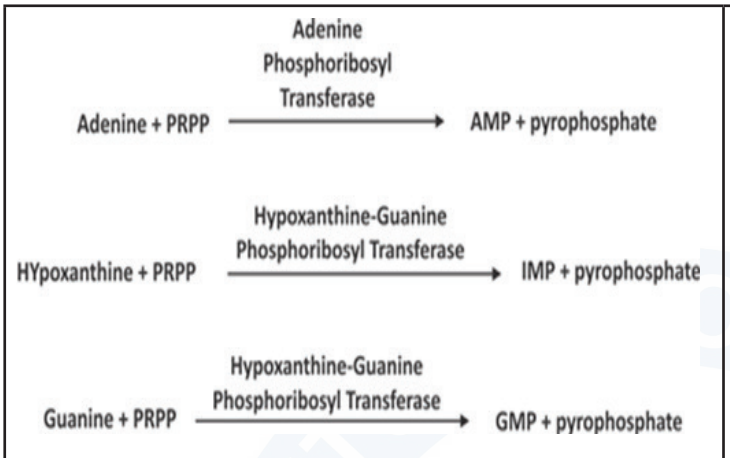
GOLDEN MCQ POINTS from DENOVO PATHWAY:

Preliminary or preparatory step:

Synthesis of PRPP is not considered as a step in the de novo synthesis of purine Nucleotides.

- ✓ Done by Multienzyme complex – Total 10 steps
- ✓ Formation of 5-phosphoribosyl 1-amine by PRPP Glutamyl Amido transferase is the First committed step & Rate limiting
- ✓ The de novo synthesis of one molecule of purine requires 6 ATP
- ✓ IMP – Inosine Mono Phosphate is the first Purine to be produced in our Liver.

Salvage pathway



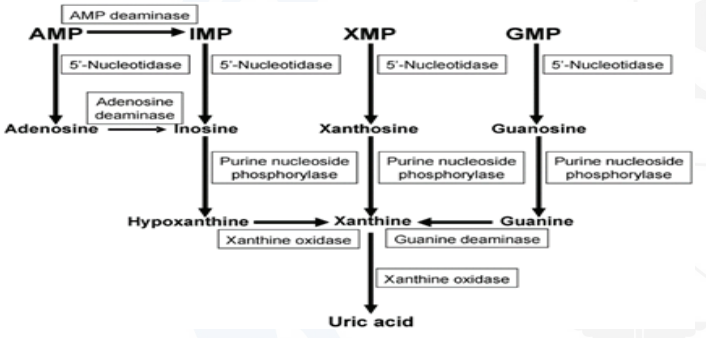
One step process and faster.

Recycling of purines formed by degradation of nucleotides.

Active in Brain & RBC primarily.

Brain tissue has a low level of PRPP glutamyl amido transferase and hence depends in part on exogenous purines.

PURINE CATABOLISM:



End product

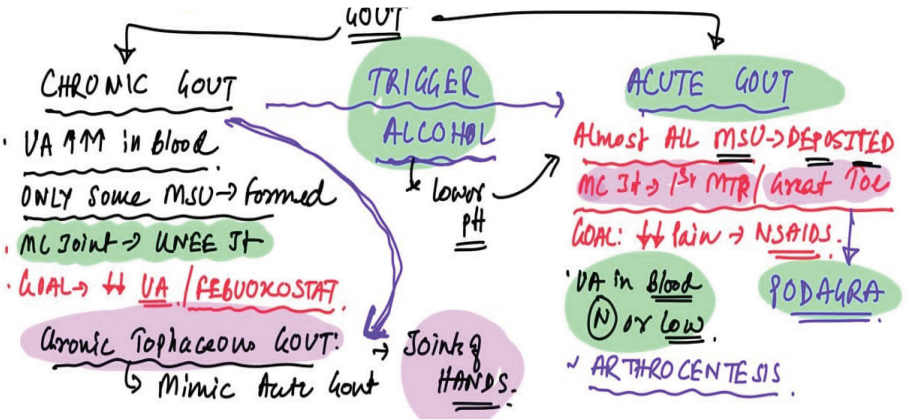
Purine catabolism - uric acid

Pyrimidine catabolism - NH_3, CO_2, beta alanine

✹ GOUT: A Disorder of Purine Metabolism (De Novo Pathway)

Mono Sodium Urate crystals accumulation in synovial fluid.

Tophi: Uric acid is deposited in cooler areas of the body due to decrease in solubility to 4.5 mg/dL at around 30°C

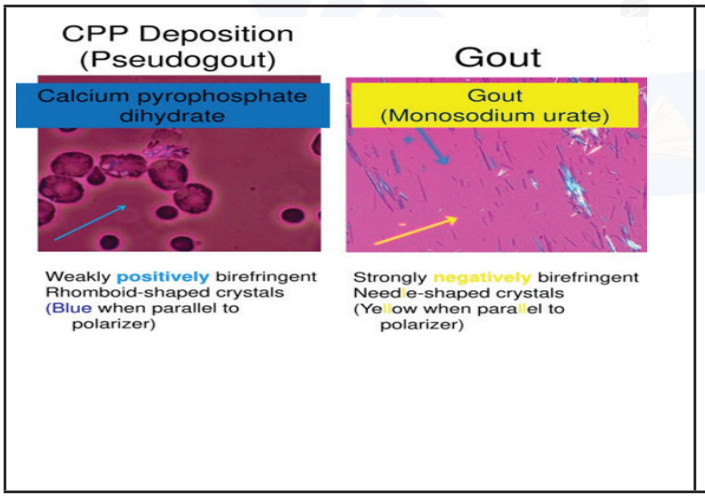


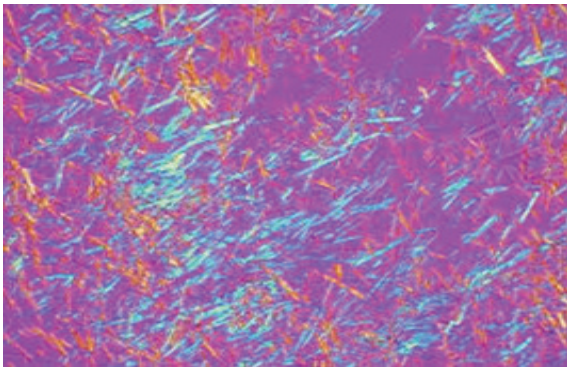
Hallmark of Gout:

- ✓ *Vast majority of Hyperuricemic patients are asymptomatic*
- ✓ *Classical Presentation:*

Patients have a few drinks, go to sleep symptomless, but are awakened during the early hours of morning by excruciating joint pain and swelling followed by abrupt fall in middle aged elderly male or postmenopausal women.

- *Alcohol leads to accumulation of lactic acid and precipitates attack of GOUT*
- *Increased Lactate → Decreases pH → Urate crystals precipitated.*
- *Typical involvement of first metatarsophalangeal joint (big toe) [PODAGRA] → Most Common *(2nd MC is Ankle joint) Lab Dx:*
- *Serum uric acid level can be normal or low at the time of acute attacks.*
- *Definitive Dx → Arthrocentesis : Synovial fluid/ Tophaceous deposits will show birefringent needle shaped MSU crystals with negative elongation under polarized light.*
- *MSU Crystals aggregate in extraarticular & articular structure causing deformity & destruction of hard & soft tissue.*





Pseudogout: Calcium pyrophosphate dihydrate (CPPD) crystal deposition disease aka pyrophosphate arthropathy. *MC affected is knee joint.

✹ Lesch Nyhan syndrome → A Disorder of Salvage pathway of Purine Metabolism due to Complete deficiency of HGPRT

Clinical features

»Gout

»Neurological disorder, Spasticity, dystonia

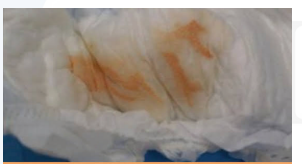
»Self-mutilation → Chewing of fingers → bleeding

»MR, Extremely hostile male child,

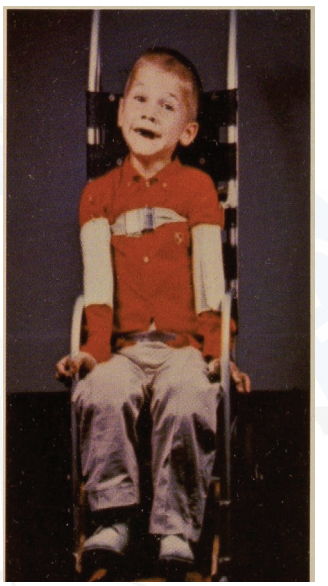
Nephrolithiasis

Orange Sands

in baby's diaper



"Orange sand" crystals in a diaper composed of sodium urate crystals, a classic finding of Lesch-Nyhan syndrome.



✹ Disorder of Pyrimidine Metabolism:

Orotic Aciduria:

Deficiency of UMP synthase functions both as orotate phosphoribosyl transferase and OMP decarboxylase.

Clinical features:

↑OA in Blood & Urine→ ORANGE / PINK crystals in diapers

Growth retardation and megaloblastic anemia (↓Pyrimidines impair DNA/RNA synthesis), Urinary Tract obstruction, Renal insufficiency.

EXTENDED SELF READING ON VITAMINS

VITAMINS:

"A vitamin is a substance that makes you ill if you don't eat it".

Comparison of two types of vitamins		
	Fat soluble vitamins	Water soluble vitamins
Water solubility	Not soluble	Soluble
Absorption	Along with lipids & requires bile salts	*Absorption simple
Carrier proteins	Present	*No carrier proteins required
Storage	Stored in liver	*No Storage
Excretion	Not excreted	Excreted
Deficiency	Manifests only when stores depleted	*Manifests rapidly as no storage
Toxicity	Hypervitaminosis may result	Unlikely, since excess is excreted
Treatment of deficiency	Single large doses	Regular dietary supply is required
Major vitamins	A, D, E and K	B Complex and C
	*Vitamin B12 is an exception	

✹ THIAMINE (VITAMIN B1)

Aneurine (it can relieve neuritis)

• Antiberberi factor

• TPP/TDP/Cocarboxylase is the active form

• Thiamine deficiency is best diagnosed by observing an INCREASED ERYTHROCYTE TRANSKETOLASE ACTIVITY by 25% on adding EXOGENOUS TPP in vitro as compared to baseline.

• Normal Erythrocyte transketolase activity is reduced

→ Physiological Role of Thiamine:

• Central role in energy yielding carbohydrate metabolism.

1. Coenzyme in Oxidative decarboxylation of a-keto acids as a part of Multienzyme complex:

• PDH Complex in Link reaction of Glycolysis & Kreb's cycle

• a-KGDH complex in the Kreb's cycle

• Branched-chain a-keto acid DH Complex involved in the BCAA (leucine, isoleucine & valine metabolism

2. Transketolase, in the HMP Shunt

3. Role in nerve conduction: activates, a chloride channel in the nerve membrane.

Deficiency Manifestations of B1

3 distinct syndromes:

1. Chronic peripheral neuritis

2. Beriberi: Singhalese word, meaning "weakness". Early symptoms are anorexia, dyspepsia, heaviness and weakness

• Wet beriberi: AKA Acute pernicious (fulminating) beriberi (shoshin beriberi).

C/F: High output Cardiac failure, EDEMA, metabolic abnormalities without peripheral neuritis

• Dry beriberi: peripheral neuritis + CNS manifestations, NO EDEMA.

• Infantile beriberi: infants born to thiamine deficient mothers.

3. Wernicke encephalopathy with Korsakoff psychosis aka cerebral beriberi aka WK Syndrome: Thiamine def. in

RIBOFLAVIN (VITAMIN B2)

• First B complex component to be isolated in a pure state. • Warburg's "yellow enzyme" of cellular respiration. • Active form: 2 Co-enzymes FMN and FAD	• Heat & Acid resistant Destroyed by Light & Alkali • Used as a food additive because of its Yellow color. • Riboflavin deficiency is detected by measurement of in vitro ERYTHROCYTE GLUTATHIONE REDUCTASE activity.	Deficiency Manifestations of B2 • Natural deficiency is uncommon, because riboflavin is synthesized by the intestinal flora. • Confined to skin and mucous membranes. i. Glossitis (Greek, glossa = tongue) ii. Magenta colored tongue iii. Cheilosis (Greek, cheilos = lip) iv. Angular stomatitis (inflammation at corners of mouth) v. Circumcorneal vascularization vi. Seborrheic dermatitis vii. Proliferation of the bulbar conjunctival capillaries is the earliest sign of riboflavin deficiency.

NIACIN (VITAMIN B3)

- Pellagra preventing factor of Goldberger.
- Two compounds, Nicotinic acid (Niacin/ vitamin) and nicotinamide/ niacinamide (active form).
- Not strictly a vitamin since it can be synthesized in the body from the essential amino acid tryptophan.
- About 60 mg of tryptophan is equivalent to 1 mg of dietary niacin.

NADPH generating reactions:

1. HMP Shunt pathway
2. Cytoplasmic isocitrate dehydrogenase
3. Cytoplasmic Malic enzyme (malate to pyruvate).

Causes for Niacin Deficiency:

i. Dietary deficiency of tryptophan:

→Maize eaters → Sorghum eaters- Sorghum, contains leucine in high quantities. Leucine inhibits the QPRT enzyme, and so niacin cannot be converted to NAD+ (Leucine pellagra).

ii. Deficient synthesis: Kynureninase, is PLP dependent. So conversion of tryptophan to niacin is not possible in pyridoxine deficiency

iii. Isoniazid (INH)

iv. Hartnup disease

v. Carcinoid syndrome

Deficiency Manifestations of B3

- Pellagra meaning "rough skin".
- Caused by the deficiency of Tryptophan as well as Niacin
- More Common in women as tryptophan metabolism is inhibited by estrogen metabolites.
- 4Ds of Pellagra

i. Dermatitis: Casal's necklace.

ii. Diarrhoea

iii. Dementia

iv. Death

- Therapeutic Use of Niacin:

i. reduces Lp(a) levels

ii. inhibits the flux of free fatty acids from adipose tissue

iii. serum cholesterol is lowered.

PANTOTHENIC ACID (B5)

• "pantos" means everywhere • Active form: Co-enzyme A; CoA is a nucleotide (Requires 4 ATP for synthesis)	→ Physiological Role of Pantothenic Acid: i. Acyl group carrier ii. Acetylation Reactions iii. ACP (acyl carrier protein), A component of FAS complex I	Deficiency Manifestations of B5 • **Gopalan's Burning Foot Syndrome** is manifested as paresthesia (burning, lightning pain) in lower extremities, staggering gait due to impaired coordination and sleep disturbances & spectacle alopecia

PYRIDOXINE (VITAMIN B6)

• 3 related precursor pyridine derivatives; pyridoxine (alcohol)- in plants, pyridoxal (aldehyde) and pyridoxamine – in animals • Active form is pyridoxal 5' phosphate (PLP) which can be obtained from all 3 precursors.	→ Physiological Role of Pyridoxine 1. Transamination reactions. 2. NON OXIDATIVE Decarboxylation reactions 3. Transsulfuration reactions 4. Heme Synthesis – ALA synthase is B6 dependent 5. Production of Niacin 6. Glycogenolysis: • Glycogen Phosphorylase requires 4 molecules of PLP per enzyme molecule → 80% of the body's total vitamin B6 is pyridoxal phosphate in muscle → • This is NOT available in deficiency, but is released in starvation 7. Synthesis of sphingolipids 8. Inhibits Lactation.	Deficiency manifestations of B6 • Can cause Homocystinuria • Sideroblastic / Sideroachrestic anemia (Microcytic hypochromic) • PELLAGRA, • XANTHURENIC ACIDURIA • peripheral neuritis → Assay of Vitamin B6: ERYTHROCYTE TRANSAMINASES ACTIVITY → Toxicity of Vitamin B6: Only water soluble Vitamin with significant toxicity. → Sensory neuropathy

BIOTIN (VITAMIN B7)

Was known as anti-egg white injury factor. AKA VITAMIN H	Physiological Role of Biotin: co-enzyme for carboxylation Biotin-Independent Carboxylation reactions i. Carbamoyl phosphate synthetase I & II, ii. Addition of CO2 to form C6 in purine ring iii. Malic enzyme	Deficiency manifestations of B7 Dietary Biotin deficiency is unknown as it is synthesized by intestinal flora. * Multiple Carboxylase deficiency

FOLIC ACID (B9)

- 'Folium' means leaf of vegetable.

- Absorbed by upper part of jejunum

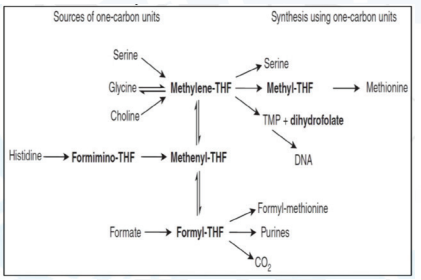
- Chemically Pteroyl Glutamic Acid (max upto seven glutamate residues are linked to the pteroyl group)

- Active form is 5,6,7,8-tetrahydro folic acid (THFA)

Physiological Role of Folic Acid:
THERE ARE MULTIPLE ACTIVE FORMS OF FOLATE IN THE DIET

- One-carbon substituted folates are the carrier of one-carbon groups.

- One carbon pool



- Methyl group in N5-methyl THFA is used for synthesis of active methionine (SAM),

→ Causes for Folate Deficiency:

MC vitamin deficiency in India.

i. Pregnancy: As requirement is increased

ii. Defective absorption iii. Anticonvulsant drugs

iv. Hemolytic anemia v. Dietary deficiency

vi. FOLATE TRAP: Vitamin B12 Deficiency Causes Functional Folate Deficiency.

B12 deficiency results in accumulation of methyl THF that cannot be used — the "folate trap."

Deficiency Manifestations of Folic acid

1. Reduced DNA synthesis
2. Macrocytic Anemia → Most characteristic feature of folate deficiency
3. Reticulocytosis
4. Neural tube defects like spina bifida.
5. Hyperhomocysteinemia & Homocystinuria – atherosclerosis, MI

→ Assessment of Folate Deficiency

i. Blood level
ii. Histidine load test or FIGLU excretion test
iii. AICAR excretion test
iv. Peripheral blood picture.

→ Folic Acid Therapy

→ Therapeutic dose is 1mg of folic acid per day oral

→ Folic acid alone should not be given in macrocytic anemia, because it may aggravate the neurological manifestation of B12 deficiency. Folic acid and vitamin B12 are given in combination to patients.

→ Toxicity of Folic Acid: >1mg masks vitamin B12 deficiency, aggravates & precipitates nerve damage.

Folate Antagonists:
Sulfonamides,
Pyrimethamine,
Methotrexate,
Aminopterin and
Amethopterin

VITAMIN B12

- AKA Cobalamin, extrinsic factor (EF) of Castle and antipernicious anemia factor. Water soluble, heat stable and red in color

→Contains one cobalt atom with 4.35% cobalt by wt.

i. Cyanocobalamin: laboratory artefact

ii. Hydroxy Cobalamin: aka vitamin B12a. Injectable preparations

iii. Adenosyl Cobalamin (Ado-B12): major storage form in liver, functional co-enzyme

iv. Methyl Cobalamin: Major form in blood circulation,

functional co-enzyme

- Absorption, Transport and Storage:

Requires two binding proteins.

1. Cobalophilin from Saliva

2. Castle's Intrinsic factor (IF): A glycoprotein secreted by the gastric parietal cells

- Physiological Role of B12

1. Methyl Malonyl CoA Mutase (Isomerase): In B12 deficiency → Methyl malonic aciduria

2. Methionine synthase or homocysteine methyl transferase folate trap → folic acid scarcity in B12 def.

Causes of B12 Deficiency

1.Nutritional: Common in strict vegetarians

2.Decrease in Absorption: gastrectomy, ileal resection and malabsorption

3.Addisonian Pernicious (fatal) Anemia: Antibodies are generated against IF

Deficiency Manifestations of B12

1. Folate trap: Discussed earlier

2. Megaloblastic anemia →MCC of pernicious anemia (Pernicious means fatal) is failure of the absorption of vitamin B12 rather than dietary deficiency

3. Hyperhomocysteinemia & Homocystinuria

4. Demyelination: d/t ↓ production of SAM

5. Subacute combined degeneration (SCD):

→ failure of methylation of one arginine residue in myelin basic protein d/t methionine deficiency in the central nervous system

→Assessment of B12 Deficiency

Schilling test: Radioactive labelled

(Cobalt-60) B12

ASCORBIC ACID (VITAMIN C)

→James Lind published 'Treatise on Scurvy' in 1753

Benedict's test →Positive in the urine sample after the vitamin C administration.

→It is a vitamin for human beings, higher primates can't produce it in vivo from glucose d/t absence of L-gulonolactone oxidase in Uronic acid pathway

→Physiological Role of Vit C:

Coenzyme for Hydroxylases

i. Cross linking of collagen: Proline and lysine hydroxylases are required for the post translational modification (PTM) of procollagen to collagen to give tensile strength

2. Iron Metabolism: enhances the iron absorption from the intestine by reducing ferric iron to ferrous state (absorbable form)

Deficiency Manifestations of Vitamin C

Gross deficiency of vitamin C results in scurvy

- Infantile Scurvy (Barlow's Disease)
- Hemorrhagic Tendency: collagen is abnormal →Subcutaneous haemorrhage viz. petechiae (mild deficiency) and ecchymoses
- Oral Cavity: Painful, swollen and spongy gum → pulp is separated from the dentine → Gum decay →Tooth loss
- Bone fracture
- Anemia: microcytic, hypochromic

→Therapeutic Use of Vitamin C:

Adjuvant in treating infections

- Wound healing: treatment of ulcer, trauma, and burns.
- Antioxidant role: Can decrease incidence of common cold, cataract & Ca bladder
- Prevents formation of Atheromatous plaque

VITAMIN A

→Active form is present only in animal tissues.

Pro-vitamin, beta-carotene is present in plant tissues.

• Retinoids: compounds with vitamin A activity.

Poly-isoprenoid compounds having a beta-ionone (cyclohexenyl) ring system viz. retinol (vitamin A alcohol), retinal (vitamin A aldehyde) and retinoic acid (vitamin A acid).

*All-trans variety of retinal, aka vitamin A1 is most common form.

*Vitamin A2 is found in fish oils and has an extra double bond in the ring.

*Mobilisation of stored Retinol from liver is ZINC dependent.

*Biologically important compound is 11-cis-retinal

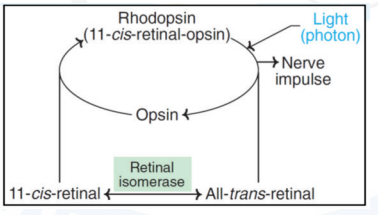
Biochemical Role of Vitamin A:

Wald's Visual Cycle (Rhodopsin Cycle):

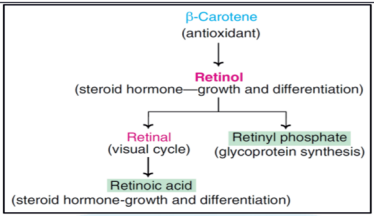
• MOA: conformational change in opsin that l/t generation of nerve impulse

• Main event: On exposure to light Photobleaching of rhodopsin l/t Conversion of 11-cis retinal + Opsin i.e. Rhodopsin to all-trans retinal.

• Membrane HYPERPOLARIZATION is the KEY EXCITATORY RESPONSE to the visual cortex in Brain in genesis of nerve impulse.



[Rods – Towards Periphery : dim light vision; Cones – Towards Center : bright light & colour vision]



Deficiency Manifestations of Vitamin A

→ REVERSIBLE:

• Night Blindness or Nyctalopia: Visual acuity ↓in dim light → cannot read or drive a car in poor light, Dark adaptation time ↑.

• Xerophthalmia: Dry, thick, keratinized, wrinkled conjunctiva → Dry, glazy, lustreless, keratinized cornea → ↓normal transparency.

• Bitot's Spots: Grayish-white triangular plaques firmly adherent to the conjunctiva d/t ↑thickness of conjunctiva.

→ NON REVERSIBLE:

• Keratomalacia: Prolonged xerophthalmia → softening of the cornea → keratomalacia → degeneration of corneal epithelium → corneal opacities → ↑ Bacterial infection corneal ulceration, perforation →total blindness.

• Preventable Blindness: Vitamin A deficiency is the most common preventable cause of blindness in Indian children <5yrs

• Earliest sign of deficiency is a loss of sensitivity to green light.

VITAMIN D (CHOLECALCIFEROL – Vitamin D3)

• "sun-shine vitamin"- Derived either from 7-dehydrocholesterol (minor pathway of cholesterol synthesis in the Malpighian layer of epidermis.) or ergosterol by the action of UV radiations.

• vitamin D3 or cholecalciferol – A Prohormone

• Ergosterol used commercially to produce ergocalciferol or vitamin D2

Synthesis:

• In LIVER: cholecalciferol produced in skin is first transported to liver for hydroxylation @ 25th position

* 25-HCC (Calcidiol) → Major storage form.

* In plasma, 25-HCC transported by "vitamin D binding protein" (VDBP), an a-2 globulin.

• FINAL ACTIVATION In KIDNEY: Hydroxylation @ 1st position by 1-alpha hydroxylase located in mitochondria of proximal convoluted tubules is considered as the Rate limiting step

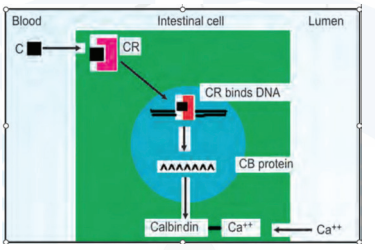
• Calcitriol – Active form, a Hormone (Calciotropic hormone).

* INACTIVE form: 24, 25-dihydroxy Cholecalciferol

→ Biochemical Effects of Vitamin D:

→ Role in Regulation of absorption of Calcium & its homeostasis:

→ Calcitriol promotes the absorption of calcium and phosphorus from the intestine.



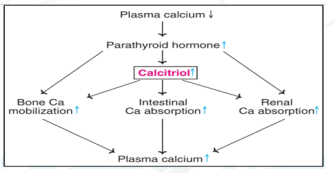
C = calcitriol; CR = calcitriol receptor complex; CB = calbindin

→ Role in Renal Tubules :

Calcitriol →reabsorption of Ca and phosphorus by DCT →both minerals are conserved. (PTH conserves only Ca)

→ Role in Bone Mineralization:

*↑ed by increasing osteoblastic activity.



→ Other Actions of Vitamin D:

◊ Vitamin D deficiency → ↑ Coronary Artery Calcium Score (Agatston Score) → ↑ed chance of Coronary artery disease

◊ Role in Innate Immunity: Modulator of immune response, Binding of Mycobacterium tuberculosis to Toll like receptors (TLR) on macrophages induces the synthesis of VDR and 1-a hydroxylase →activation of 25-HCC →upregulation of specific genes leads to production of antimicrobial proteins

◊ Role in Metabolic Syndrome: Low serum calcitriol is known to perturb cellular function in the endocrine pancreas, which is involved in the pathogenesis of obesity and type 2 diabetes

Deficiency of Vitamin D – anti-rachitic vitamin

Rickets in children and Osteomalacia in adults.

<15 ng/mL is associated with cardiovascular events

→ Clinical Features of Rickets

• Soft and pliable bones in Children d/t insufficient mineralization.

• bone deformities: Weight bearing bones are bent.

C/F: bow legs, knock knee, bossing of frontal bones, and pigeon chest.

• Rickety rosary

• Harrison's sulcus

→ Clinical Features of Osteomalacia

Bones softening d/t insufficient mineralization and ↑ osteoporosis.

• Biochemical parameters: slightly ↓ serum calcium, ↓ serum phosphate.

Vitamin D deficiency never produces severe hypocalcemia. No Tetany is seen.

*Serum alkaline phosphatase, bone isoenzyme, is markedly ↑ed.

Hypervitaminosis D

Therapeutic doses: 2000–4000 IU/day or weekly.

Toxicity → >10,000 IU/day for longer period.

Most toxic in overdose among all vitamins

VITAMIN E (Tocopherol)

Anti-infertility vitamin. (tokos = child birth; pheros = to bear; ol = alcohol), Least Toxic.

◊ Most potent biological (naturally occurring) antioxidant.

• Alpha tocopherol (5, 7, 8-trimethyl tocol) has greatest biological activity.

◊ Selenium has been found to decrease the requirement of vitamin E and vice versa. They act synergistically to minimize lipid peroxidation

→ Biochemical Role of Vitamin E

→ DOES NOT HAVE A PRECISELY DEFINED METABOLIC FUNCTION.

* Most powerful natural lipid-soluble antioxidant in cell membranes & Plasma Lipoproteins. [Protects the polyunsaturated fatty acids (PUFA) from peroxidation]

* Quench lipid peroxidation chain and protect the plasma membranes from the attack of free radicals →chain-breaking, free-radical-trapping antioxidant→ Prevents Rancidity.

• Protects RBC from hemolysis

• Slows down aging process, boosts immunity

• Reduces the risk of atherosclerosis by reducing oxidation of LDL.

Deficiency Manifestations of Vitamin E:

Dietary deficiency in human beings is unknown,

• Seen classically in premature infants that causes Retrolental Fibroplasia

• Neurological disorders d/t poor nerve conduction → spinocerebellar ataxia, retinopathy [Age-related macular degeneration (AMD)], peripheral neuropathy

• Hemolytic anemia of the newborn*

• Resorption of fetuses, Testicular atrophy

VITAMIN K

→Letter "K" is "koagulation vitamin"

• Only fat soluble vitamin with a specific coenzyme function- Needed for production of blood clotting factors.

There are 3 forms of naphthoquinone derivatives, with a long isoprenoid side chain. All 3 are heat stable.

◊ Phylloquinone in plants.

◊ Menaquinone produced by the intestinal bacteria and also found in animals.

◊ Menadione is a water-soluble synthetic form.

Absorption takes place along with fat (chylomicrons) and is dependent on bile salts→ transported with LDL and is stored mainly in liver→ transported in plasma along with beta lipoproteins

→ Biochemical Role of Vitamin K : Vitamin K Dependent Carboxylation

• Vitamin K is necessary for post-translational modification of coagulation factors (inactive zymogens) viz. Factor II (prothrombin); Factor VII (SPCA); Factor IX (Christmas factor); Factor X (Stuart Power factor), Protein C & Protein S by gamma carboxylation of glutamic acid residues in them to form unusual AA gamma carboxy glutamic acid

(Gla Residues) that serves as binding sites for calcium ions (Chelation).

Vitamin K antagonist: →

• Therapeutically used as anticoagulants

* Warfarin: Synthetic analogue of Vit K.

* Dicoumarol: Natural analogue

*A high dose of vitamin K is the antidote to an overdose of warfarin

• Causes for Vitamin K Deficiency

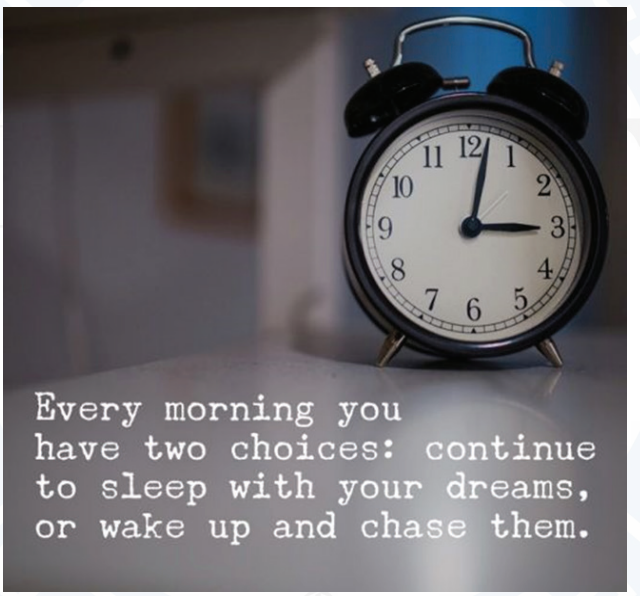
*Malabsorption of lipids d/t obstructive jaundice, chronic pancreatitis, sprue.

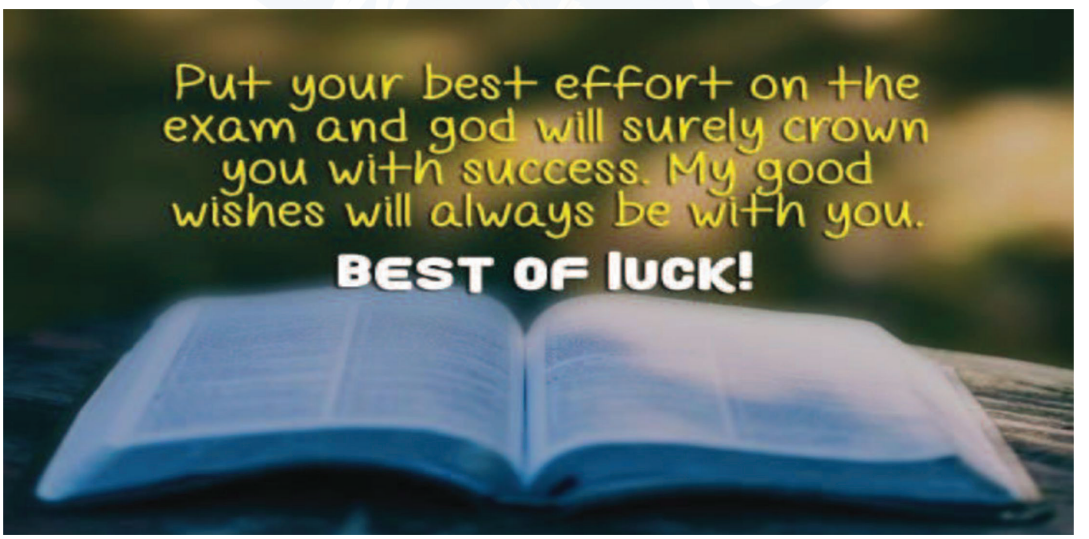
*Prolonged antibiotic therapy and GI infections with diarrhea that destroys normal bacterial flora.

Deficiency Manifestations of Vit K:

• Hemorrhagic disease of the newborn particularly in premature infants

• Prolonged PT (index of liver function)


Every morning you
have two choices: continue
to sleep with your dreams,
or wake up and chase them.


Put your best effort on the
exam and god will surely crown
you with success. My good
wishes will always be with you.
BEST OF LUCK!